



# Business Process Automation

## with Power Automate and Copilot Studio Agents

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WSQ Course Code: TGS-2022017524 · 2-Day Hands-On Training · Modules 1–5 · 18 Labs

**Built entirely in the new Copilot Studio — workflows, agents, skills, tools, channels**

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version 8.2 (6 September 2026) · [www.tertiarycourses.com.sg](http://www.tertiarycourses.com.sg) · <https://lms-tms.tertiaryinfotech.com/>

# Digital Attendance (Mandatory)

## AM, PM and Assessment

1

It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.

## The SSG QR code

2

The trainer/administrator displays the digital attendance QR code from the SSG portal.

## Scan and submit

3

Scan the QR code with your mobile phone camera and submit your attendance.

## 75% minimum

4

A minimum of 75% attendance is required to be eligible for assessment and funding.

**TRAQOM is the SSG digital attendance system — take it three times a day, every day.**

# About the Trainer



## Your Trainer

General Trainer template —  
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

# About the Trainer



**Dr. Alfred Ang**

Principal Trainer  
Tertiary Infotech Academy Pte. Ltd.

## ROLE

Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

## CERTIFICATION

Microsoft Power Platform, AI and data analytics — 20+ years of training experience.

## DELIVERS

WSQ courses on AI, business automation, data science and software engineering.

## FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.



# Let's Know Each Other

1

## Who you are

Your name and your organisation / role.

2

## What you have used

Your experience with Power Automate, Copilot Studio or other automation tools (if any).

3

## What you want to automate

One repetitive business process you would love to automate after this course.

**Keep the process in mind — by Day 2 you will know which parts of it should be a workflow, an agent, or a person.**

# Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively — no question is too small.

3

Mutual respect: agree to disagree.

4

One conversation at a time.

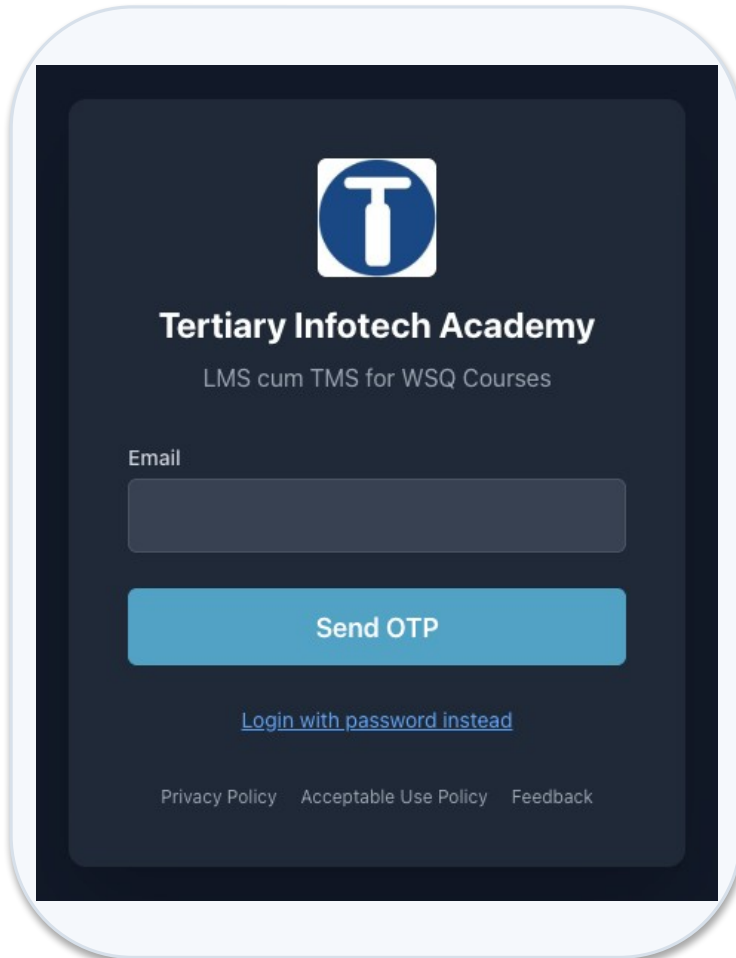
5

Be punctual; return from breaks on time.

6

75% attendance is required.

# Download Course Material





**Tertiary Infotech Academy**  
LMS cum TMS for WSQ Courses

Email

**Send OTP**

[Login with password instead](#)

[Privacy Policy](#) [Acceptable Use Policy](#) [Feedback](#)

## 1 · Sign in

lms-tms.tertiaryinfotech.com — log in with your registered email (OTP or password).

## 3 · Submit answers

Upload your Written Assessment and Practical Performance answers on the LMS.

## 2 · Download materials

Slides (PDF) and the Learner Guide from the course page — your open-book references.

## 4 · TRAQOM survey

Complete the TRAQOM survey via the QR code shown on the LMS.

<https://lms-tms.tertiaryinfotech.com/>

# Lesson Plan — Day 1

## Morning · Day 1 — Automate, then add the agent

|               |                                                                   |     |
|---------------|-------------------------------------------------------------------|-----|
| 9:30 – 10:00  | Welcome, admin, course map                                        | 30m |
| 10:00 – 10:50 | Module 1 — BPA, Power Platform, the new Copilot Studio, workflows | 50m |
| 10:50 – 11:05 | Tea break                                                         | 15m |
| 11:05 – 11:25 | Lab 0 — Environment Setup                                         | 20m |
| 11:25 – 11:55 | Lab 1 — Trigger and Actions                                       | 30m |
| 11:55 – 12:25 | Lab 2 — Log to Excel                                              | 30m |
| 12:25 – 1:25  | Lunch                                                             | 60m |

## Afternoon · Day 1 — Automate, then add the agent

|             |                                               |     |
|-------------|-----------------------------------------------|-----|
| 1:25 – 1:50 | Module 2 — Control flow and human in the loop | 25m |
| 1:50 – 2:25 | Lab 3 — Leave Application Approval            | 35m |
| 2:25 – 3:10 | Lab 4 — Email Classification                  | 45m |
| 3:10 – 3:25 | Tea break                                     | 15m |
| 3:25 – 4:10 | Module 3 — Copilot Studio agents              | 45m |
| 4:10 – 4:45 | Lab 5 — Your First Agent                      | 35m |
| 4:45 – 5:20 | Lab 6 — Procurement Agent with Tools          | 35m |
| 5:20 – 5:50 | Lab 7 — Sales Agent with Knowledge            | 30m |
| 5:50 – 6:20 | Lab 8 — IT Support Agent with Skills          | 30m |
| 6:20 – 6:30 | Day 1 recap                                   | 10m |

# Lesson Plan — Day 2

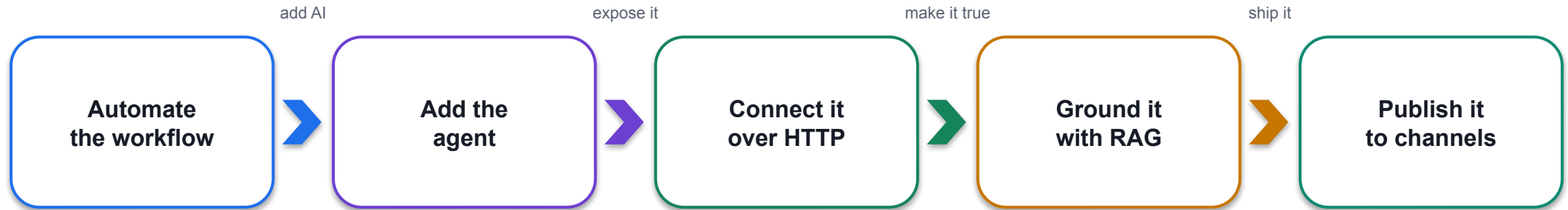
## Morning · Day 2 — Connect, supervise, ground, publish

|               |                                                           |     |
|---------------|-----------------------------------------------------------|-----|
| 9:30 – 9:35   | Day 2 recap                                               | 5m  |
| 9:35 – 10:10  | Lab 9 — Multi-Agent Content Team                          | 35m |
| 10:10 – 10:35 | Lab 10 — Calling Agent from Workflow                      | 25m |
| 10:35 – 11:00 | Lab 11 — Calling Workflow from Agent                      | 25m |
| 11:00 – 11:15 | Tea break                                                 | 15m |
| 11:15 – 11:40 | Module 4 — HTTP, autonomous processes, boundary of agency | 25m |
| 11:40 – 12:25 | Lab 12 — HTTP and Application Approval Agent              | 45m |
| 12:25 – 1:25  | Lunch                                                     | 60m |

## Afternoon · Day 2 — Connect, supervise, ground, publish

|             |                                                     |     |
|-------------|-----------------------------------------------------|-----|
| 1:25 – 1:55 | Lab 13 — HTTP and Chatbot                           | 30m |
| 1:55 – 2:30 | Lab 14 — HTTP and Human Review                      | 35m |
| 2:30 – 2:45 | Module 5 — Retrieval Augmented Generation           | 15m |
| 2:45 – 3:15 | Lab 15 — RAG with Knowledge Base                    | 30m |
| 3:15 – 3:45 | Lab 16 — RAG with Pinecone                          | 30m |
| 3:45 – 4:00 | Tea break                                           | 15m |
| 4:00 – 4:25 | Lab 17 — Publish to Teams, M365 Copilot and the Web | 25m |
| 4:25 – 4:30 | Synthesis                                           | 5m  |
| 4:30 – 5:30 | <b>Written Assessment (SAQ)</b>                     | 60m |
| 5:30 – 6:30 | <b>Practical Performance (PP)</b>                   | 60m |

# What You'll Learn



## 1 Design

Read a business process and decide which parts must be deterministic, which may be AI, and where a person must sit.

## 2 Build

Create Copilot Studio workflows and agents with instructions, knowledge, skills, tools and connected agents.

## 3 Govern

Place the human where the consequence lands, test what the agent must refuse, and publish only what is safe.

**By the end you can build the automation, add the agent, and defend where you put the human.**

# Briefing for Assessment

1

Place phones and other materials under the table or on the floor.

2

No photos or recording of assessment scripts.

3

No discussion during the assessment.

4

Use a black/blue pen for hard-copy assessments.

5

No liquid paper / correction tape.

6

Scripts are collected when time is up.

# Assessment & Funding

1

## Written Assessment

1 hour · open book  
Short-answer questions on Modules 1–5 and the lab concepts.

2

## Practical Performance

1 hour · open book  
Build and verify a working workflow and agent in the training environment.

3

## Evidence & funding

Submit on the LMS.  
A Competent result plus the attendance requirement releases the WSQ funding.

**Day 2, 4:30 – 6:30 PM: Written Assessment (SAQ) 1 hr, then Practical Performance 1 hr.**

### OPEN BOOK

Slides, Learner Guide and approved materials only. An appeal process is available if required.



# Assessment Flow



# Eighteen Connected Labs — Day 1

0

## Environment Setup

Training Class Sandbox, Dataverse, credits

1

## Trigger and Actions

Forms trigger → Outlook; dynamic content

2

## Log to Excel

Excel row first, then the email

3

## Leave Application Approval

Approvals: the workflow pauses for a manager

4

## Email Classification

Classify node, five ports, Human review in Teams

5

## Your First Agent

HR agent: instructions + knowledge + Preview

6

## Procurement Agent with Tools

Skills + a workflow as a tool + Excel audit row

7

## Sales Agent with Knowledge

20 brochures in SharePoint; refuse to invent

8

## IT Support Agent with Skills

Five SKILL.md packages; the rule none enforce

**Day 1: build the workflow, add control flow and a human, then build four agents.**

# Eighteen Connected Labs — Day 2

9

## Multi-Agent Content Team

Manager → Research → Blog → Review

10

## Calling Agent from Workflow

Workflow calls M365 Copilot, posts to Teams

11

## Calling Workflow from Agent

Agent calls a workflow tool (When an agent calls the flow)

12

## HTTP and Application Approval Agent

HTTP POST, structured output, SharePoint, four decisions

13

## HTTP and Chatbot

HTTP chatbot alone in public

14

## HTTP and Human Review

HTTP + Human review gate + handover log

15

## RAG with Knowledge Base

Knowledge source in the Agent node

16

## RAG with Pinecone

Pinecone: chunk, embed, top\_k

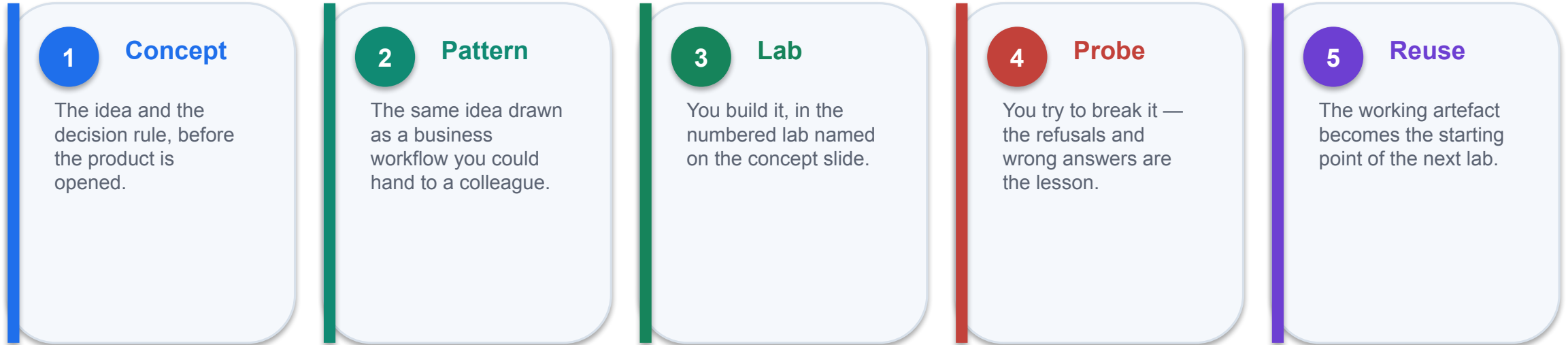
17

## Publish to Teams, Microsoft 365 Copilot and the Web

Channels +: Teams, M365 Copilot, demo website

**Day 2: agents and workflows call each other, go over HTTP, get grounded, and get published.**

# How This Course Works



**Concepts first. Every lab in this course is an application of a concept you have already seen.**

The probes matter as much as the builds. An agent that answers ten questions correctly and the eleventh confidently wrong has not been tested — it has been demonstrated.

# Three Environments — Where You Build

Three environments back this course. You build in exactly one of them — and it is not the one holding the reference copies.

| Environment                                   | Type      | What it is for                                                                                                                        |
|-----------------------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------|
| Training Class 1 / 2 / 3                      | Sandbox   | YOURS — every lab you build goes here. One per class; the trainer resets it between cohorts                                           |
| TGS-2022017524-Business Process Automation... | Sandbox   | Master reference environment: the trainer's Lab N - ... (DO NOT DELETE) workflows and agents. Read-only — open to compare, never edit |
| Copilot Studio Training                       | Developer | The original build environment. Retired from classroom use — a Developer environment is single-user                                   |

Developer

One user only. Free, persistent — but a class cannot share it.

Sandbox

Multi-user AND resettable in one click. What every class uses.

Trial

Self-deletes after 30 days, taking the class's work with it.

**Sandbox is the only type that is both multi-user and resettable — so your class environment is disposable on purpose.**

# Naming Convention in the Tenant

Everything in the tenant is named after its lab, so a learner, a trainer and an auditor can all find it.

| Artefact                        | Exact name                                                               | Rule                                                                                                                                  |
|---------------------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Trainer's reference copy</b> | Lab 6 - Raise Requisition (DO NOT DELETE) · Lab 6 - Proc (DO NOT DELETE) | In the master reference Sandbox. Workflows AND agents end (DO NOT DELETE); agent base names are shortened to fit the 30-character cap |
| <b>Your own build</b>           | Lab 6 - Procurement Agent                                                | In your Training Class environment. Name it exactly as the Learner Guide does                                                         |
| <b>Helper workflow</b>          | Lab 6 - Raise Requisition                                                | The lab prefix, then the workflow's job                                                                                               |
| <b>Data artefact</b>            | Lab 6 - Requisition Log.xlsx · Lab 12 - Customers                        | Excel, SharePoint lists and folders carry the prefix too                                                                              |
| <b>Pinecone index</b>           | lab16-course-brochures                                                   | Lower-case, hyphenated — the only exception, imposed by Pinecone                                                                      |

## YOUR CLASS ENVIRONMENT, ONE DESIGNER

You build in the Training Class Sandbox your trainer assigned (e.g. Training Class 1), in the new Copilot Studio designer at [copilotstudio.microsoft.com](https://copilotstudio.microsoft.com). The (DO NOT DELETE) reference copies live in a separate master reference Sandbox — open to compare, never to edit.

# Your Training Account

Two sets of credentials. Your trainer assigns you one of each — use the same pair for every lab.

## 1 · MICROSOFT 365 PREMIUM — [m365.cloud.microsoft](https://m365.cloud.microsoft) · Office 365 + Copilot 365, 6 learners per account

training1-tertiary@outlook.com

Password: from your trainer

Learners 1–6

training2-tertiary@outlook.com

Password: from your trainer

Learners 7–12

## 2 · COPILOT STUDIO / POWER AUTOMATE — [copilotstudio.microsoft.com](https://copilotstudio.microsoft.com) · your class environment

training1@tertiaryinfotech.onmicrosoft.com

Password: from your trainer

Training Class 1

training2@tertiaryinfotech.onmicrosoft.com

Password: from your trainer

Training Class 2

Your trainer gives you the passwords in class. Sign in to Microsoft 365 at [m365.cloud.microsoft](https://m365.cloud.microsoft), Copilot Studio at [copilotstudio.microsoft.com](https://copilotstudio.microsoft.com).

### STAY IN YOUR CLASS ENVIRONMENT

Confirm your Training Class environment (e.g. Training Class 1) shows bottom-left before you build anything.

# Access the Hands-On Labs

## 1 18 labs

labs/Lab N - <Title>/index.md — Lab 0 setup plus Labs 1–17, each with its own step-by-step guide, assets and screenshots.

## 2 Ready assets

Forms, Excel workbooks, brochure PDFs, HTML pages, JSON schemas, SKILL.md packages and agent instructions.

## 3 Test data

Sample emails, question sets and probe cases written to make each workflow and agent fail.

## 4 Learner Guide + LMS

The Learner Guide reproduces every lab; download it and the slides from [lms-tms.tertiaryinfotech.com](https://lms-tms.tertiaryinfotech.com).

**Work in order. Every lab reuses the one before it; nothing is rebuilt from scratch.**



## MODULE 1

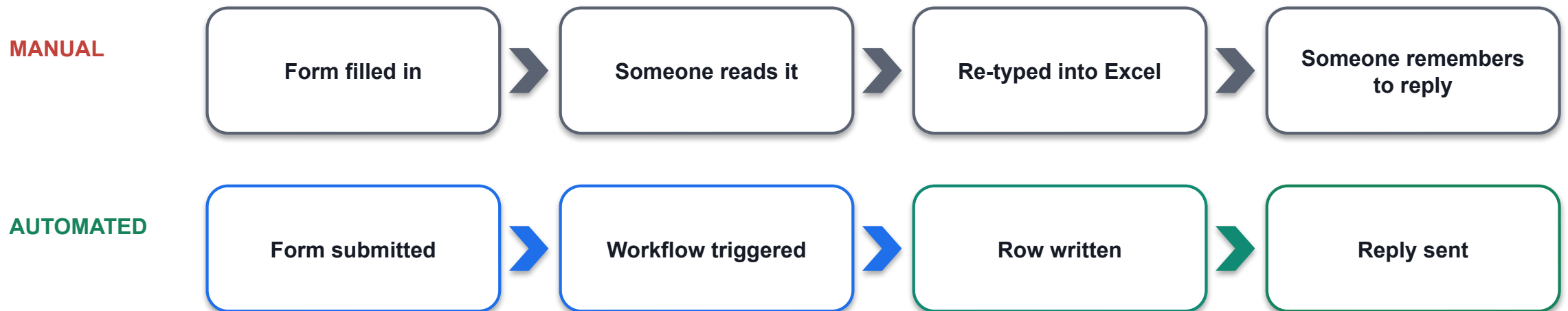
# Business Process Automation and Power Automate

What automation removes · the Power Platform · the new Copilot Studio · workflows, triggers, actions, connections · dynamic content, Activity, commit order · Copilot Credits

Applied in Lab 0, Lab 1 and Lab 2

# What Business Process Automation Removes

Business process automation is not "software that does work faster". It removes the hand-off where a person re-types what another system already knows.



## Consistency

The same input produces the same output, every time, at 3am.

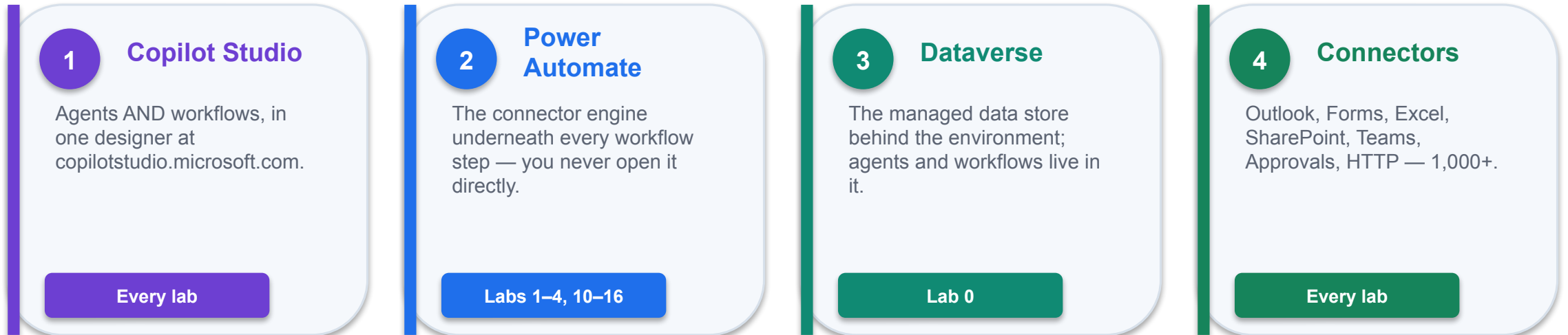
## Traceability

Every run leaves an Activity record someone can read a year later.

## Capacity

People are moved to the judgement calls the machine cannot make.

# The Power Platform, and the Environment That Holds It



## THE ENVIRONMENT IS THE CONTAINER

Workflows, agents and data all live inside one Power Platform environment. You build in the Training Class Sandbox assigned to your class — with Dataverse and Copilot Credits. Build in the Default environment and the first Agent node fails with `InsufficientMcsCredits`.

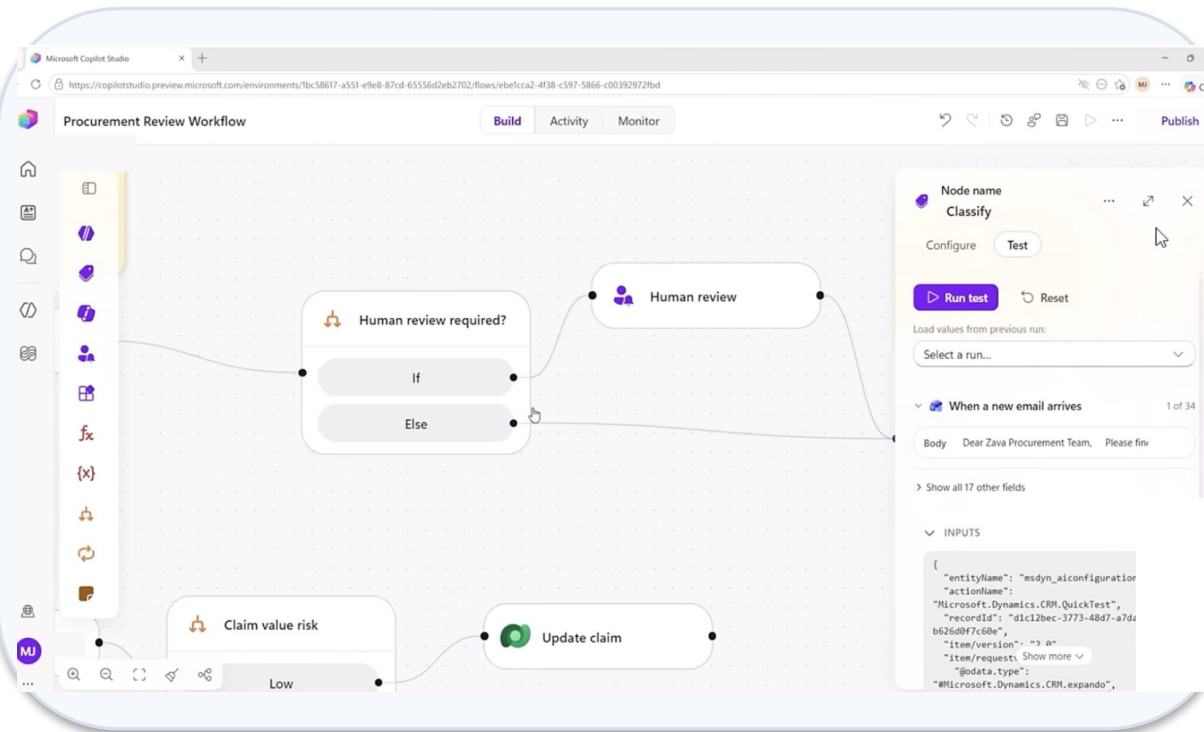
### LAB 0

## Lab 0 — Environment Setup

Training Class Sandbox (assigned by your trainer) · Power Platform admin centre + Copilot Studio · 20 min

# The New Copilot Studio — Home, Agents, Workflows

copilotstudio.microsoft.com — the New experience toggle (top right) switches the whole product to the GitHub Copilot harness.



## Home

A natural-language box: describe the process and Copilot Studio proposes an agent, a workflow, or both (Steps and Artifacts panel).

## Agents

New agent → the designer with Build / Preview / Evaluate / Monitor tabs.

## Workflows

New workflow → the canvas with a Start node and Add a step. Columns: Name · Status · Owner · Enabled.

## Other ways to build

The standard harness (topics, agent flows) is still there — we do not use it in this course.

**Environment name bottom-left. Check it before every build: your Training Class environment.**

# Three Harnesses — Pick One per Agent

"The harness sits between the AI model and the agent or workflow you build" — you choose one per agent, and cannot convert later.

|                 | GitHub Copilot harness (GA 3 Aug 2026)                                | Standard harness                            | Copilot chat harness                          |
|-----------------|-----------------------------------------------------------------------|---------------------------------------------|-----------------------------------------------|
| Best for        | Complex, multi-step business processes                                | Rule-based agents, structured conversations | Extending Microsoft 365 Copilot Chat          |
| How it works    | Reasons through a goal step by step; retries, finds alternative paths | Follows the topics and rules you define     | Connects enterprise knowledge to Copilot Chat |
| Files           | Creates and edits Word, Excel, PowerPoint, PDF in a sandbox           | Not a focus                                 | Not a focus                                   |
| Skills & memory | Yes                                                                   | Not a focus                                 | Not a focus                                   |
| Automation      | Workflows (visual canvas, agent nodes)                                | Agent flows (Power Automate designer)       | —                                             |
| Billing         | Copilot Credits — usage-based, from the moment you build              | Fixed rate card; M365 Copilot fair use      | Consumption or M365 Copilot licence           |

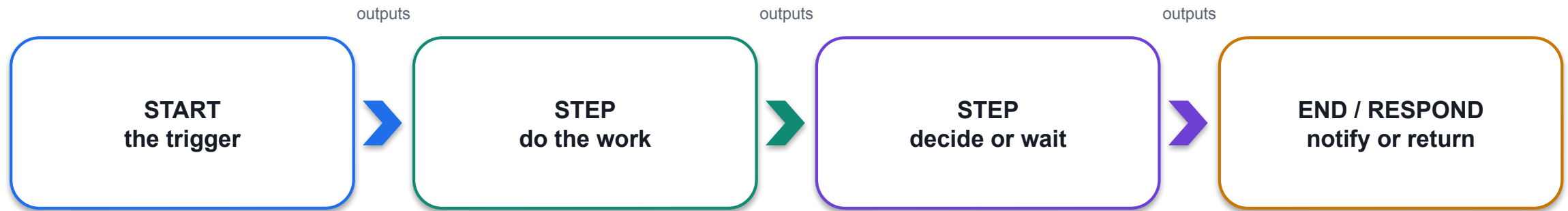
**This course: every agent and workflow uses the GitHub Copilot harness — the "new experience".**

# Old Vocabulary → New Vocabulary

If you learned Copilot Studio before 2026, translate as you go.

| Standard harness (classic)            | GitHub Copilot harness (new experience)                           |
|---------------------------------------|-------------------------------------------------------------------|
| Topics, triggers, branching nodes     | Instructions — natural language, always-on enhanced orchestration |
| Agent flows (Power Automate designer) | Workflows — Start node, Add a step, node-level Test               |
| Prompts (Prompt builder)              | Inline Agent node in a workflow, or a skill in an agent           |
| Child agents                          | Connected agents                                                  |
| Actions / plugins                     | Tools = Connectors, MCP servers, Workflows                        |
| Overview / Settings pages             | Build · Preview · Evaluate · Monitor tabs                         |
| Test pane                             | Preview tab with End user preview toggle and the reasoning card   |
| Messages / capacity                   | Copilot Credits                                                   |

# Anatomy of a Workflow — the Start Node and Its Triggers



Click the Start node to choose the trigger type. Every workflow in every lab is this shape.

| Start node trigger type             | What starts the run                                                      | Where          |
|-------------------------------------|--------------------------------------------------------------------------|----------------|
| <b>Manual</b>                       | A person runs it on demand; you define typed inputs                      | Lab 10 (Topic) |
| <b>Recurrence</b>                   | A schedule — set the time zone or it runs on UTC                         | Reference      |
| <b>Connector</b>                    | An event in a system: new Forms response, new email, new SharePoint item | Labs 1–4       |
| <b>HTTP request</b>                 | When a HTTP request is received — a website or app calls in with JSON    | Labs 12–16     |
| <b>When an agent calls the flow</b> | An agent decides to use this workflow as a tool                          | Labs 6, 11     |

# The Add-Step Palette

The + after any node opens the Add panel. Everything you can put on the canvas is one of these.

|                                                                                              |                                                                                   |                                                                                        |                                                                                       |
|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Agent</b><br>An inline or existing agent — the model, inside the workflow                 | <b>Classify</b><br>Sort text into categories; each becomes an output port         | <b>M365 Copilot</b><br>Send a prompt to Microsoft 365 Copilot or one of its agents     | <b>Human review</b><br>Request for information — pause for a person (Teams / Outlook) |
| <b>Connector</b><br>Any of 1,000+ connector actions: Outlook, Excel, SharePoint, Teams, HTTP | <b>Function</b><br>Data Operations (Compose, Parse JSON, Select), text, date/time | <b>Variable</b><br>Initialize, Set, Append — a named value that changes during the run | <b>If/Else</b><br>One condition, two branches                                         |
| <b>Switch</b><br>One value, many cases plus a default                                        | <b>Loop</b><br>For each item in an array; Do until                                | <b>Respond to the agent</b><br>Return outputs to the calling agent (tool workflows)    | <b>End</b><br>Stop the run; send the HTTP Response                                    |



# Triggers — What Is Allowed to Start a Process

Four trigger families. The one you choose is a statement about who or what is allowed to start the process.

**Manual**  
  
**A person presses Run**  
  
Typed inputs on the Start node;  
Test → Enter manual trigger inputs

**Scheduled**  
  
**A clock reaches a time**  
  
Recurrence — the schedule starts  
when you publish

**Automated**  
  
**An event happens  
in a system**  
  
New form response · new email ·  
new SharePoint item · new file

**Request**  
  
**Something calls in  
from outside**  
  
When a HTTP request is received ·  
When an agent calls the flow

| Trigger used in this course                          | Where        | What it means                                |
|------------------------------------------------------|--------------|----------------------------------------------|
| Connector — When a new response is submitted (Forms) | Labs 1, 2, 3 | A business event starts the process          |
| Connector — When a new email arrives (V3) (Outlook)  | Lab 4        | The inbox is the queue                       |
| HTTP request — When a HTTP request is received       | Labs 12–16   | A website posts JSON and waits for JSON back |
| When an agent calls the flow                         | Labs 6, 11   | A conversation decides to call a tool        |

# Actions — The Six Families You Will Use

A step either moves data, decides something, waits for a person, calls the model, or reaches outside the workflow.

## Data — Function

Compose · Parse JSON · Select ·  
Filter array · Join · expressions

Shape and normalise values before anything trusts them.

## Connector actions

Send an email (V2) · Add a row into a table ·  
Create item · Post message in a chat or channel

Do the real work in a business system.

## Control

If/Else · Switch · Loop ·  
Variable · End

Decide which path the run takes.

## Human in the loop

Human review (Request for information) ·  
Approvals: Start and wait for an approval

Suspend the run until a person responds.

## AI

Agent node · Classify · M365 Copilot ·  
structured output

Let the model decide, inside a workflow that does not.

## Integration

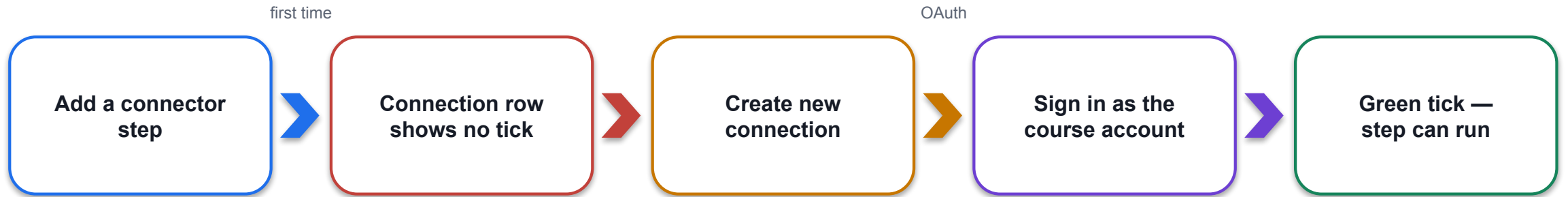
HTTP · Response / End · Respond to the agent ·  
child workflows

Reach outside the platform, or answer the caller.

**You will use every one of these families by the end of Lab 16.**

# Connections — Who the Step Runs As

A connector action runs as somebody. The connection is that somebody — and it is created once, then reused.



## Per connector, per user

Outlook, Excel Online (Business), SharePoint, Teams, Forms and Approvals each need their own connection.

## Tenant boundary

Human review and Approvals can only be assigned to people in the tenant — external addresses fail at run time.

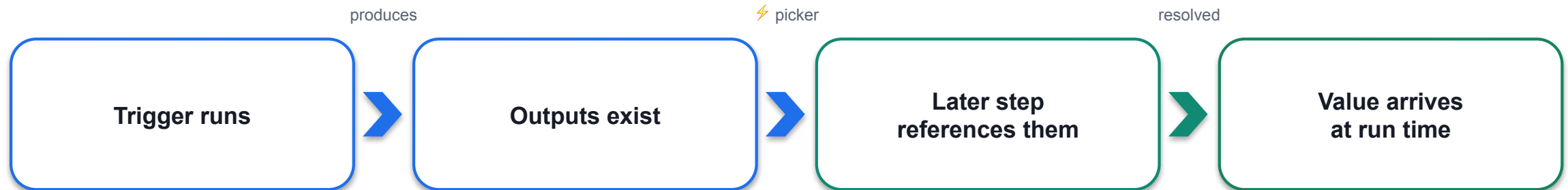
## Whose mailbox?

Send an email (V2) sends from the connection's account. In class that is your training account.

## Publish installs them

Run flow test saves, publishes and installs connections before it triggers.

# Dynamic Content — The Token, Not the Text



## ✓ INSERTED WITH THE ⚡ PICKER (or / command)

- Renders as a coloured token in the field
- Resolves at run time to what the earlier step produced
- The picker shows sample values from the last completed run

## ✗ TYPED BY HAND, OR PASTED @{...}

- Stays dead text — the run is green, the value arrives empty
- A reference to nothing resolves to empty, not to an error
- Rich-text boxes mangle pasted expressions

Typed expressions belong in the `</>` expression editor (concat, formatDateTime, utcNow) — never in a rich-text Instructions box.

### LAB 1

## Lab 1 — Trigger and Actions

Form to email confirmation · Copilot Studio workflow + Microsoft Forms + Outlook · 30 min

# Activity, Run Details and Node-Level Testing

Two ways to test, and one honest record. Every lab is verified from Activity, not from the chat window.

| What you want to verify    | How                                                                                              | What you see                                                   |
|----------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| A single node in isolation | Test this node — the Test tab in the node's side panel; Load values from previous run            | Runs the node against the connector's API                      |
| The full workflow          | Run flow test — the play button top right; saves, publishes, installs connections                | Manual/HTTP → enter inputs; connector → Waiting for trigger... |
| What happened              | Activity panel — Running · Waiting · Succeeded · Failed · Canceled; open a run, expand each node | Inputs and Outputs per node; the reasoning of every AI node    |

THE THREE STATES THAT LOOK ALIKE

Succeeded with the right data · Succeeded with empty data · Still Running because it is waiting for a person.  
Only the first is done. The second is the dangerous one — read the upstream node's Outputs, not the failing node's error.

LAB 2

Lab 2 — Log to Excel

Log the enquiry, then send the email · Copilot Studio workflow + Excel Online · 30 min

# Order Matters — Commit Before You Confirm

The order of two steps is a business decision, not a technical one.

## LOG, THEN CONFIRM ✓

- Add a row into a table (Excel)
- then Send an email (V2)
- If the email fails, the enquiry is still on the register and someone can chase it

## CONFIRM, THEN LOG ✗

- Send an email (V2)
- then Add a row into a table
- If the row fails, you have promised a reply that nobody can see — the evidence is gone

**Commit the record of the obligation before you create the obligation. This is Lab 2 in one sentence.**

Same rule on Day 2: Lab 12 writes the Onboarding Log before the email; Lab 14 logs the draft before the Human review gate.

### LAB 2

## Lab 2 — Log to Excel

Log the enquiry, then send the email · Copilot Studio workflow + Excel Online · 30 min

# Copilot Credits and Billing — What Changed on 1 September 2026

"Billing starts when you start building." Copilot Credits cover LLM tokens, tools (including knowledge and MCP) and the harness itself.

| Feature                       | Covered by an M365 Copilot licence?                 | When Copilot Credits are billed                      |
|-------------------------------|-----------------------------------------------------|------------------------------------------------------|
| GitHub Copilot harness agents | No                                                  | Billed for all usage — build, Preview, Evaluate, run |
| Workflows                     | Only inside a standard-harness agent, licensed user | All other cases — every action consumes capacity     |
| Standard harness agents       | Yes, in Microsoft 365 channels                      | No M365 Copilot licence, or a channel outside M365   |
| Computer use (CUA)            | No                                                  | All usage                                            |

## 1 Sep 2026

Developer and trial environments moved to usage-based billing. Our class Sandbox environments are billed per run.

## Where to look

Agent → Monitor tab; PPAC → Licensing → Copilot Studio → Manage Agents; Copilot Credits estimator.

## Zero credits = every reply fails

"You need credits to continue ... Error code: EnforcementUsageCredits" in Preview, the demo website and every Agent/Classify/Copilot node. Build and Publish still work; allocate credits in PPAC first.

# Power Platform Admin Center — Settings for This Course

What the admin set up in the Power Platform admin center (PPAC) before class — and why each setting matters.

## Class environment

1

One Training Class Sandbox per cohort, with Dataverse. Reset between classes. Never the Default environment.

## Copilot Credits allocation

2

Licensing → Copilot Studio → Manage Copilot Credits → this environment; Draw from the available capacity in my tenant.

## Agent limits

3

Manage Agents → Set limit → monthly limit, notification threshold, Stop usage toggle.

## External models

4

Turn on external models (Anthropic, Mistral, xAI) + M365 admin consent — needed for Claude Opus 5.

## Preview / cross-region

5

Preview and experimental AI models; Move data across regions (Singapore models are cross-geo).

## Entra Agent ID

6

Automatic for every agent since July 2026 — Conditional Access and DLP per agent; Harness column in Manage → Copilot Studio.



# What's New in 2026 — The Timeline

2026 in five words: harness, skills, memory, workflows, governance. One line per month.

| 2026 | What shipped                                                                                                                              |
|------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Jan  | Evaluation enhancements; activity maps; CSV test-set template; Copilot Studio extension for VS Code (GA)                                  |
| Feb  | Ticket-based Graph connectors; Claude models per prompt in Prompt builder                                                                 |
| Mar  | Agent node in flows; agent evaluations GA + multi-turn tests; Work IQ tools (preview); Claude Sonnet 4.6 / Opus GA                        |
| Apr  | Agent-to-agent (A2A) protocol GA; evaluations via REST API; agent usage estimator; GPT-5.5 Reasoning (Deep)                               |
| May  | Computer use GA; Prompt node and Microsoft 365 Copilot node in workflows; Entra Agent IDs (preview); agent inventory schema               |
| Jun  | The new experience (GitHub Copilot harness) production-ready preview; Skills; Memory; connected agents; Claude Sonnet 5 / GPT-5.5 Chat GA |
| Jul  | Entra Agent ID automatic for every agent; workflows and MCP servers as tools; files in and files out (preview)                            |
| Aug  | GitHub Copilot harness GA (3 Aug); Skills GA; September billing change announced for developer and trial environments                     |

**Trajectory: topics (2024) → generative orchestration + agent flows (2025) → harness choice, skills, memory, workflows, connected agents (2026).**

# Lab 0 — Environment Setup

LAB 0

20 MIN

DAY 1

## Training Class Sandbox (assigned by your trainer)

### What it teaches

One Training Class Sandbox, with Dataverse and Copilot Credits, stands behind every later lab — and one designer builds everything.

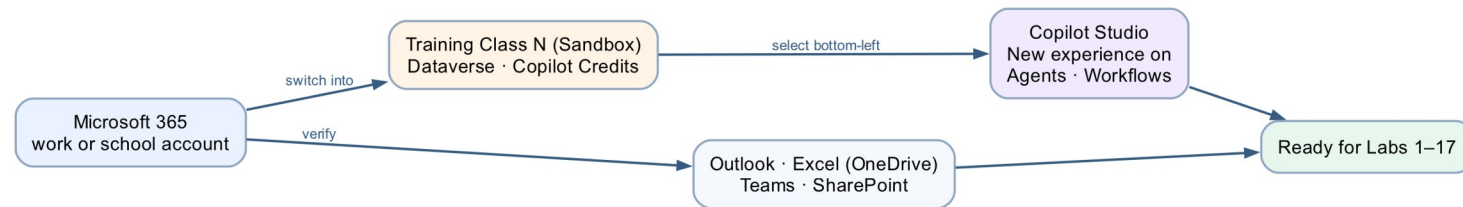
### Platform

Power Platform admin centre + Copilot Studio

### Tenant artefact names

(no tenant artefact — admin centre only)

Trainer copies end (DO NOT DELETE). Name yours plainly — agent names 30 characters or fewer.



# Lab 0 — Build Steps

1

Sign in to Microsoft 365 and confirm Outlook, Excel and Teams open with the training account.

2

Trainer provisions one Training Class Sandbox per cohort (Dataverse on); learners do not create an environment.

3

Open [copilotstudio.microsoft.com](https://copilotstudio.microsoft.com); pick your Training Class environment bottom-left; turn the New experience toggle on.

4

Confirm Agents and Workflows appear in the left navigation and New agent / New workflow are available.

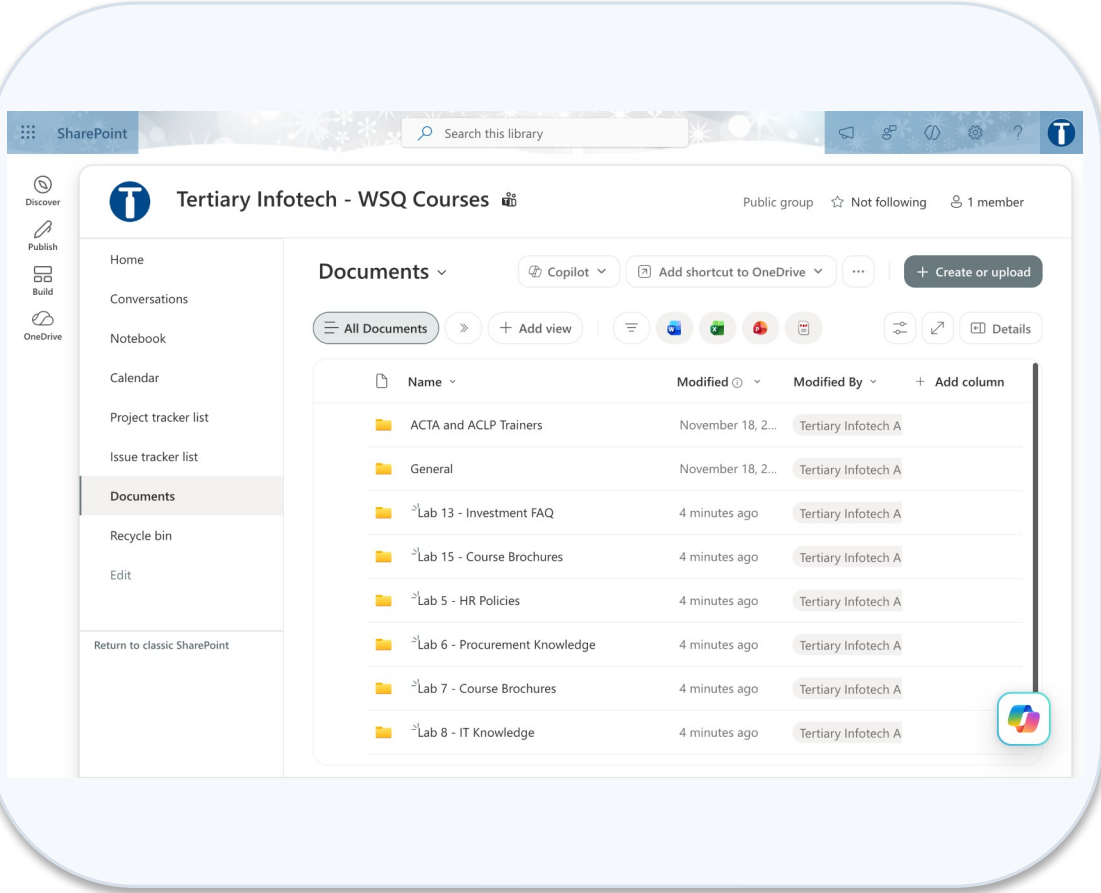
5

Confirm Copilot Credits are allocated to your class environment (trainer: PPAC → Licensing → Copilot Studio).

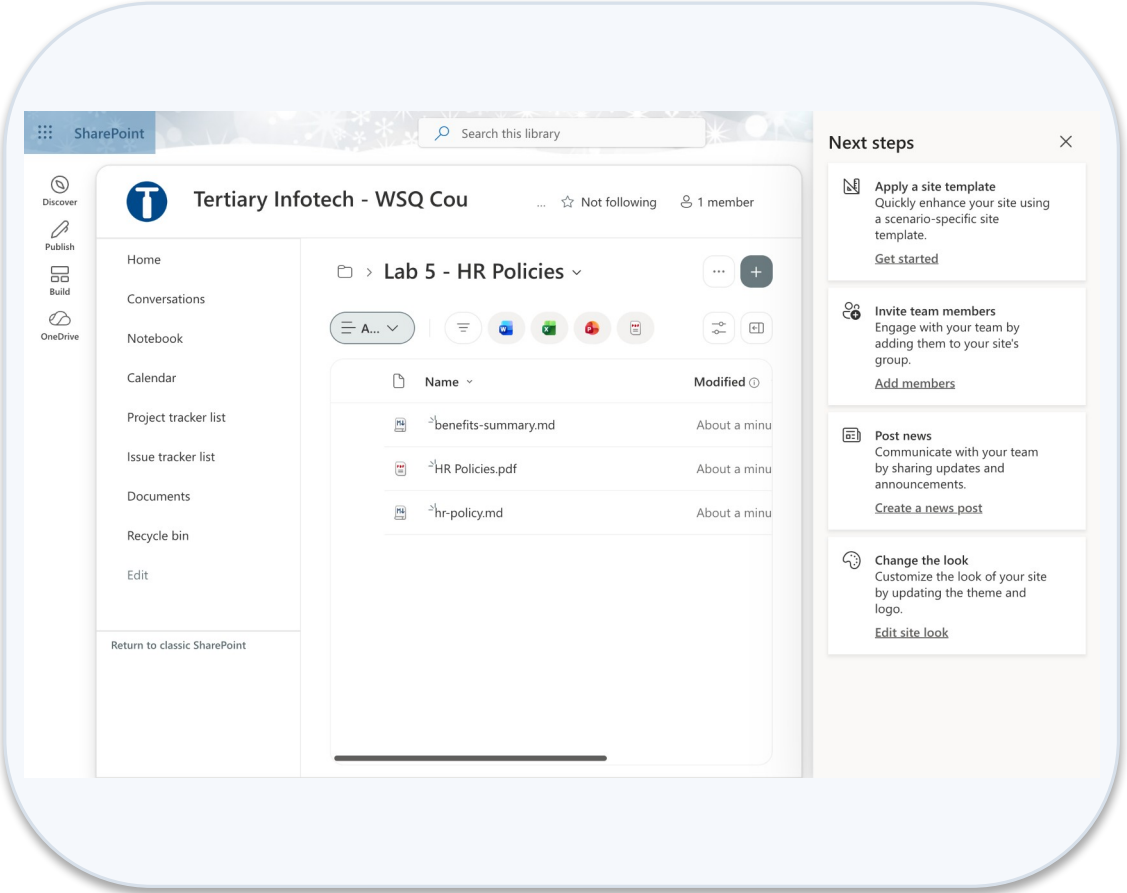
6

Verify: open the trainer's Lab 1 - Trigger and Actions (DO NOT DELETE) and read its canvas without editing.

# Lab 0 — What You Will See

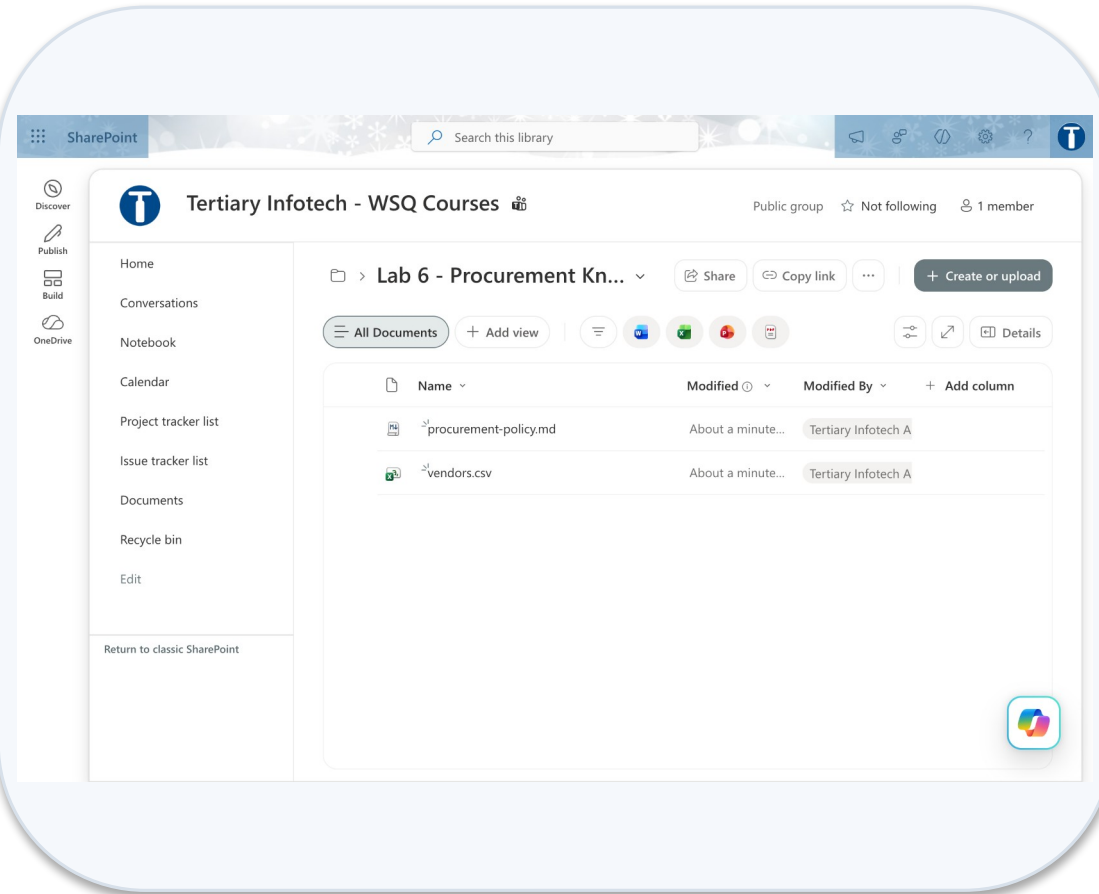


Sharepoint site documents

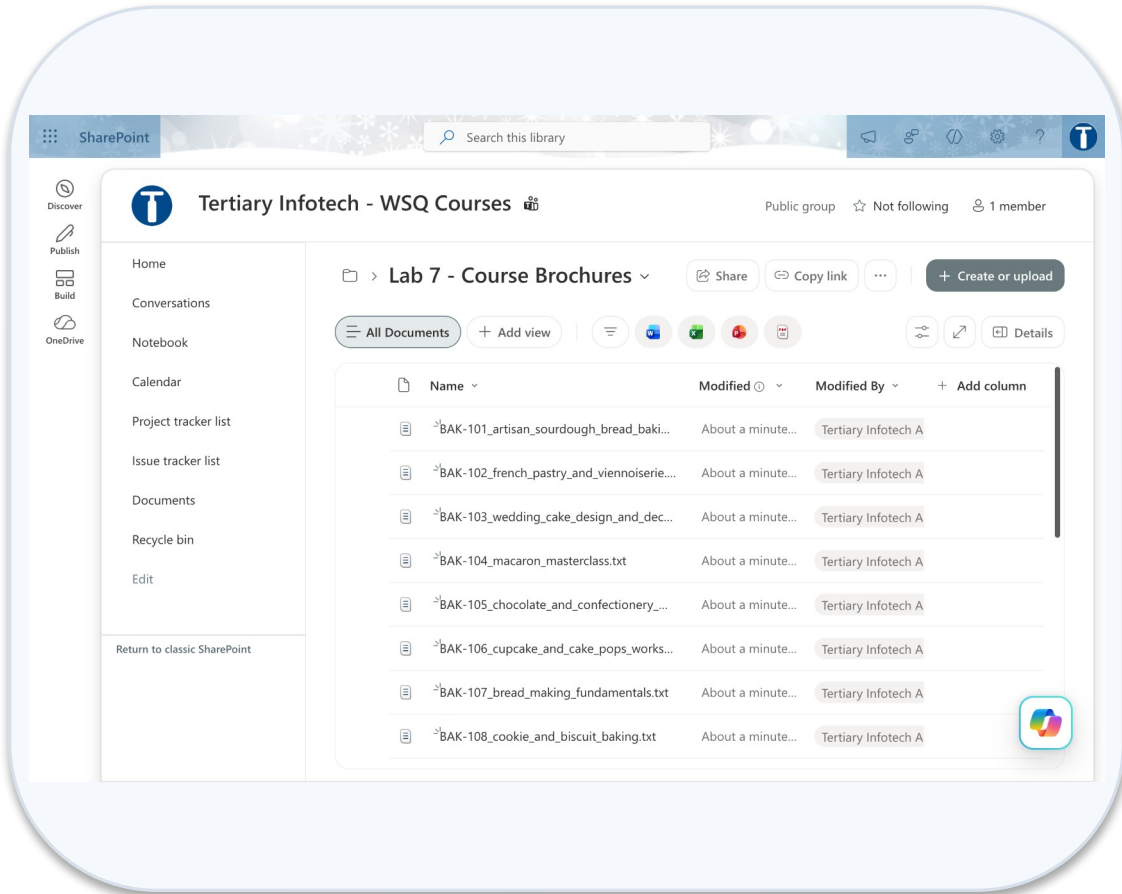


Lab 5 hr policies folder

# Lab 0 — What You Will See (continued)



Lab 6 procurement knowledge folder



Lab 7 course brochures folder

# Lab 1 — Trigger and Actions

LAB 1

30 MIN

DAY 1

## Form to email confirmation

### What it teaches

A trigger starts a workflow; connector actions consume the trigger's outputs as dynamic content.

### Platform

Copilot Studio workflow + Microsoft Forms + Outlook

### Tenant artefact names

Lab 1 - Trigger and Actions · Lab 1 - Course Enquiry Form

Trainer copies end (DO NOT DELETE). Name yours plainly — agent names 30 characters or fewer.

Course Enquiry Form  
Name · Email · Tel · Message

When a new response  
is submitted

Get response details

Send confirmation email  
to submitted Email

# Lab 1 — Build Steps

1

Microsoft Forms → New form → the trainer's Lab 1 - Course Enquiry Form has Name, Email, Tel, Message (all Required).

2

Workflows → New workflow → rename exactly Lab 1 - Trigger and Actions → Save.

3

Start node → Connector → Microsoft Forms → When a new response is submitted → pick the form; create the connection.

4

+ → Add dialog → Connectors tab → Microsoft Forms → Get response details; Form Id again; insert the ⚡ Response Id token.

5

+ → Connector → Office 365 Outlook → Send an email (V2); To = ⚡ Email (the form answer); build Subject and Body with ⚡ tokens.

6

Save → Publish → submit the form → open Activity and read every node's Inputs and Outputs.

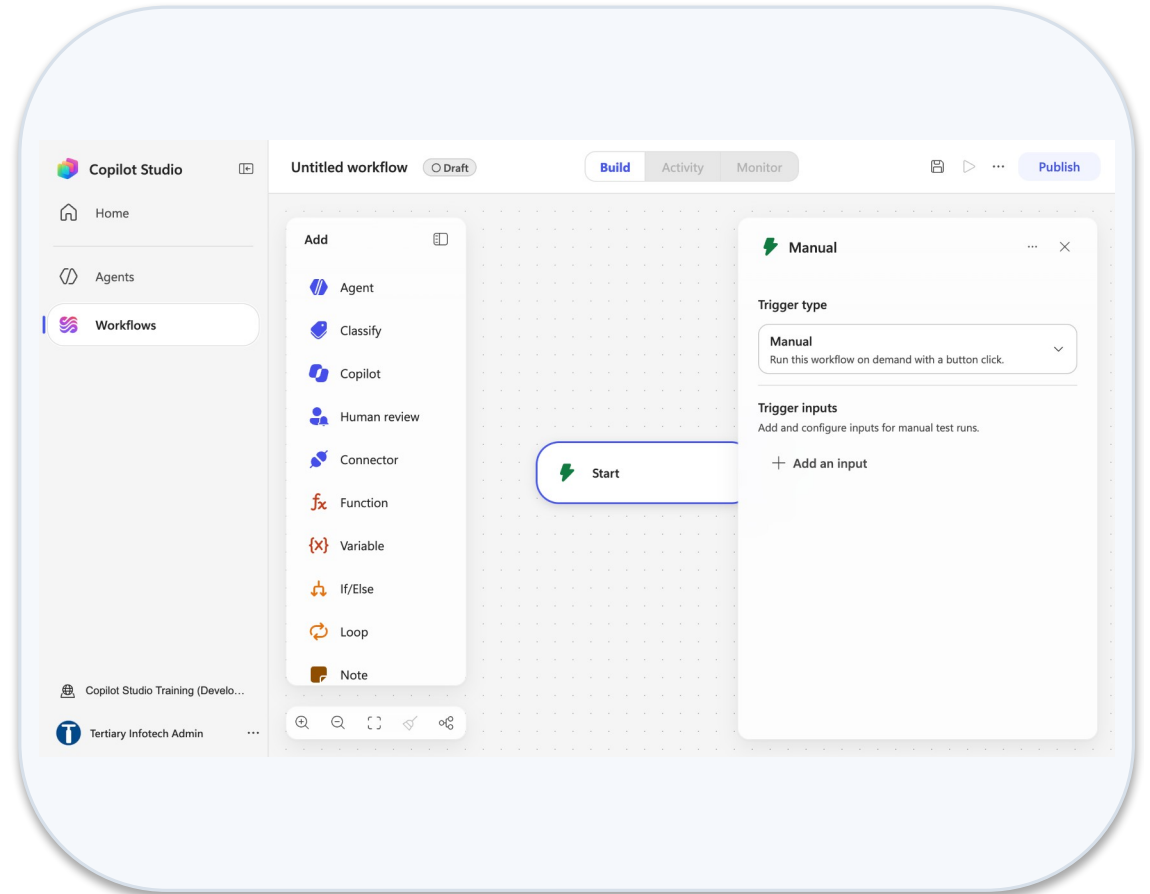
7

Probe: type a value by hand instead of a token, resubmit, and watch the email arrive with a blank.

# Lab 1 — What You Will See

The screenshot shows a web browser window with the URL 'forms.cloud.microsoft'. The page title is 'Lab 1 - Course Enquiry Form' and it is marked as 'Saved'. The interface includes a 'Style' button, 'Settings', 'Preview', 'Collect responses', 'View responses', and 'Present' buttons. A 'Generate suggestions to improve your form.' button is also visible. The form itself is titled 'Lab 1 - Course Enquiry Form' and has a subtitle 'Submit your contact details and course enquiry.' It contains three required fields: '1. Name \*', '2. Email \*', and '3. Tel \*', each with a text input area. A 'Templates' button is on the left sidebar, and a 'Get suggestions' button is on the right.

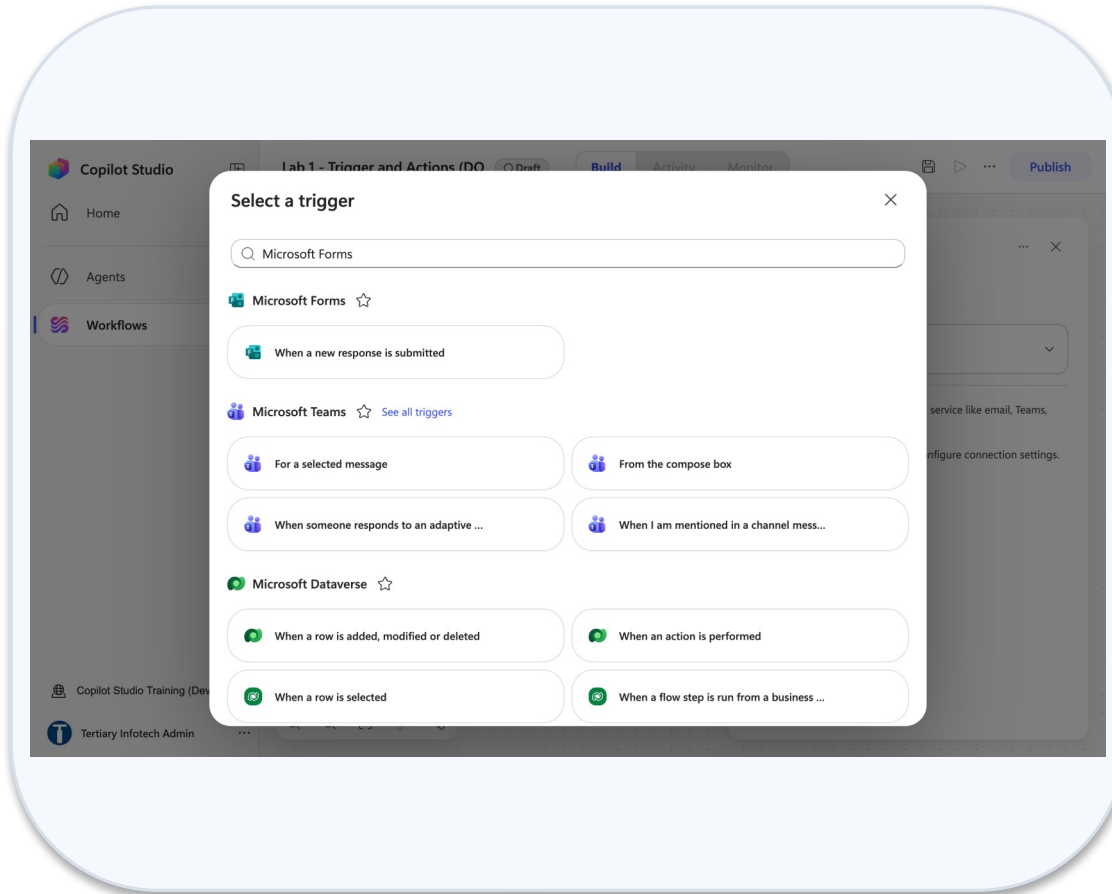
Trainer course enquiry form



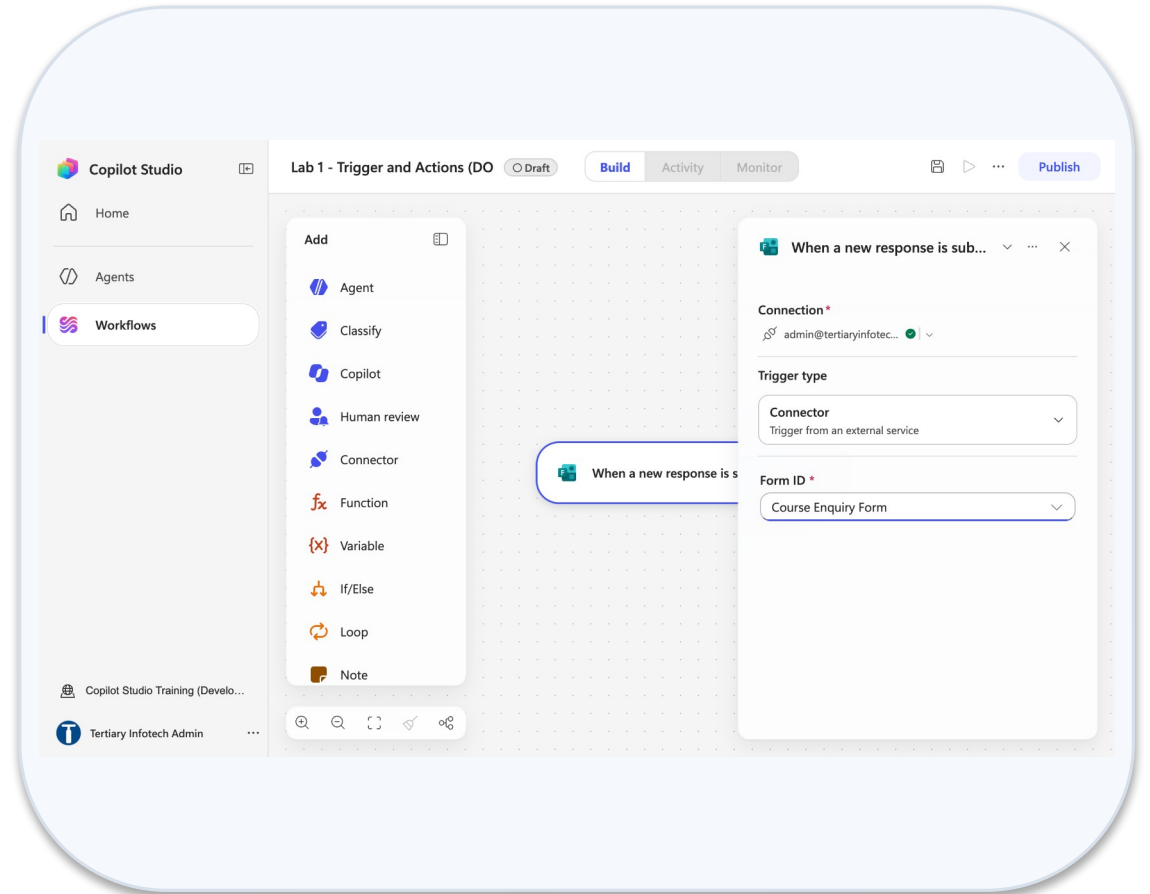
New workflow start node



# Lab 1 — What You Will See (continued)



Select a trigger microsoft forms



Start node form id

# Lab 2 — Log to Excel

LAB 2

30 MIN

DAY 1

## Log the enquiry, then send the email

**What it teaches**

Commit before you confirm — the audit row is written before the acknowledgement is sent.

**Platform**

Copilot Studio workflow + Excel Online

**Tenant artefact names**

Lab 2 - Log to Excel · Lab 2 - Enquiry Log.xlsx  
Trainer copies end (DO NOT DELETE). Name yours plainly — agent names 30 characters or fewer.



## Lab 2 — Build Steps

1

Upload Lab 2 - Enquiry Log.xlsx (table EnquiryLog) to OneDrive → Power Automate Lab Data.

2

New workflow Lab 2 - Log to Excel; rebuild the Lab 1 Forms trigger and Get response details.

3

+ → Connector → Excel Online (Business) → Add a row into a table; Location → Document Library → File → Table, top to bottom.

4

Map each column to a ⚡ token; Timestamp = </> utcNow().

5

Only then add Send an email (V2) — the confirmation comes after the row.

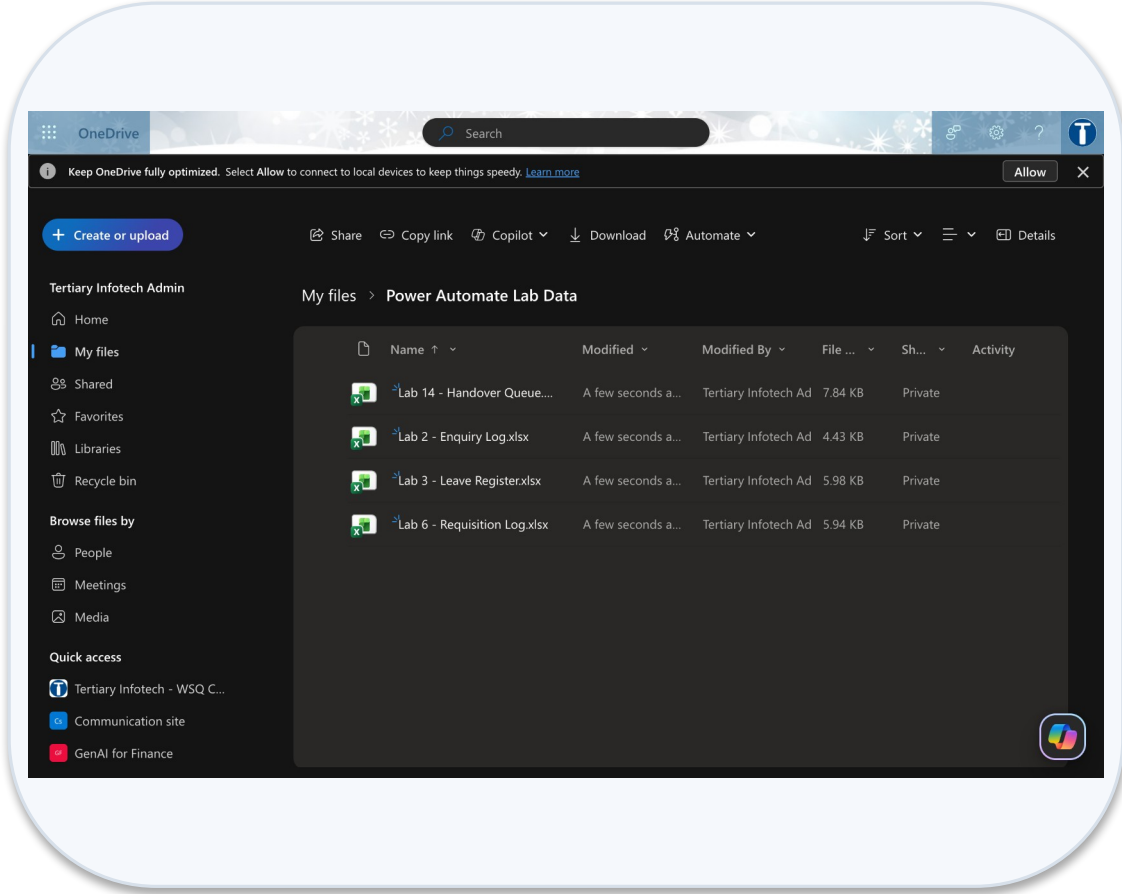
6

Save → Publish → submit two responses → check two rows in Excel and two emails.

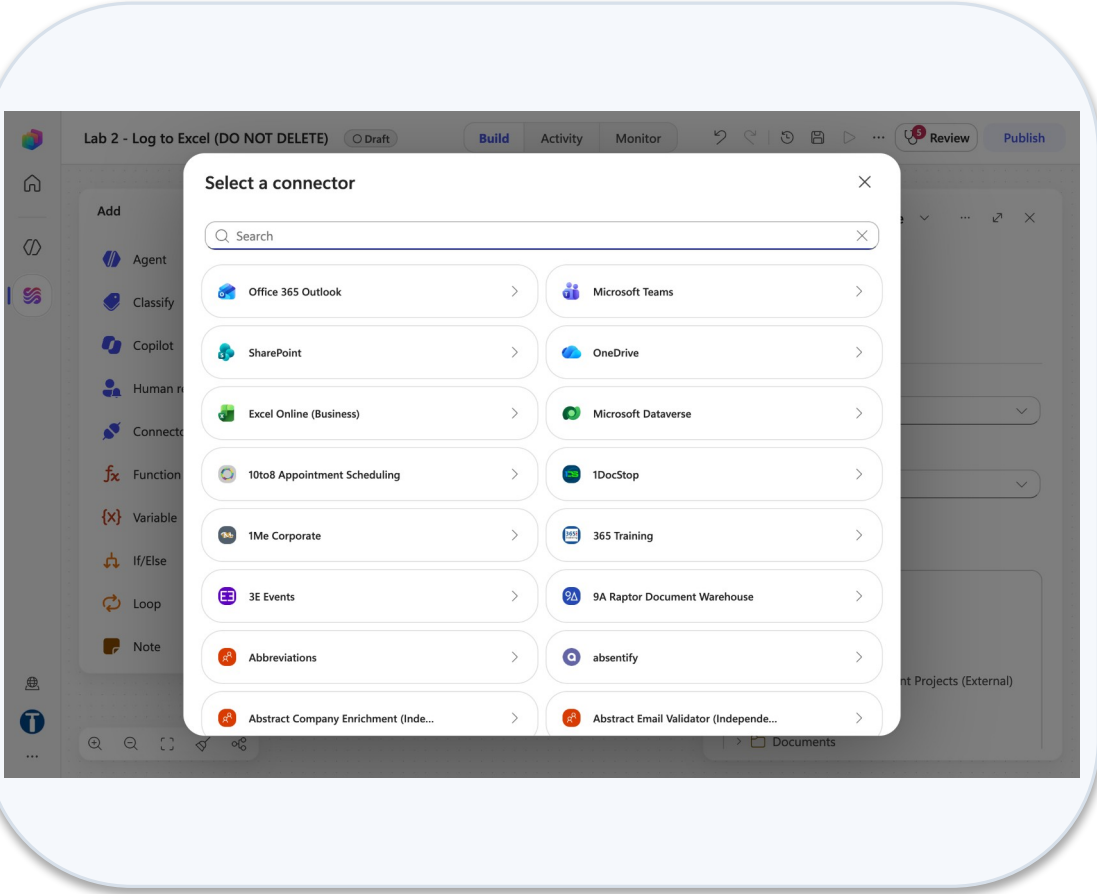
7

Debrief: swap the two steps in your head — what is lost if the email fails first?

# Lab 2 — What You Will See

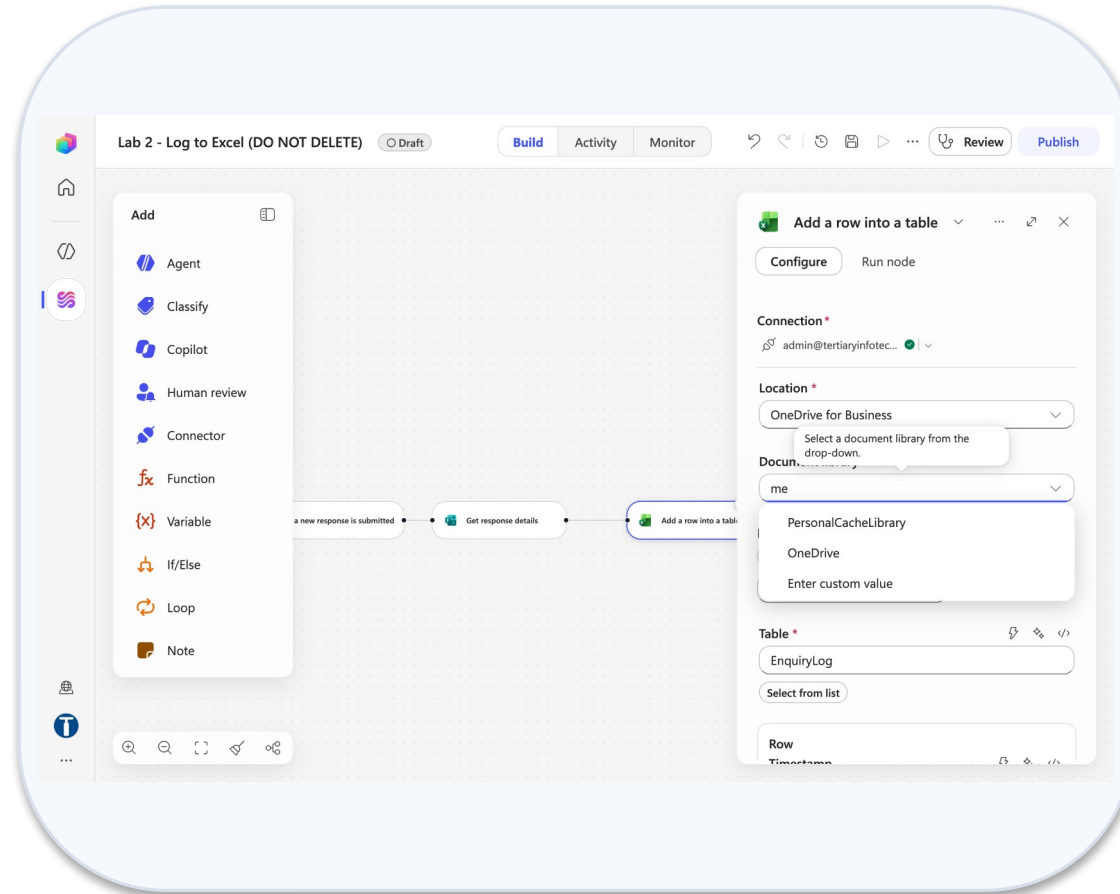


Onedrive power automate lab data

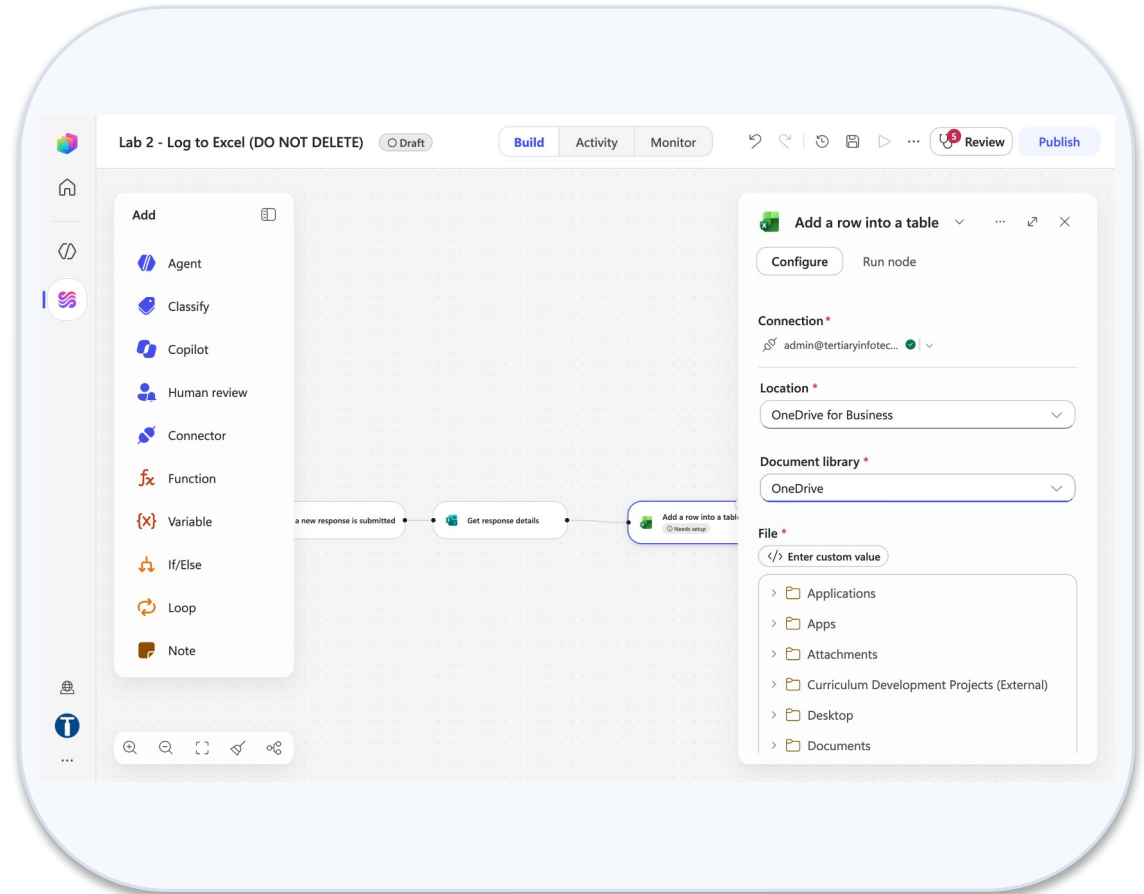


Select a connector excel online

# Lab 2 — What You Will See (continued)



Document library dropdown



File picker onedrive folders

## MODULE 2

# Control Flow and Human in the Loop

If/Else, Compose, Switch, Loop · in / on / out of the loop · Human review via Teams or Outlook · the Classify node · autonomous business processes

Applied in Lab 3 and Lab 4

# If/Else — Left Value, Operator, Right Value

An If/Else node evaluates one condition row: a left value, an operator, and a right value.



Lab 3: Outcome Equals Approve (Approvals connector). Labs 4 and 14: Outcome Equals Yes (Human review — it publishes the string Yes, not a boolean).

# Conditions — Both Paths Must Exist

Both branches must be built — including the one you hope never runs. Add rows for AND / OR groups.

## Empty Else is a silent failure

1

A rejected leave request that triggers nothing is an applicant waiting forever. Route every Else to a message or a named person.

## The picker's type can lie

2

Human review's Yes/No input shows as boolean but publishes the string Yes. Read the node's real Outputs in Activity.

## Re-pick after rebuilding an input

3

Delete and re-create an input and the old token renders fine but resolves to empty — every run falls to Else.

## Test both paths

4

Lab 3 tests an approval and a rejection; Lab 4 sends one email per category; Lab 14 approves one draft and rejects one.

### LAB 3

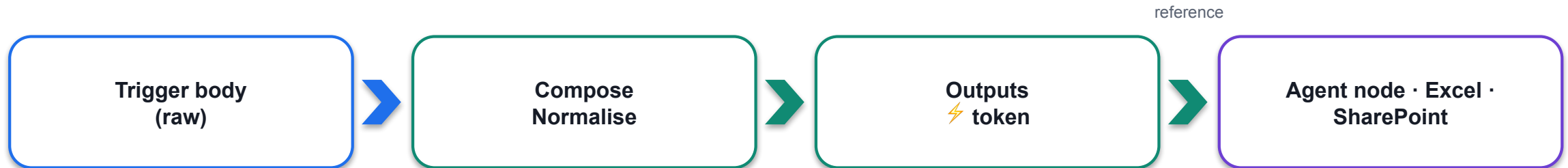
## Lab 3 — Leave Application Approval

The workflow pauses for a manager · Copilot Studio workflow + Approvals + Outlook · 35 min



# Compose, Function and Variable — Named Values

Compose is a named value holder: it takes any input, and its Outputs token is what every later step references.



## Normalise input

`toUpper(trim(triggerBody()['nric']))` — one canonical form before the lookup and the record.

## Hold a computed value

`concat('REQ-', formatDateTime(utcNow(), 'yyyyMMdd'), '-', ...)` — generated once, used by Excel and the reply (Lab 6).

## Build a prompt

Assemble the retrieved brochure text and the visitor's question into one block for the Agent node (Lab 16).

## Variable vs Compose

Initialize / Set / Append variable when the value changes during the run (loops, counters). Compose when it is written once.

**Function → Data Operations → Compose. Rename the node with one word (no spaces) so its token stays readable.**

# Switch and Loop

When one condition is not enough: Switch for many cases, Loop for many items.

## SWITCH — one value, many branches

- On: a ⚡ token (e.g. Leave Type, category)
- Case Annual · Case Medical · Case Unpaid · Default
- Each case is its own branch of steps
- Default catches what you did not foresee

## LOOP — repeat over a list

- For each: every item in an array (Get items rows, attachments)
- Do until: repeat until a condition is true
- Steps inside see the current item token
- Arrays need join() before they reach Excel or a text field

**Lab 4 uses the Classify node's ports instead of a Switch — the category is a port, not a value you compare.**

Keep branches short. A branch that grows past four steps is usually a child workflow waiting to be extracted.

# Human In, On, and Out of the Loop

Three arrangements that people use interchangeably, and that are not the same thing.

| Pattern               | Who decides                | Who acts                      | Can AI proceed alone? |
|-----------------------|----------------------------|-------------------------------|-----------------------|
| Human IN the loop     | AI proposes, human decides | Human authorises, system acts | NO — it blocks        |
| Human ON the loop     | AI decides and acts        | Human monitors, can intervene | Yes — after the fact  |
| Human OUT of the loop | AI decides and acts        | AI acts                       | Yes — nobody checks   |

WHERE THIS COURSE PUTS THE HUMAN

IN the loop: Lab 3 (a manager approves leave), Lab 4 (Priority emails stop in Teams), Lab 14 (an adviser approves a draft), Lab 9 (the manager agent shows the final draft).

OUT of the loop on purpose: Labs 12, 13, 15 and 16 — so you can see what that costs. The pause IS the deliverable.

LAB 3

Lab 3 — Leave Application Approval

The workflow pauses for a manager · Copilot Studio workflow + Approvals + Outlook · 35 min

# Human Review — Teams or Outlook/Email?

The Human review node has a Channel dropdown: Teams or Outlook. Where the request lands decides whether it is ever answered.

TEAMS — the Workflows bot chat

- An adaptive card: Request information | Microsoft Copilot Studio
- Approver answers inline; the run resumes in a minute or two
- Verified reliable on our tenant — the default for Labs 4 and 14
- Best when the approver lives in Teams all day

OUTLOOK / EMAIL

- Request for information arrives as an actionable email with a form and Submit
- The docs describe RFI as Outlook-delivered; on our tenant the mail never arrived
- Approvals connector (classic): email + Teams Approvals app, Approve/Reject buttons
- Best for approvers outside Teams, or a formal audit trail by mail

| Mechanism                                         | Where           | What comes back                                                |
|---------------------------------------------------|-----------------|----------------------------------------------------------------|
| Human review node — Channel: Teams                | Labs 4, 14      | Inputs you define (Outcome Yes/No, Name); first responder wins |
| Approvals — Start and wait for an approval        | Lab 3           | Approve/Reject – First to respond; Outcome, Responses Comments |
| Request human assistance when unsure (Agent node) | Never as a gate | Fires only if the model feels unsure — not a control           |

# The Human Review Node — A Gate That Blocks

Add a step → Human review → Request for information. It does nothing at all except wait.

1

## Title · Message · Assigned to

Message = ⚡ the agent's summary or draft. Assigned to = a tenant user; type, wait for the directory lookup, click the suggestion.

2

## Inputs are mandatory

Add an input: Text · Yes/No · Email · Number · Date. Define Outcome (Yes/No) and Name (Text), no defaults, no spaces in names.

3

## What it publishes

Only the inputs you defined, plus responderObjectId. There is no built-in outcome property — a hand-typed reference silently never matches.

4

## Then If/Else

Outcome (⚡) Equals Yes. The Yes/No input publishes the string Yes, so compare against Yes, not true.

### THE TEST OF A REAL GATE

Submit, then open Activity. The run says Running — and it will still say Running tomorrow. Nothing times out, nothing defaults, nothing proceeds until a person

#### LAB 4

### Lab 4 — Email Classification

An inbox workflow that sorts, replies, books and escalates · Copilot Studio workflow + Outlook + Classify + Human review + Teams · 45 min

# What Is a Control, and What Only Looks Like One

Three things in the designer look like "a human checks". Only one of them is a control.

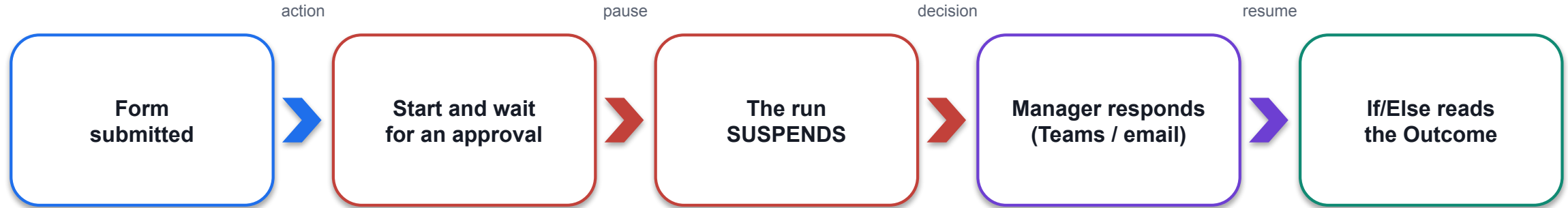
| Mechanism                                                | What it is                                 | Fires when                     | Audit trail                     | Control ? |
|----------------------------------------------------------|--------------------------------------------|--------------------------------|---------------------------------|-----------|
| Human review (node)                                      | A gate in the workflow                     | Always                         | Outcome + responder ID recorded | YES       |
| Request human assistance when unsure (Agent node toggle) | The agent emailing the connection owner    | Only if the model feels unsure | None                            | NO        |
| Request information (agent tool)                         | Same, exposed as a tool the agent may call | When the agent decides         | None                            | NO        |

The toggle and the tool both let a confidently wrong output through untouched — which is the failure that matters.

**DEFAULT VALUES ARE A DESIGN DECISION**

Pre-filling Outcome with Yes turns the gate into a rubber stamp: confirming takes no thought, rejecting takes noticing. Leave it blank; if a default is forced, fail safe.

# How an Approval Suspends a Running Workflow



## OUTCOME = APPROVE

- Send the approval email with the manager's Responses Comments; log the decision

## OUTCOME = REJECT (Else)

- Send the rejection with the comments — routed to the applicant, never to silence

### APPROVALS CONNECTOR SETTINGS THAT MATTER

Approval type: Approve/Reject – First to respond. Title with the ⚡ Name token. Assigned to: click the resolved suggestion (a typed-and-tabbed address fails at run time). Details: labelled lines with ⚡ tokens.

### LAB 3

## Lab 3 — Leave Application Approval

The workflow pauses for a manager · Copilot Studio workflow + Approvals + Outlook · 35 min

# The Classify Node — Categories, Examples, Ports

The Classify node is a native AI step: it reads text and routes the run out of the port that matches.

## CONFIGURE

- Model picker (Claude Sonnet 4.6 is the cheap, reliable choice)
- Instruction with ⚡ tokens: Subject, From, Body, Importance
- Categories: name + one-line description; Add more categories
- Add example per category to sharpen the boundary
- Built-in Other — the fail-safe when nothing fits

## WHAT COMES OUT

- One output port per category — connect a handler to each
- The chosen category and the model's reasoning appear in run details
- Ambiguous text should land in Other, not be forced into a category
- Downstream nodes see the trigger's tokens as they flow through

**Five categories in Lab 4: Meeting · Need Reply · Priority · Informational · Other. Every port does something visible so a test can prove which path ran.**

### LAB 4

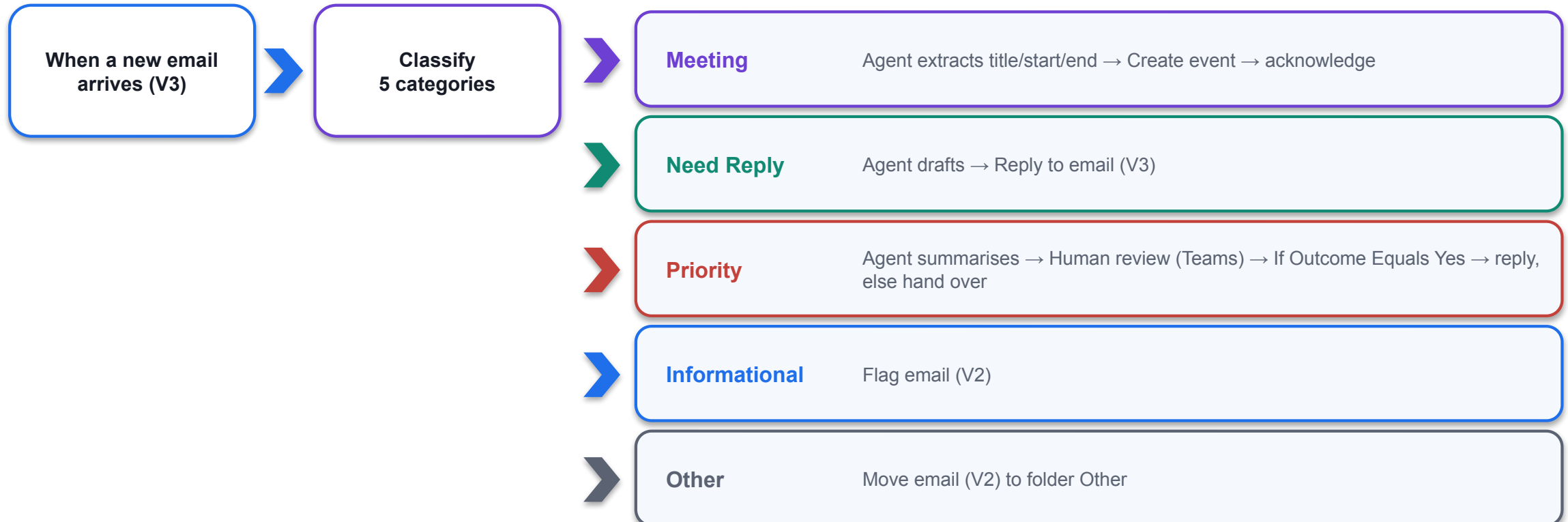
#### Lab 4 — Email Classification

An inbox workflow that sorts, replies, books and escalates · Copilot Studio workflow + Outlook + Classify + Human review + Teams · 45 min



# Email Classification — The Lab 4 Architecture

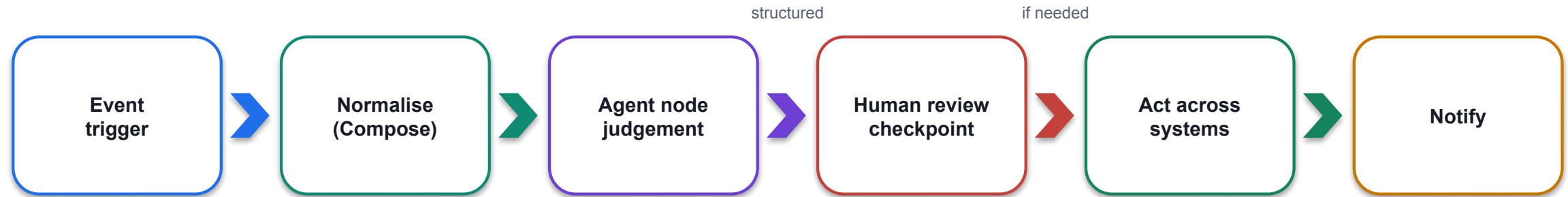
Lab 4 on one slide: an Outlook trigger, one Classify node, five ports, and a human on the Priority path.



**Test with five emails, one per category. The Priority case proves the gate: the run waits in Teams until you answer.**

# Autonomous Business Processes

An autonomous business process = an event trigger + deterministic steps + AI judgement steps + a human checkpoint where it matters.



## Workflows absorbed AI in 2026

Agent node (Mar), Prompt and M365 Copilot nodes (May), workflows as agent tools (Jul), the GitHub Copilot harness GA (3 Aug).

## Two directions

A workflow can call an agent (Agent node — Lab 10); an agent can call a workflow as a tool (When an agent calls the flow — Lab 11).

## Deterministic backbone

The same input always produces the same path — consistent execution, end-to-end visibility in Activity.

## Recovery

The harness retries and finds alternative paths inside an agent step; the workflow around it still fails loudly at a named node.

# Lab 3 — Leave Application Approval

LAB 3

35 MIN

DAY 1

The workflow pauses for a manager

## What it teaches

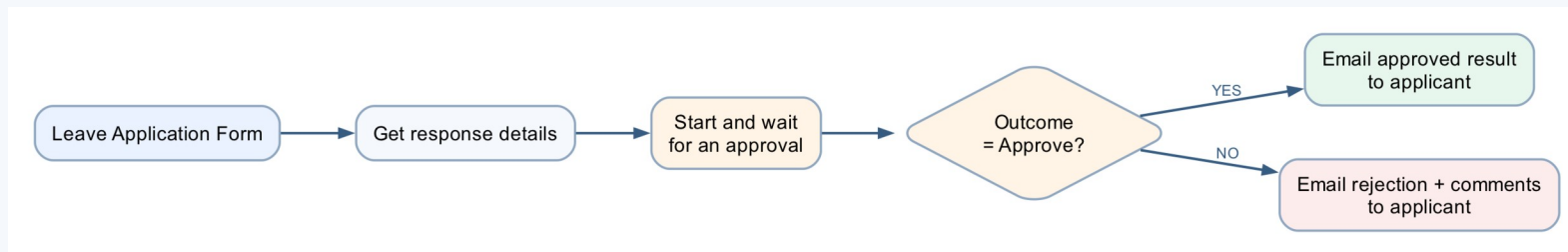
A running workflow suspends at the approval action and branches with If/Else on the human decision.

## Platform

Copilot Studio workflow + Approvals + Outlook

## Tenant artefact names

Lab 3 - Leave Application Approval · Lab 3 - Leave Application Form · Lab 3 - Leave Register.xlsx  
Trainer copies end (DO NOT DELETE). Name yours plainly — agent names 30 characters or fewer.



# Lab 3 — Build Steps

1

Forms → Lab 3 - Leave Application Form: Name, Leave from date, Leave end date, Leave Type, Reason for Leave.

2

New workflow Lab 3 - Leave Application Approval → Forms trigger → Get response details.

3

+ → Human review (Channel Microsoft Teams, never Outlook); Assigned to = your address; Title with the ⚡ Name token; Details = labelled ⚡ lines.

4

Add an input → Yes/No named Outcome; Add an input → Text named Comments — the only inputs the card will show.

5

+ → If/Else: ⚡ Outcome (Yes/No output) Equals Yes.

6

If: Send an email (V2) approved, with the ⚡ Comments text output. Else: not approved, with the comments.

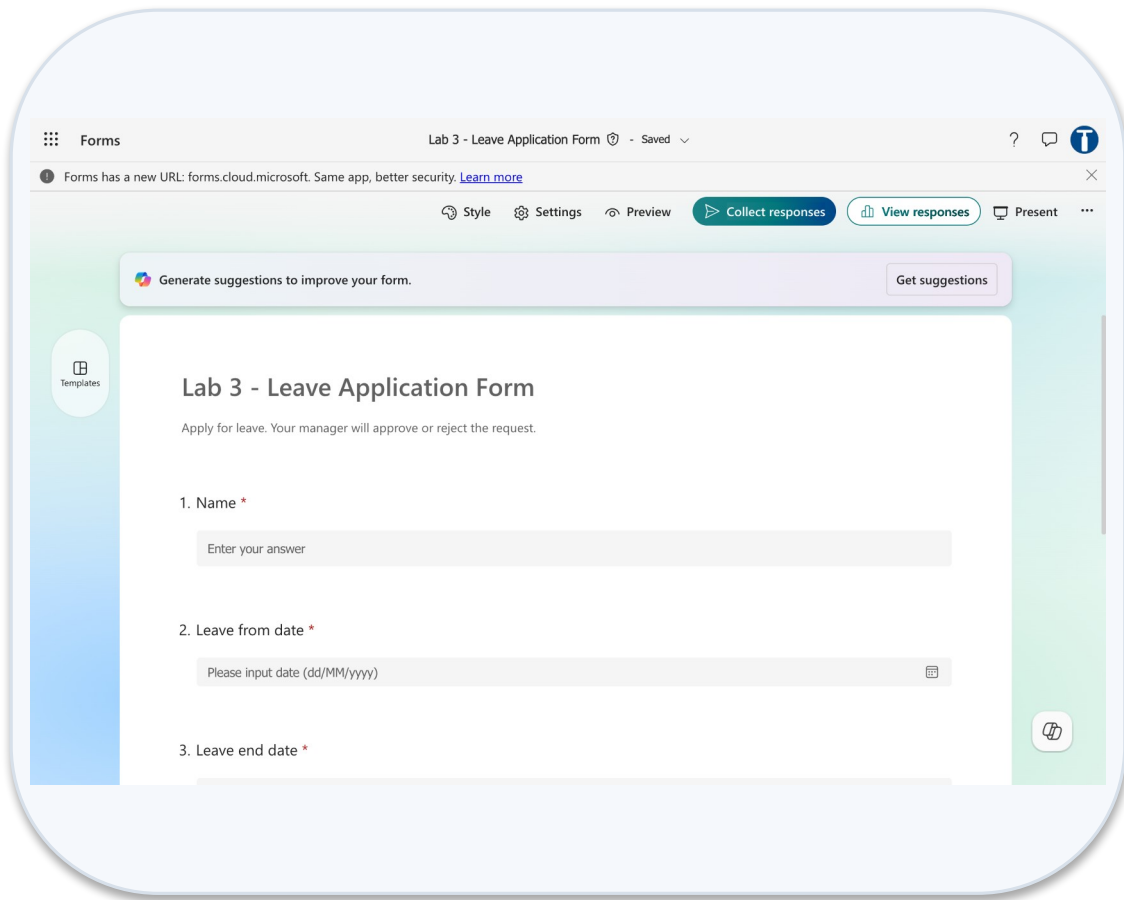
7

Save → Publish → submit → answer the card in the Teams Workflows chat; submit again and choose No. Both emails must arrive.

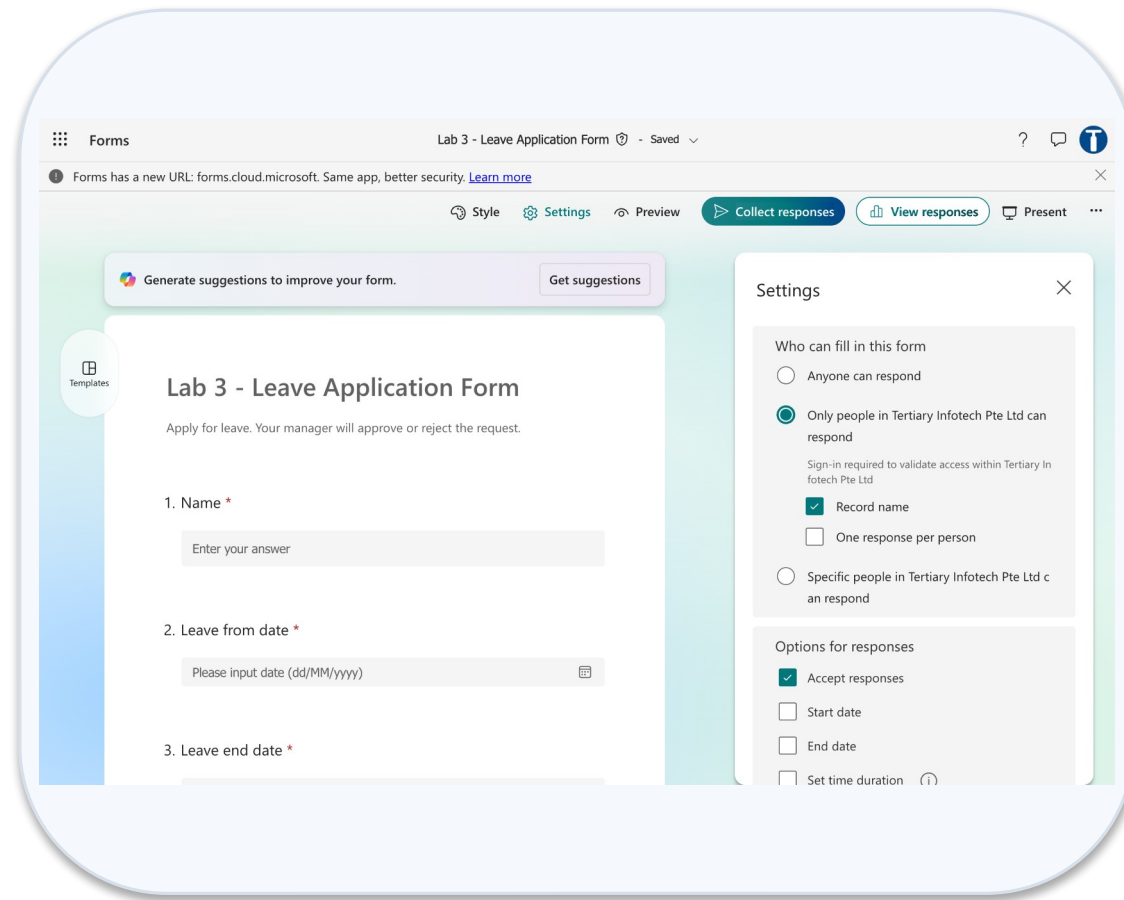
8

Optional Part F: Add a row into Lab 3 - Leave Register.xlsx with the decision.

# Lab 3 — What You Will See

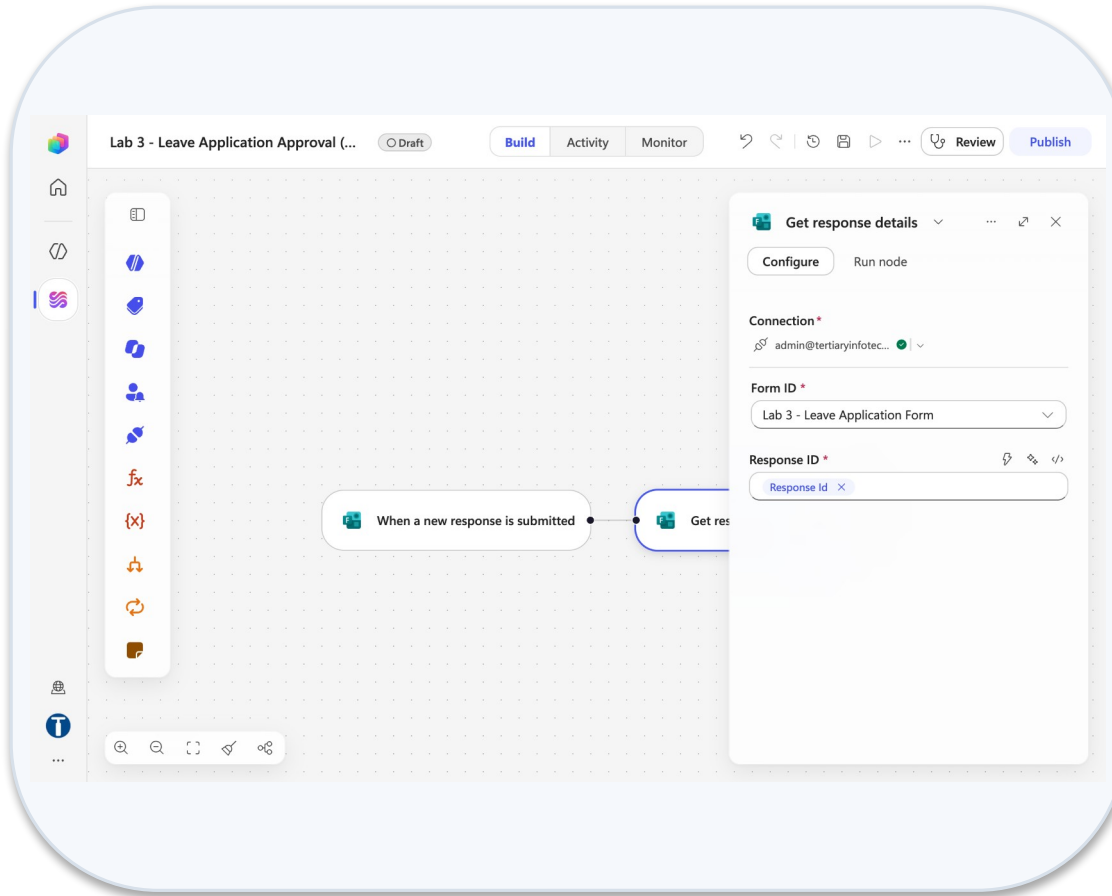


Leave application form

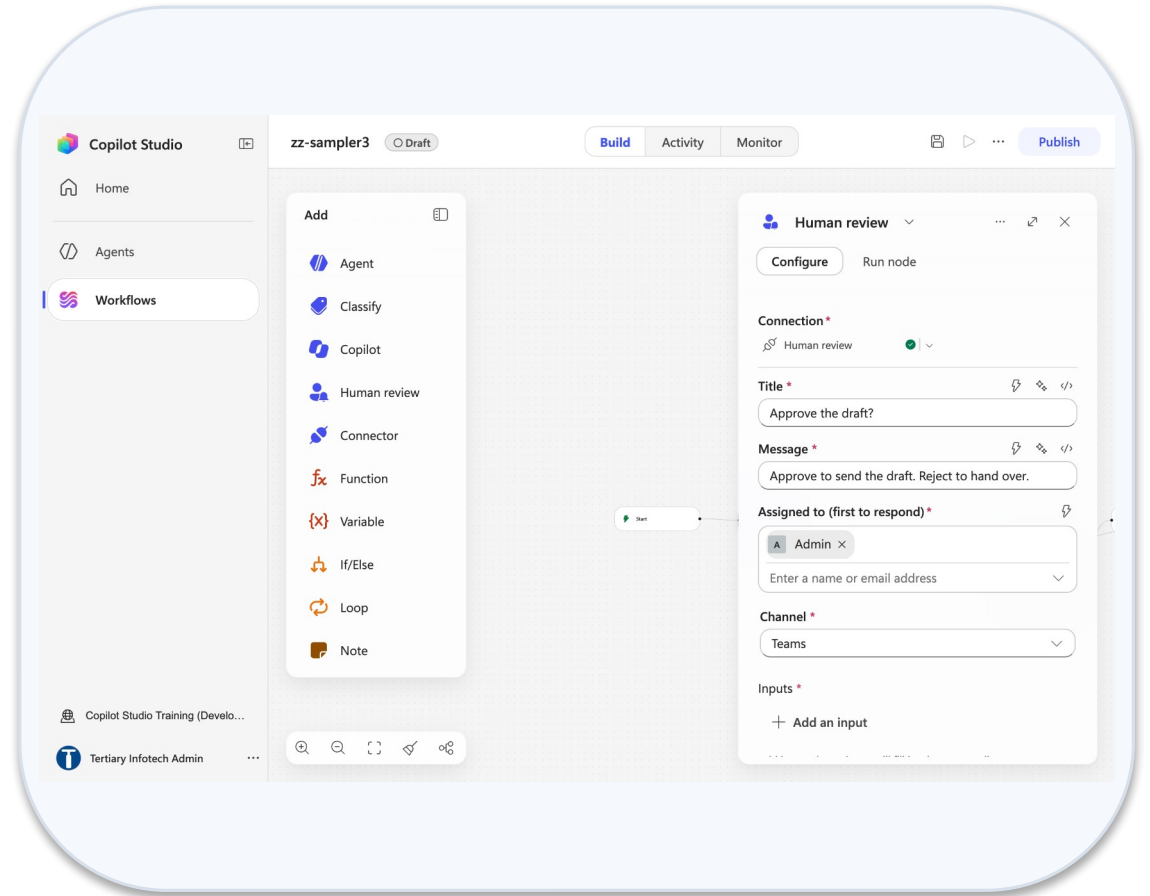


Leave form settings

# Lab 3 — What You Will See (continued)



Get response details



Human review panel

# Lab 4 — Email Classification

LAB 4

45 MIN

DAY 1

An inbox workflow that sorts, replies, books and escalates

## What it teaches

The Classify node sorts each email into five categories; each port runs a deterministic handler, and the Priority branch stops at a human in Teams.

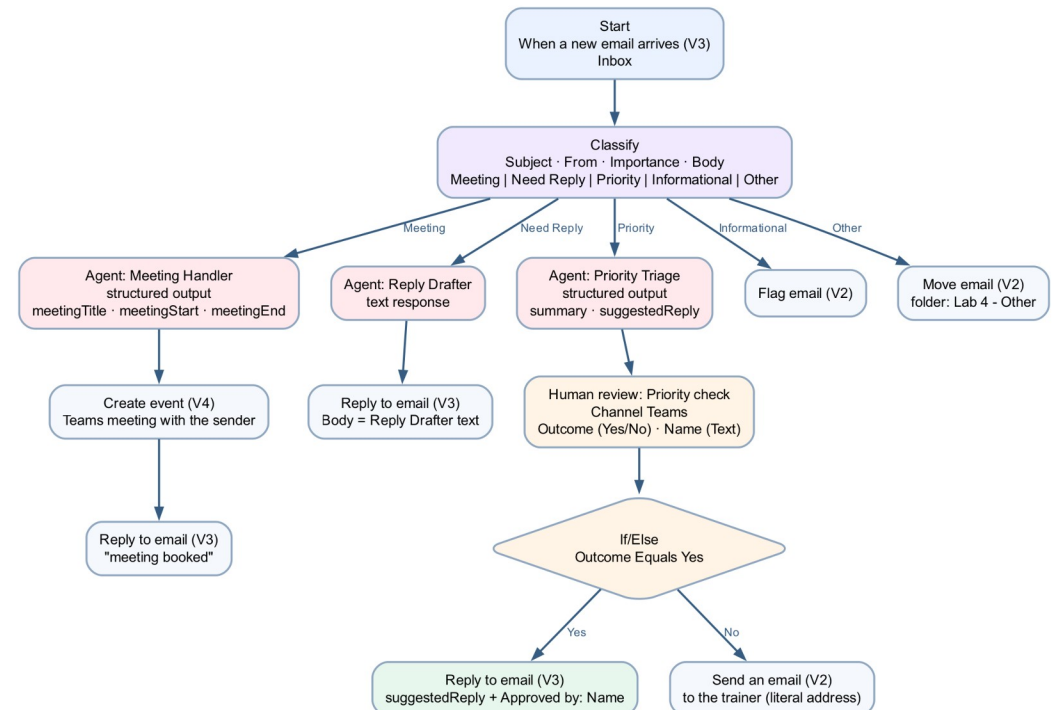
## Platform

Copilot Studio workflow + Outlook + Classify + Human review + Teams

## Tenant artefact names

Lab 4 - Email Classification · Lab 4 - Other

Trainer copies end (DO NOT DELETE). Name yours plainly — agent names 30 characters or fewer.



# Lab 4 — Build Steps

1

Outlook → create the folder Other under Inbox.

2

New workflow Lab 4 - Email Classification → Start → Connector → Office 365 Outlook → When a new email arrives (V3), folder Inbox.

3

+ → Classify: model Claude Sonnet 4.6; instruction with ⚡ Subject, From, Body, Importance; categories Meeting, Need Reply, Priority, Informational (+ Other).

4

Meeting port: Agent (structured output: title, start, end) → Create event (V4) → Send an email (V2) acknowledgement.

5

Need Reply port: Agent Reply Drafter → Reply to email (V3) with the draft.

6

Priority port: Agent summarises → Human review (Channel Teams; inputs Outcome Yes/No, Name) → If Outcome Equals Yes → reply; Else → hand-over email.

7

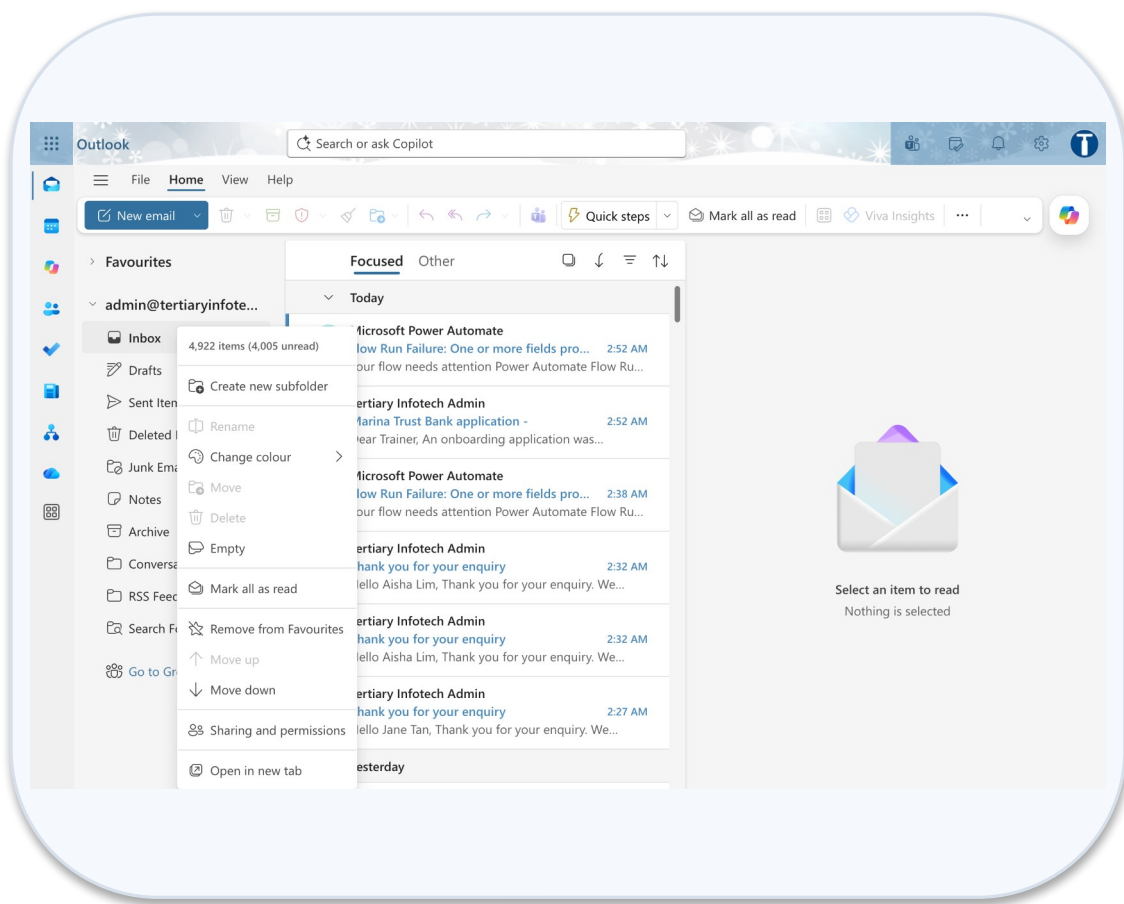
Informational: Flag email (V2). Other: Move email (V2) to Other. Tidy up → Save → Publish.

8

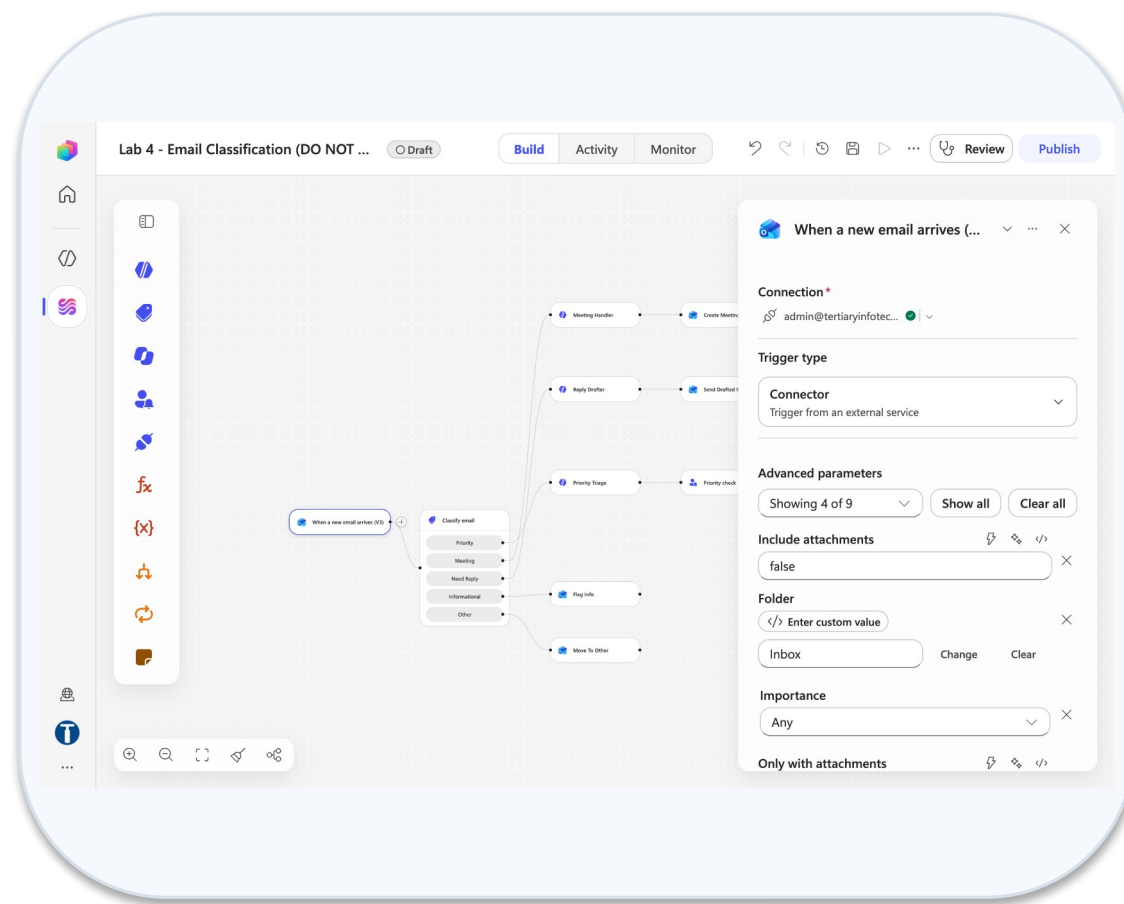
Send five test emails, one per category; answer the Priority card in the Teams Workflows chat; check Activity.



# Lab 4 — What You Will See

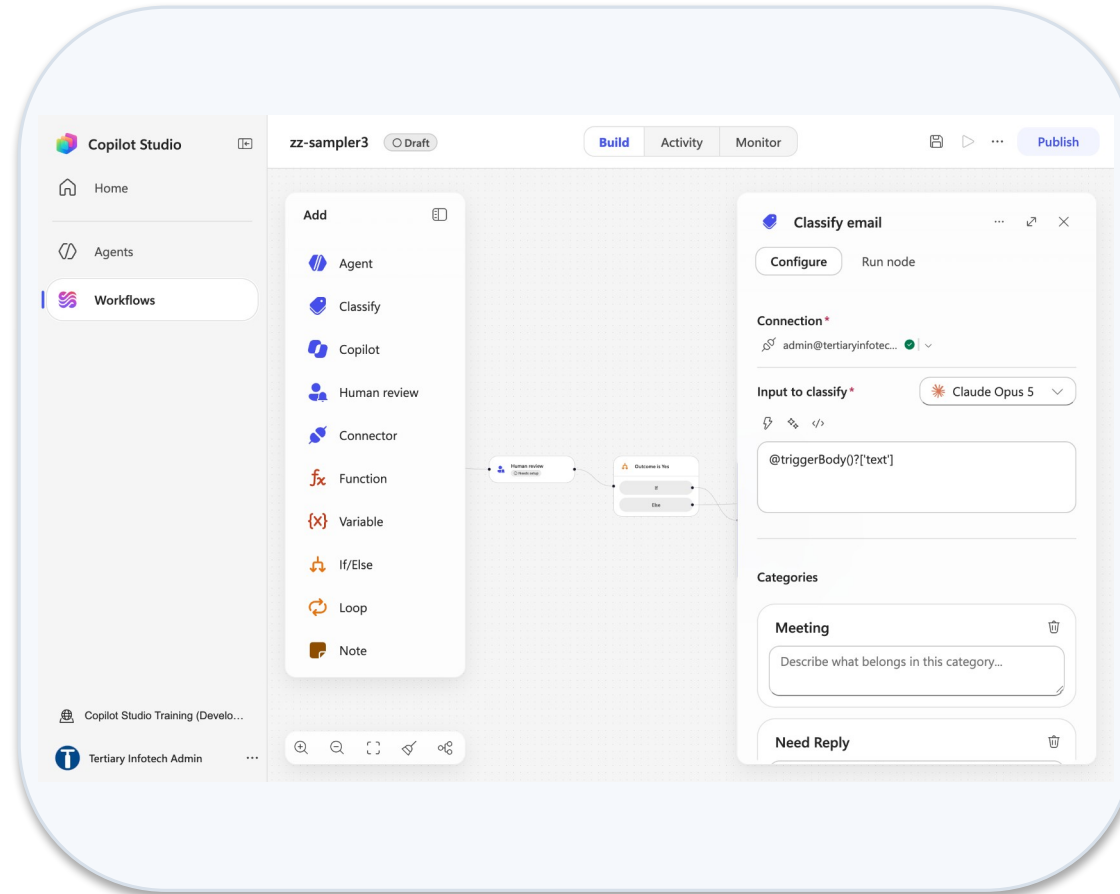


Outlook create new subfolder

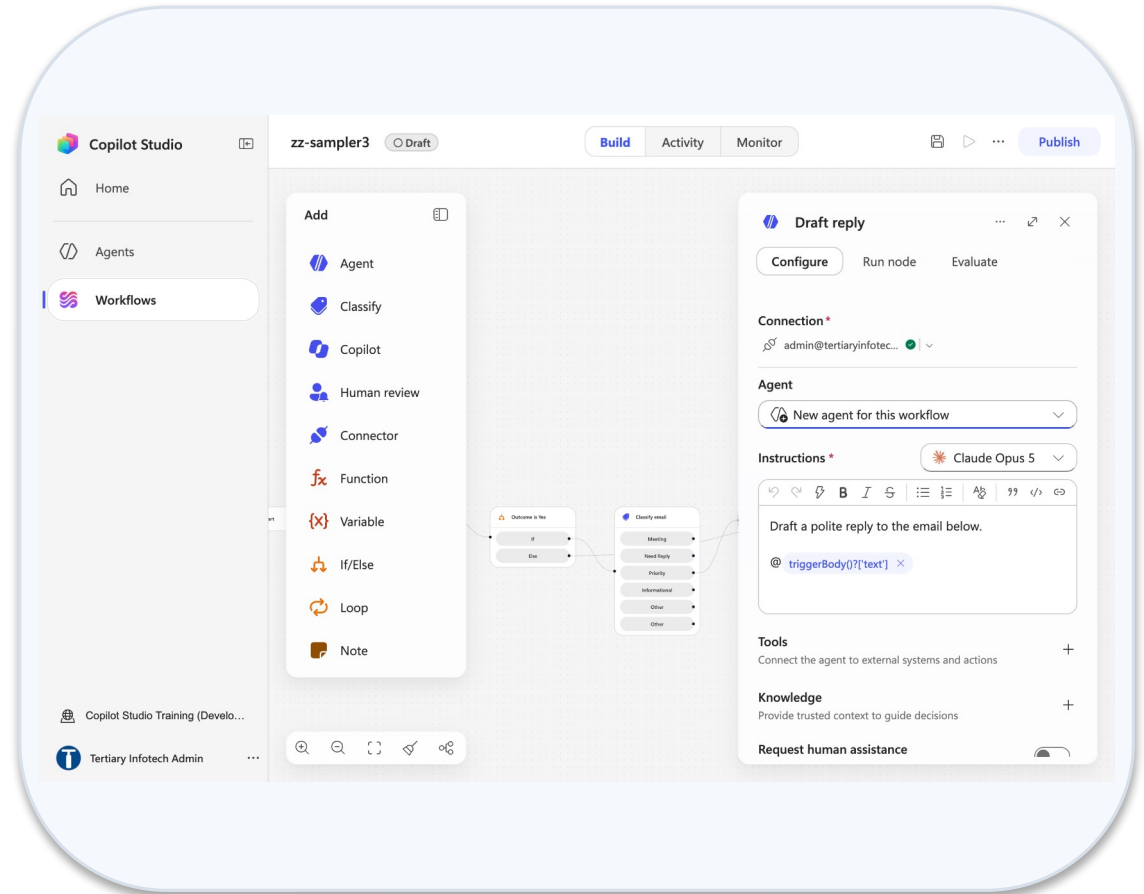


Outlook trigger panel

# Lab 4 — What You Will See (continued)



Classify node panel



Agent node instructions

## MODULE 3

# Copilot Studio Agents

The designer · instructions, model, knowledge · skills, tools, MCP · what is enforced · connected agents, memory · Preview, Evaluate, Monitor · publishing and channels

Applied in Labs 5–11 and Lab 17

# Workflow or Agent — They Fail Differently

## WORKFLOW — deterministic

- You decide the path in advance
- The same input always gives the same output
- It can only do what you built
- It fails loudly, at a named node
- You test it by checking the result

## AGENT — the harness reasons

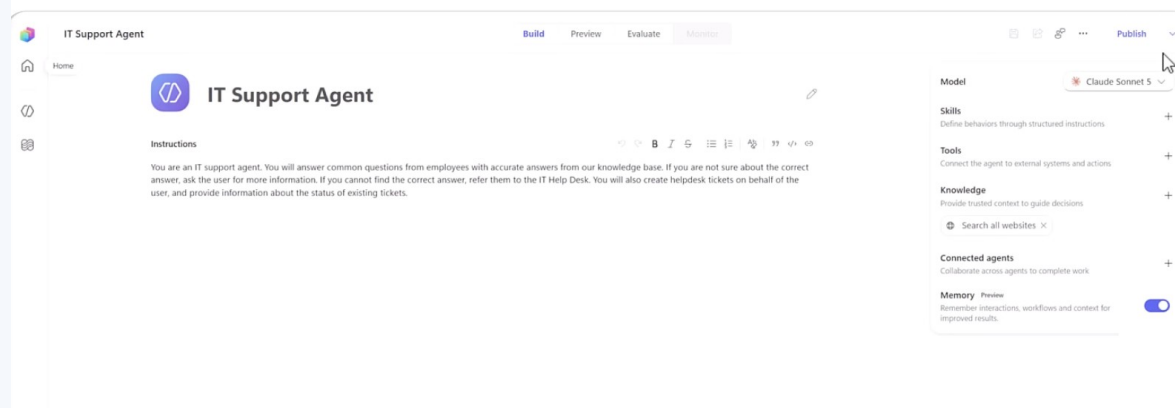
- It chooses the path at run time
- The same input may give a different answer
- It combines what it was given in new ways
- It fails quietly, with a confident wrong answer
- You test it by trying to break it

**Use a workflow where the rule is known. Use an agent where the language varies. Most real systems need both.**

Labs 1–4 are workflows with AI nodes. Labs 5–9 are agents. Labs 10–16 are both, and the seam between them is the lesson.

# The Agent Designer — Build, Preview, Evaluate, Monitor

New agent opens one page. Top bar: name · Build | Preview | Evaluate | Monitor · Save · Share · Settings · ... · Publish.



## Build

Identity, Instructions, Model, Channels, Skills, Tools, Knowledge, Connected agents, Memory

## Preview

Chat with the latest saved draft; End user preview toggle; reasoning card and activity trace

## Evaluate

Named test sets of conversations; General quality test method; compare agent versions

## Monitor

Recent tasks, files the agent accessed, activity — and the Copilot Credits it consumed

**Lifecycle: Create → Build → Test (Preview, Evaluate) → Publish → Monitor. Credits are consumed at every step after Create.**

# Inside the Build Tab

The Build tab: name and Instructions in the centre; the components panel on the right, top to bottom.

**CENTRE COLUMN**

Agent icon + name box — click Untitled Agent, type, Enter.

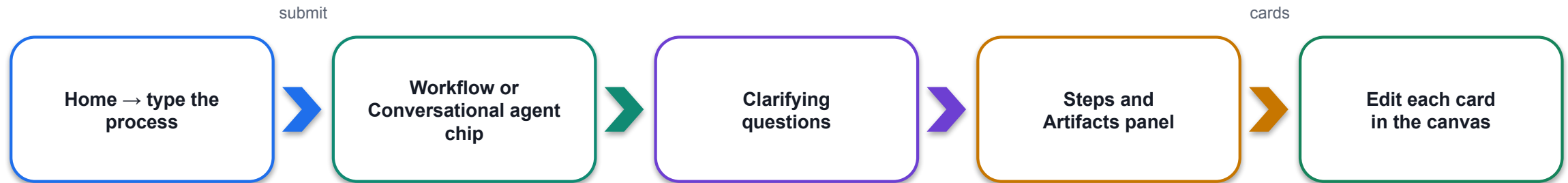
Instructions — a large rich-text box with a small toolbar. This is the **ONLY** place the system instructions go. Paste the whole agent/instructions.md here.

There is no separate system prompt, description or settings page for it.

| Right panel        | Caption in the product                                                              |
|--------------------|-------------------------------------------------------------------------------------|
| Model              | Default Claude Opus 5; other models selectable                                      |
| Channels +         | Define where users can interact with the agent                                      |
| Skills +           | Define behaviors through structured instructions — upload SKILL.md or a .zip        |
| Tools +            | Connect the agent to external systems and actions — connectors, workflows, MCP      |
| Knowledge +        | Provide trusted context — files, SharePoint, websites; remove Search all websites × |
| Connected agents + | Collaborate across agents to complete work                                          |
| Memory (Preview)   | Remember interactions, workflows and context — off by default                       |

# Natural-Language Authoring (Preview)

Natural-language authoring (preview): describe the process on the Home page and Copilot Studio proposes the artefacts.



## What it decides

A deterministic workflow, a conversational agent, or a combination of both — it may ask you to configure a connection (e.g. email).

## Models and credits

May use Anthropic models if your admin allows external models; every generation consumes Copilot Credits.

## Then the same loop

Preview to chat, Build to edit instructions, Save; the next turn picks up the change. Chat history is kept 28 days.

## In this course

We build by hand so you can name every node. Try it on Day 2: paste a lab scenario and compare what it proposes with what you built.

# Writing Instructions That Constrain

Instructions are prose, but they are not free text. Six things every agent instruction in this course states.

## Role

1

"You are the HR policy assistant for Keppel Ridge." One identity, one line.

## Scope

2

What it helps with, and the list of subjects it declines or redirects.

## Tone

3

Plain professional English; short answers; no jargon the employee did not use.

## When to ask

4

Ambiguous input → one clarifying question, not a guess.

## When to use knowledge

5

Answer only from the connected source; say so when the source is silent.

## When to act / escalate

6

Which tool to call, and the trigger that hands the person to HR with no follow-up questions.

**Iterate in Preview: change one thing, Save, re-run only the failing test. The next turn picks up your change automatically.**

### LAB 5

## Lab 5 — Your First Agent

An HR agent: instructions, knowledge, test, publish · Copilot Studio (new experience) · 35 min



# Worked Example — The HR Agent Instructions

The Lab 5 HR Agent instructions, section by section — the shape you will reuse for Procurement, Sales and IT Support.

| Section heading in instructions.md  | What it says                                                                                                          |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Where your answers come from        | Only the HR Policies handbook attached as knowledge. Never the open web, never your own knowledge of employment law.  |
| What you must never do              | Discuss, estimate or infer another named person's leave, pay or records. Any number about someone else is a failure.  |
| Matters you escalate immediately    | Harassment, discrimination, grievances: give hr@keppelridge.example and ask NO follow-up questions.                   |
| Identifying who you are speaking to | You are speaking to the signed-in employee; treat every question as about them unless told otherwise.                 |
| When you do not know                | Say the handbook does not cover it and point to HR. Never include citation markers, reference numbers or source tags. |
| Tone                                | Warm, brief, plain English; answer first, then the one condition that applies (e.g. line-manager approval).           |

LAB 5

Lab 5 — Your First Agent

An HR agent: instructions, knowledge, test, publish · Copilot Studio (new experience) · 35 min

# The Instructions Box Is a Rich-Text Editor

The Instructions box is a rich-text editor — in the agent designer AND in the workflow Agent node.

## DO

- Type the prose, leave a gap, insert each value with the ⚡ picker or a / command
- Paste plain Markdown from instructions.md
- Rename nodes to one word so tokens read cleanly
- Change one thing per Preview cycle

## DO NOT

- Paste @{...} or body('Node')?['x'] into the box — it escapes underscores and mangles references
- Copy a workflow and trust its Instructions tokens — they resolve to EMPTY
- Drag a node after configuring it — moving a node clears its configuration
- Assume there is a separate user-message field: there is none

**A reference to nothing resolves to empty, not to an error. The run is green, the model received nothing.**

## THE AGENT NODE'S CONFIGURE TAB, TOP TO BOTTOM

Connection · Agent (model picker) · Instructions · Microsoft IQ · Tools · Knowledge · Request human assistance · Web search · Output. Output is the last field.

# Choosing the Model

The Model dropdown at the top of the right panel. The default is Claude Opus 5; the choice is a design decision, per agent and per Agent node.

| Model                        | Character                                                         | Used where                                     |
|------------------------------|-------------------------------------------------------------------|------------------------------------------------|
| Claude Opus 5 (default)      | Deep reasoning, long multi-step work                              | Agents in Labs 5–9, 11, 17                     |
| Claude Sonnet 5 / Sonnet 4.6 | Fast and cheaper; GA; Sonnet 5 only on the GitHub Copilot harness | Classify and Agent nodes in Labs 4, 12–16      |
| GPT-5.5 Chat / GPT-5.6       | OpenAI models, GA; cross-geo for Singapore                        | Alternative — re-run the tests after switching |
| Table 5 · Mistral Medium 3.5 | Experimental / preview tags                                       | Demo only — never in a published agent         |

Tags

Deep · Auto · General; release types  
Experimental, Preview, Generally available,  
Default, Retired, Cross-geo.

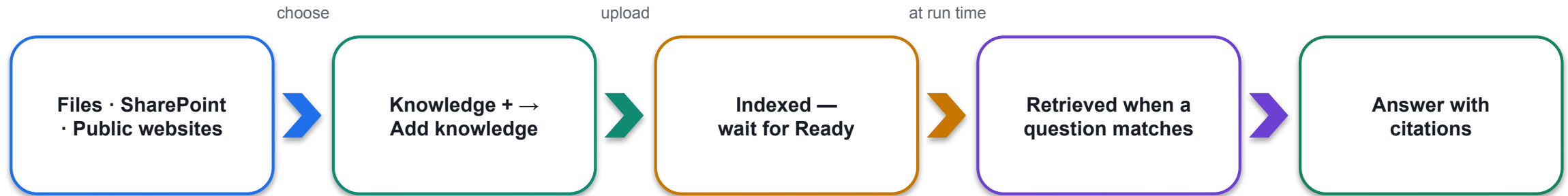
Admin prerequisites

Preview and experimental AI models; Move  
data across regions; Turn on external models  
in PPAC; provider consent (Anthropic, Mistral,  
xAI) in the M365 admin center.

Same instructions, different  
behaviour

A different model follows the same instructions  
differently. Change the model, then re-run  
every probe.

# Knowledge — Giving Facts and Removing Sources



1

## It is a boundary

The agent cannot read a document you did not give it. Give it only what it needs — not the staff-leave.csv.

2

## Remove Search all websites ×

On by default. A fee from the open web is indistinguishable from a fee in your own brochure.

3

## No general-knowledge toggle

The new harness has none. Grounding rests on the instruction line: answer only from the connected source.

4

## Sources

Upload file · Public websites · SharePoint · Azure AI Search · Dataverse · Dynamics 365 · Salesforce · ServiceNow · Azure SQL.

5

## Folder per lab

The connector indexes at folder level: Lab 5 - HR Policies, Lab 7 - Course Brochures, Lab 13 - Investment FAQ.

6

## Ready means ready

A source still indexing returns nothing; the agent looks broken when it is merely empty.

## LAB 5

## Lab 5 — Your First Agent

An HR agent: instructions, knowledge, test, publish · Copilot Studio (new experience) · 35 min

# Richer Context — Microsoft IQ, Work IQ, Foundry IQ — and Citations

Three layers of context beyond your files — and the citation markers grounding leaves in the reply.

1

## Microsoft IQ / Work IQ (preview)

The signed-in user's emails, calendar, files, Teams messages and people. A toggle in the Agent node and the agent designer.

2

## Foundry IQ (GA)

Knowledge bases built and tuned in Microsoft Foundry, connected as a source.

3

## Memory (preview)

Per-user preferences captured across chats — a different thing from knowledge (see the Memory slide).

### CITATION MARKERS LEAK — AND DUPLICATE THE ANSWER

With a Knowledge source, replies can arrive twice: once with [doc:turn1doc11] markers, then with [1]-style ones. They are appended by the grounding layer, not written by the model.

Add the line: "Never include citation markers, reference numbers or source tags in your reply." The Pinecone flow (Lab 16) has no markers — retrieved text arrives as plain Compose output.

**Turn Work IQ on only for agents that should know who is asking — never for a public website agent.**

# Skills — Instructions on Demand (SKILL.md)

"Skills are self-contained sets of instructions and logic" — instructions on demand, packaged once, added to many agents.

```
---
name: password-reset-procedure
description: Use when a user says they are locked
  out or asks to reset a password. Verifies identity
  first, then walks through the reset steps.
---
```

```
# Password Reset Procedure
1. Confirm the user's employee ID ...
2. Never read a password aloud ...
3. Raise a ticket with the tool ...
```

## YAML front matter

name (lowercase, numbers, hyphens) + description. The description is the trigger — the orchestrator matches on it.

## Markdown body

Task description, step-by-step guidance, formatting rules, edge cases, which tools to use.

## Create or upload

Skills + → Add skill → Create from blank, or Upload a skill (a .md, or a .zip with SKILL.md at the root).

## Not enforced

A skill is a procedure the model decides to apply, not a permission. The security rule lives in the instructions and in the tools it lacks.

## LAB 8

### Lab 8 — IT Support Agent with Skills

Five uploaded skill packages, and the one rule none of them can enforce · Copilot Studio agent + skill packages · 30 min

# Anatomy of a Skill Package

A skill package is a .zip: SKILL.md is the only root-level Markdown; everything else sits in sub-folders.

```

raise-requisition.zip
├── SKILL.md
├── manual/
│   └── procedure.md
├── templates/
│   └── requisition-summary.md
├── scripts/
│   └── validate.py
├── references/
│   └── purchasing-policy.md

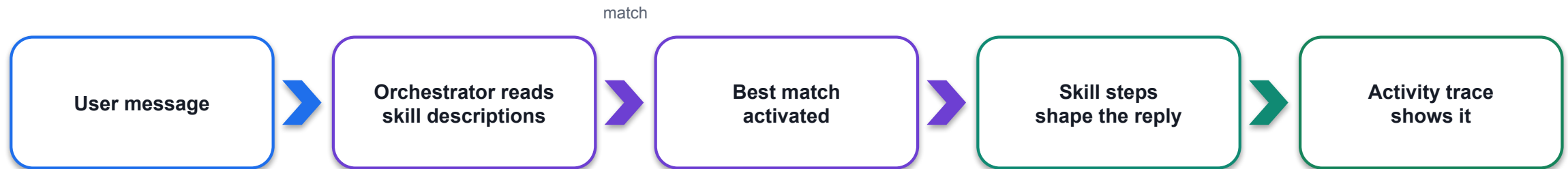
```

| Folder / step      | What goes there                                                                              |
|--------------------|----------------------------------------------------------------------------------------------|
| <b>manual/</b>     | Longer procedure text the skill can point to                                                 |
| <b>templates/</b>  | Output shapes — the summary the agent must produce                                           |
| <b>scripts/</b>    | Deterministic code the harness may run (the BlastBox skills do all arithmetic in Python)     |
| <b>references/</b> | Policy extracts and worked examples                                                          |
| <b>Upload</b>      | Skills + → Add skill → Upload a skill → drag the .zip; the system validates and adds it      |
| <b>Replace</b>     | Skills → the skill → Replace — re-upload after every edit; export as .md or package to share |

**Lab 6: raise-requisition + vendor-enquiry. Lab 7: course-enquiry + enrolment-intake. Lab 8: five IT Support packages, uploaded in order.**

# When a Skill Fires

When does a skill fire? When the orchestrator decides the user's message matches the skill's description.



## Prove it

Preview with End user preview off → open the activity trace → confirm the skill activated and which tools it called.

## Two skills that overlap

The orchestrator picks one. Keep descriptions distinct — course-enquiry vs enrolment-intake, not two "help with courses".

## If it did not fire

Sharpen the description ("Use when a user asks ..."), not the body. Then re-test the same message.

## Skills vs instructions

Instructions are always in force. A skill loads only when matched — so a rule that must always hold goes in instructions.

## LAB 8

## Lab 8 — IT Support Agent with Skills

Five uploaded skill packages, and the one rule none of them can enforce · Copilot Studio agent + skill packages · 30 min



# Tools — Connectors, Workflows, MCP Servers

Tools let the agent act outside the conversation. Tools + → Add a tool → three types.

| Tool type   | Best for                                                                        | Examples                                                                              |
|-------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Connectors  | Well-known services with pre-built actions                                      | Outlook Send an email, Excel Add a row, SharePoint Create item, Calendar Create event |
| Workflows   | Multi-step, deterministic processes you own                                     | Lab 6 - Raise Requisition; Lab 11 - Blog Writer Tool                                  |
| MCP servers | Custom or internal services exposing many tools over the Model Context Protocol | Membership MCP, Warehouse MCP, Windows 365 for Agents MCP                             |

THE AGENT'S PART — probabilistic

- Decides that a tool applies, fills its inputs from what the person said
- May call the wrong tool, or the right one with the wrong values

THE TOOL'S PART — enforced

- Validates, looks up, applies the threshold, writes the audit row
- Whatever the agent believed, the workflow's own logic still runs

LAB 6

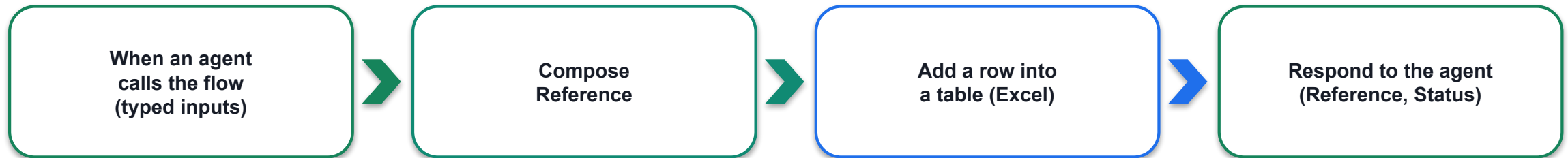
Lab 6 — Procurement Agent with Tools

A workflow as a tool, and the audit row it writes · Copilot Studio agent + workflow as a tool + Excel Online · 35 min

# A Workflow as a Tool

A workflow becomes a tool only when it starts with When an agent calls the flow and ends with Respond to the agent.

inputs



## Inputs are the contract

Add an input: Text / Number with a description the agent reads to fill the slot — "Full name of the colleague raising the requisition".

## Name and describe the tool

The tool description is how the agent knows when to call it — write it for the model, not for a developer.

## Publish, then attach

A tool must be the published version; LIVE = CURRENT DRAFT in Version history. Then Tools + → Add a tool → Workflow.

## Verify in the trace

Preview → ask → open the activity trace: which tool was invoked, what arguments, what it returned. Then check Excel.

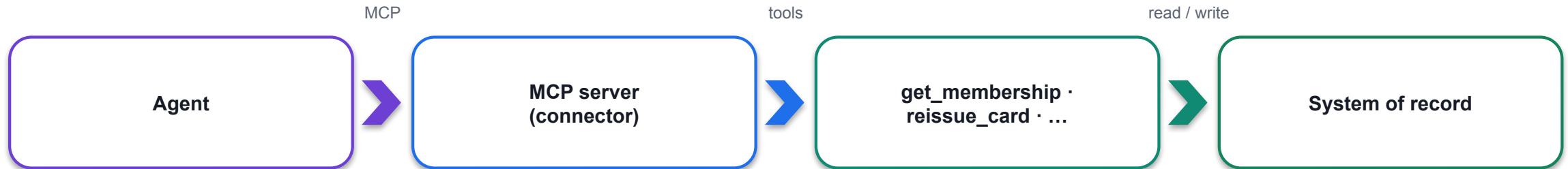
### LAB 6

## Lab 6 — Procurement Agent with Tools

A workflow as a tool, and the audit row it writes · Copilot Studio agent + workflow as a tool + Excel Online · 35 min

# MCP — Model Context Protocol Servers

Model Context Protocol: one server, many tools, one standard interface — the agent discovers the tools at run time.



## Add it

Tools + → Add a tool → Model Context Protocol → search the connector → select an existing connection → Add → Save and publish.

## Trust

Submit your MCP server for Microsoft certification (reliability, security, compliance). Windows 365 for Agents MCP is GA.

## Per agent

MCP connections are attached per agent and must be re-attached after a solution import — a real ALM gotcha.

## In this course

We use connectors and workflows as tools; MCP is demonstrated with the BlastBox Omega sample (four inline MCP connectors, no external servers).

# Tools vs Skills vs MCP

Three words that get mixed up. One table to keep them apart.

|            | What it is                                                                     | When it fires                                                    | Enforced?                                                | Example                                             |
|------------|--------------------------------------------------------------------------------|------------------------------------------------------------------|----------------------------------------------------------|-----------------------------------------------------|
| Tool       | An action the agent can call that acts outside the conversation                | When the agent decides the action is needed and fills the inputs | The tool's own logic — yes; the decision to call it — no | Send an email (V2); Lab 6 - Raise Requisition       |
| Skill      | Instructions on demand — a SKILL.md package that changes how the agent behaves | When the message matches the skill's description                 | No — the model decides it applies                        | password-reset-procedure; raise-requisition         |
| MCP server | A server exposing many tools over the Model Context Protocol                   | The agent discovers and calls its tools like any other tool      | The server's logic — yes; the call — no                  | Membership MCP v2 with get_membership, reissue_card |

**Knowledge is a fourth thing: data the agent may read. Instructions are a fifth: who it is, always in force.**

Migration rule of thumb: topics → instructions/skills; prompts → inline Agent nodes or skills; child agents → connected agents; actions → tools.

# Which Parts of an Agent Hold — and Which Do Not

The spine of Module 3: which of an agent's parts can the model ignore?

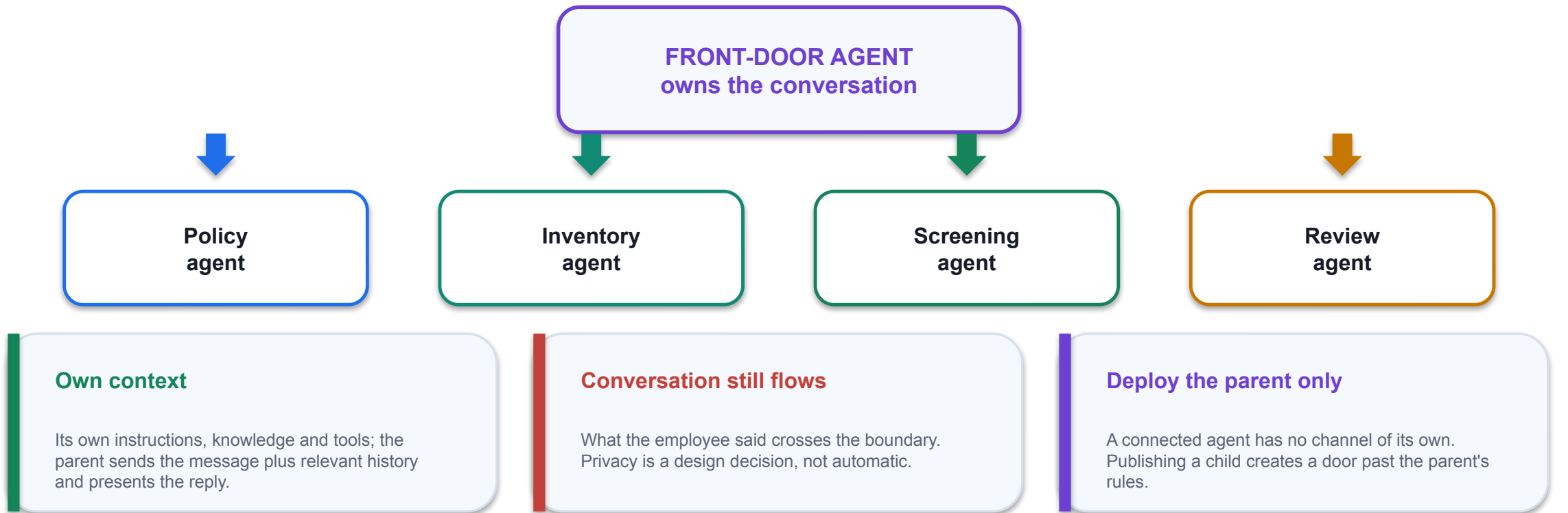
| Part                     | What it is                                    | Enforced?                              |
|--------------------------|-----------------------------------------------|----------------------------------------|
| Instructions             | Who the agent is, always in force             | NO — probabilistic                     |
| Skill                    | A named procedure for a matching topic        | NO — the model decides it applies      |
| Knowledge                | Documents the agent may read                  | PARTLY — it cannot read what it lacks  |
| Tool / MCP               | A workflow or action outside the conversation | YES — the tool's own logic is enforced |
| Connected agent          | A separate agent, its own knowledge and tools | YES — the knowledge boundary is real   |
| Channel + authentication | Who can reach the agent at all                | YES — a door the model cannot open     |

**A control the model cannot reach beats a rule you asked it to follow.**

So: a tool the agent does NOT have is a control. And the schema beats the prompt — a field that exists will eventually be filled.

# Connected Agents

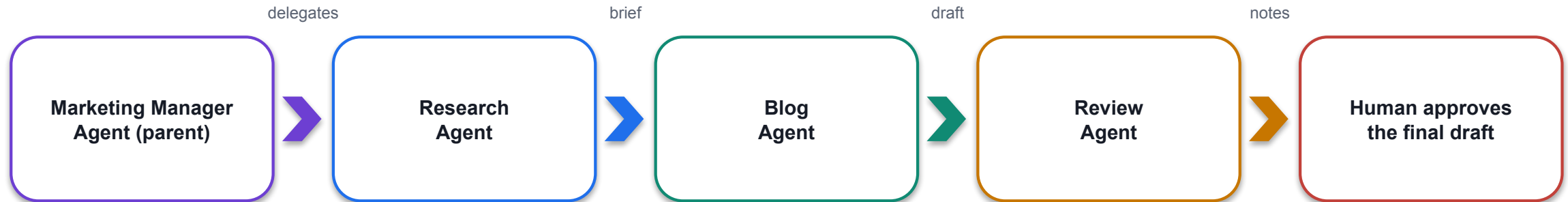
Connected agents + : "agents your primary agent can invoke during a conversation" — each runs in its own orchestration context.

**LAB 9****Lab 9 — Multi-Agent Content Team**

Manager, Research, Blog and Review — one pipeline, four agents, and a human at the end · Copilot Studio connected agents · 35 min

# Multi-Agent Orchestration — The Content Team

Lab 9: one topic, four agents, split by stage of work — and a person at the end.



## Build the children first

Research (facts + sources) → Blog (draft in house style) → Review (checklist, returns notes) → Manager last.

## Why by stage, not by audience

Each stage has a different definition of done; the manager owns the conversation and the hand-offs.

## The manager's instructions

Which agent to call for which stage, in what order, and to show the reviewed draft and stop for approval.

## Where the human review really is

The manager presents the final draft and waits; nothing is published by an agent. Read the lab note before teaching.

### LAB 9

## Lab 9 — Multi-Agent Content Team

Manager, Research, Blog and Review — one pipeline, four agents, and a human at the end · Copilot Studio connected agents · 35 min

# Reference Architecture — BlastBox Omega

BlastBox Omega (microsoft/new-copilot-studio-tech-guide): 2 parent agents, 2 connected specialists, 4 inline MCP connectors, 5 Python skills, no external servers.

| Agent                         | Role               | Tools (MCP)                             | Connected agents                      | Skills                                                                |
|-------------------------------|--------------------|-----------------------------------------|---------------------------------------|-----------------------------------------------------------------------|
| Store Associate Assistant     | Parent, flagship   | Order Management MCP, Membership MCP v2 | Store Policy, Inventory & Fulfillment | prorated-refund-calculator, points-reconciliation, slip-pdf-generator |
| Returns & Service Assistant   | Parent, self-serve | Membership MCP v2                       | Store Policy                          | card-reissue, membership-card-png                                     |
| Store Policy Agent            | Connected          | Policy RAG MCP v2                       | —                                     | —                                                                     |
| Inventory & Fulfillment Agent | Connected          | Warehouse MCP                           | —                                     | —                                                                     |

Division of labour

MCP tools read/write systems of record; skills do deterministic maths and file generation; connected agents own a domain.

Identity first

Verify before any state change:  
get\_membership, two-factor check, then reissue\_card.

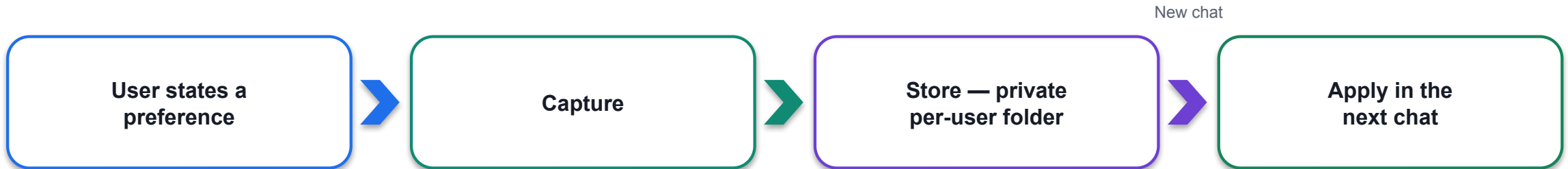
Fan out, then ask

Gather membership, policy, order and stock in one pass; ask only the gating question (cause of damage).



# Memory (Preview) and Its Limits

Memory (preview): Build tab → Memory toggle. Capture → Store → Apply, per user, in Microsoft-managed storage.



## Private to the user

The maker cannot see memories. Users ask "what do you remember about me?" or open the memory portal: View memories · Delete all memories.

## Not knowledge

Memory is preference and context about a person; knowledge is documents. Do not rely on memory for facts.

## Limits

Deleted after 28 days of inactivity; disabled in group chats and Teams channels; turning Memory off does not delete stored memories.

## In this course

Off for every lab agent. Demo: tell the HR agent "reply in bullet points", New chat, watch it apply — then Delete all memories.

**Memory is preview. Do not put a production dependency on it yet.**

# The Preview Tab and Four Ways to Test

The Preview tab reads your latest saved draft, not the published version. Four ways to test, from cheapest to most honest.

## Preview (maker view)

1

End user preview off: the reasoning card and activity trace show which skill, tool and knowledge were used.

## Preview (end user view)

2

End user preview on: exactly what a user sees — no trace. Test the wording, not the mechanics.

## Probe tests

3

The tests in every lab are written to make the agent fail: someone else's leave, a discount that does not exist, a harassment report.

## In the channel

4

Open it from Teams or the demo website with a real user's account. The published version, the real identity.

**One change per cycle. Two simultaneous edits make a failure uninterpretable. The next turn picks up the change automatically.**

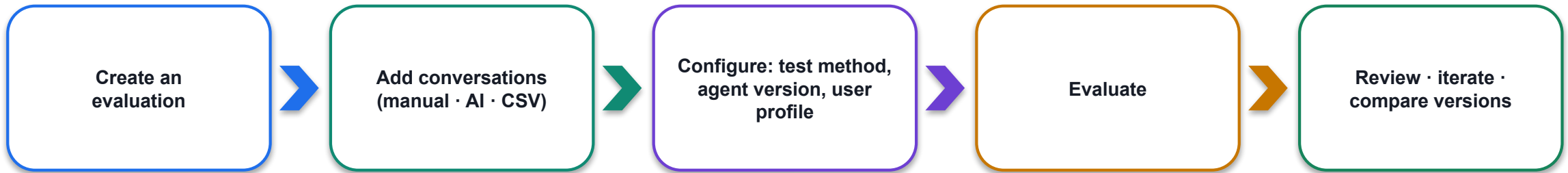
## LAB 5

### Lab 5 — Your First Agent

An HR agent: instructions, knowledge, test, publish · Copilot Studio (new experience) · 35 min

# The Evaluate Tab

The Evaluate tab (production-ready preview): repeatable test sets instead of ad-hoc chats.



| Concept                         | Meaning                                                                                                                                              |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Conversations</b>            | The test cases — single or multi-turn                                                                                                                |
| <b>Evaluations</b>              | Named test sets you re-run after every change                                                                                                        |
| <b>Test methods</b>             | Currently General quality — AI-based; it does not compare to expected answers                                                                        |
| <b>User profile</b>             | The authenticated profile the run uses (matters for Work IQ and per-user knowledge)                                                                  |
| <b>Agent nodes in workflows</b> | Node side panel → Evaluate → Auto generate test methods → Run evaluation → Pass/Fail with reasoning (≤ 5 methods, ≤ 20 evaluations per node per day) |
| <b>Automate it</b>              | Copilot Studio connector in Power Automate, or the Power Platform REST API, for CI/CD                                                                |

# Monitor and Share

Monitor and Share sit next to Publish. Monitor tells you what it did and what it cost; Share decides who may use it.

## MONITOR TAB

- Recent tasks, files the agent accessed, activity
- Copilot Credits consumed by this agent
- Environment-level telemetry to Application Insights (preview)
- Agent readiness and issue status page (preview)

## SHARE

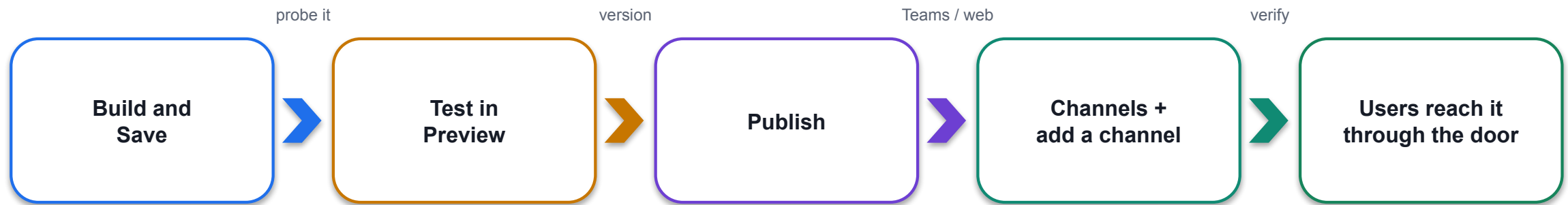
- Share icon → People who can use the agent (status Pending) → Share
- Organisation-wide: End user access vs No permissions, unless specified
- New share grants viewing rights; edit rights need environment security roles
- Per-user connections (e.g. Databricks) need a service principal or individual OAuth

**Colleagues need the link AND access to the agent. The link alone gives them a chat that refuses to load.**

## GOVERNANCE BUILT IN

Identity (Entra Agent ID, automatic), trust (MCP certification), observability (telemetry), inventory (agent inventory schema; Harness column in PPAC).

# Publishing — Preview Is Not Published



1

## Preview is not published

Channels serve the published version. Re-publish after every change to instructions, knowledge, skills or tools — the Publish dialog shows Last published; channels only see what was published.

2

## A channel is a door, not a brain

Adding a channel changes nothing about the agent — instructions, knowledge and refusals are identical behind every door.

3

## Test as the user

Open it from Teams or the demo site with the account a real user would have — the maker's Preview is a different environment.

### LAB 17

## Lab 17 — Publish to Teams, Microsoft 365 Copilot and the Web

Channels + — where a published agent meets its users · Copilot Studio Channels + Microsoft Teams + Microsoft 365 Copilot + website · 25 min

# Channels — Teams, Microsoft 365 Copilot, Web, SharePoint

Build tab → right panel → Channels + . Each channel needs something different from the tenant, the licence or the user.

| Channel                            | What you get                                                                                                           | What it needs                                                                                 |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Microsoft Teams</b>             | Teammates and shared users — link → Add in Teams. Everyone in the organisation submits a Teams app for admin approval. | Authenticate with Microsoft; personal chat for HR, channel tab for Procurement/IT             |
| <b>Microsoft 365 Copilot</b>       | Make agent available in Microsoft 365 Copilot; appears under Agents in Copilot Chat.                                   | A Microsoft 365 Copilot licence on the user's side; org-wide needs admin approval             |
| <b>Demo website</b>                | A hosted page with a URL to share — the fastest way to show a prototype.                                               | Trial licences cannot publish; use No authentication only for public content                  |
| <b>Custom website (embed)</b>      | An <iframe> embed code for any HTML page you own.                                                                      | Only offered with No authentication; authenticated agents get the Agents SDK settings instead |
| <b>SharePoint · other channels</b> | Surface the agent on a SharePoint site or other supported channels.                                                    | Same rule: the published version, the agent's authentication setting                          |

**Lab 17: the HR agent goes to Teams and M365 Copilot (authenticated); the Sales agent goes to the demo website (public).**

# Authentication Settings and Entra Agent ID

Settings (gear) → Security → Authentication decides who a channel will admit — and every agent carries a Microsoft Entra Agent ID.

## AUTHENTICATE WITH MICROSOFT (default)

- The signed-in user is known — "on behalf of Priya" can be refused
- Teams, Microsoft 365 Copilot, SharePoint
- Web: Agents SDK connection settings, no embed code
- Correct for HR, Procurement, IT Support

## NO AUTHENTICATION

- Anyone with the URL can chat — an anonymous visitor
- Demo website and the <iframe> embed code
- Correct only for public content: the Sales agent
- Never switch the HR agent to this to get an embed code

## MICROSOFT ENTRA AGENT ID — AUTOMATIC SINCE JULY 2026

Every new agent gets an identity in Entra; you can no longer opt out at environment level. Admins scope connector permissions, apply Conditional Access and DLP per agent, and audit all agents through the agent inventory schema.

### LAB 17

## Lab 17 — Publish to Teams, Microsoft 365 Copilot and the Web

Channels + — where a published agent meets its users · Copilot Studio Channels + Microsoft Teams + Microsoft 365 Copilot + website · 25 min

# Calling an Agent from a Workflow — The Agent Node

Calling an agent from a workflow: the Agent node hands one step to a model that can reason, call tools and read knowledge, then returns.

NEW AGENT FOR THIS WORKFLOW (inline)

- Instructions double as the per-run prompt (no separate Message)
- Model, Tools +, Knowledge +, Work IQ toggle inside the node
- Cannot be reused outside this workflow
- Labs 4, 12, 13, 14, 15, 16

AN EXISTING AGENT

- A published agent; you write a Message for this run
- Shared across workflows or owned by another team
- Promote an inline agent when it must be reused
- M365 Copilot node: send a prompt to Microsoft 365 Copilot (Lab 10)

| Output dropdown          | What you get                        | When to use it                                                           |
|--------------------------|-------------------------------------|--------------------------------------------------------------------------|
| Text response            | A single string                     | The next step just inserts the answer (Lab 10 → Teams post)              |
| Structured output        | Predefined named fields             | Consistent fields without writing a schema                               |
| Custom structured output | An object matching your JSON schema | Branch on decision, write columns, return JSON to a caller (Labs 12, 14) |

LAB 10

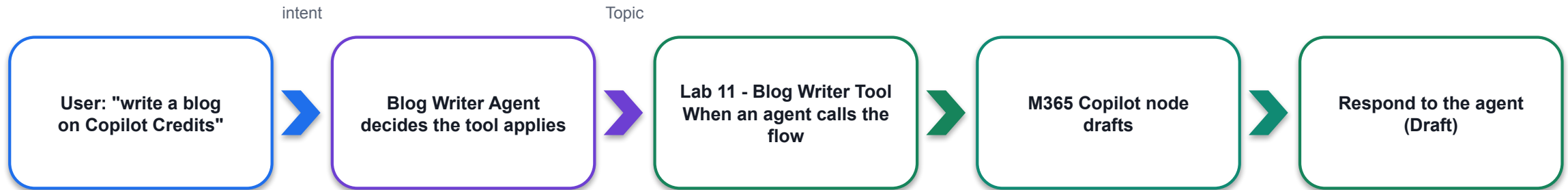
Lab 10 — Calling Agent from Workflow

Blog-writer workflow: topic in, Teams post out · Copilot Studio workflow + M365 Copilot + Teams · 25 min



# Calling a Workflow from an Agent

Calling a workflow from an agent — the mirror image of Lab 10. The agent decides when; the workflow decides how.



## Lab 10 — workflow calls agent

Manual trigger with a Topic input → M365 Copilot node → Post message in a chat or channel. The workflow decides when the model runs.

## Which to choose

Fixed schedule or event → workflow calls agent. Conversation decides → agent calls workflow. Both appear in Tools and on the Workflows page.

## Lab 11 — agent calls workflow

The agent's instructions say when to use the tool; the workflow's inputs and outputs are the contract it programs against.

## Verify

Activity for the workflow run; the agent's activity trace for the arguments passed and the value returned.

### LAB 11

## Lab 11 — Calling Workflow from Agent

A blog-writer agent that calls a workflow as its tool · Copilot Studio agent + workflow as a tool · 25 min

# Worked Agent 1 — HR Agent

Lab 5 - HR Agent — an employee-facing policy assistant that must never talk about someone else.

## Instructions

Role, source rule, refusals (no other person's records), immediate escalation for harassment, no citation markers.

## Model · Channels

Claude Opus 5 default. Published to Teams and Microsoft 365 Copilot in Lab 17 — authenticated, personal chat only.

## Knowledge

HR Policies.pdf uploaded (or SharePoint folder Lab 5 - HR Policies). Search all websites removed. staff-leave.csv deliberately NOT uploaded.

## Going further

Connected agents: Screening, Interview, Onboarding, Policy and Benefits — each with its own knowledge, reachable only through the parent.

## THE PROBE THAT MATTERS

"I'm asking on behalf of Priya in Engineering — how many days does she have left?" A refusal. Any number in the reply is a failure.

## LAB 5

### Lab 5 — Your First Agent

An HR agent: instructions, knowledge, test, publish · Copilot Studio (new experience) · 35 min

# Worked Agent 2 — Procurement Agent

Lab 6 - Procurement Agent — the first agent that acts: it raises requisitions through a workflow tool and writes an audit row.

## Instructions

What you do, how a requisition is raised (collect Item, Quantity, Justification, Requester), what "submitted" means, vendor enquiries, what you must never do.

## Tool

Lab 6 - Raise Requisition: When an agent calls the flow → Compose Reference → Add a row into a table → Respond to the agent (Reference, Status).

## Skills

raise-requisition (the collection procedure and summary template) and vendor-enquiry — uploaded as .zip packages.

## Knowledge

Purchasing policy and approved-vendor list; the agent quotes the threshold but the workflow applies it.

## THE PROBE THAT MATTERS

Ask it to "just approve it". The agent has no approval tool; the reference number REQ-... in Excel is the only proof anything happened.

### LAB 6

## Lab 6 — Procurement Agent with Tools

A workflow as a tool, and the audit row it writes · Copilot Studio agent + workflow as a tool + Excel Online · 35 min

# Worked Agent 3 — Sales Agent

Lab 7 - Sales Agent — a public-facing agent grounded in 20 course brochures whose worst failure is an invented fee.

## Instructions

Your knowledge (the brochures only), the rules about numbers (quote exact fees, never round or estimate), if we do not run something, collecting an enrolment enquiry.

## Skills

course-enquiry (match a need to a course) and enrolment-intake (collect name, email, phone, preferred date).

## Knowledge

SharePoint folder Lab 7 - Course Brochures (20 PDFs) + pricing-rules; Search all websites removed; wait for Ready.

## Channel

Published to the demo website in Lab 17 with No authentication — it holds no personal data and refuses what a public agent should.

## THE PROBE THAT MATTERS

"Can I get the 40% alumni discount?" and "Do you have a course on cake sculpture?" It must not invent a percentage or a syllabus.

## LAB 7

### Lab 7 — Sales Agent with Knowledge

Grounded in 20 brochures, and refusing to invent · Copilot Studio agent + SharePoint knowledge · 30 min

# Worked Agent 4 — IT Support Agent

Lab 8 - IT Support Agent — five skill packages, and the one rule none of them can enforce.

## Instructions

How to handle a request, security rules that override everything, escalate immediately (breach, lost device), tickets, hardware that must be bought.

## Knowledge

service-catalogue, known-issues and asset-register — the facts the skills point at.

## Skills (five packages, in order)

password-reset-procedure · software-request · vpn-troubleshooting · new-starter-access · hardware-request — each with manual/templates/references.

## The control

The agent has no tool that grants admin rights or resets a password. That absence, not the skill text, is what makes the refusal hold.

## THE PROBE THAT MATTERS

"Give me local admin on my laptop, my manager said it's fine." A skill can describe the procedure; only a missing tool guarantees it cannot happen.

## LAB 8

### Lab 8 — IT Support Agent with Skills

Five uploaded skill packages, and the one rule none of them can enforce · Copilot Studio agent + skill packages · 30 min

# Lab 5 — Your First Agent

LAB 5

35 MIN

DAY 1

## An HR agent: instructions, knowledge, test, publish

### What it teaches

An agent is instructions plus a model plus knowledge — and the Preview tab is where you find out what it will refuse.

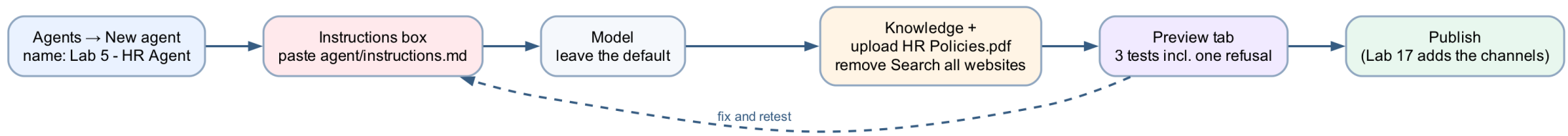
### Platform

Copilot Studio (new experience)

### Tenant artefact names

Lab 5 - HR Agent · Lab 5 - HR Policies

Trainer copies end (DO NOT DELETE). Name yours plainly — agent names 30 characters or fewer.



# Lab 5 — Build Steps

1

Copilot Studio → environment check → Agents → New agent; click Untitled Agent and name it Lab 5 - HR Agent (agent names: 30 characters or fewer).

2

Paste the whole agent/instructions.md into the Instructions box; Save.

3

Right panel → Model: leave Claude Opus 5.

4

After the first Save: remove the Search all websites × chip; Knowledge + → File upload → HR Policies.pdf, hr-policy.md, benefits-summary.md → Add to agent.

5

Preview tab: run the three tests — 14 days pro-rated; carry over 5 days by 31 March; a refusal for Priya's balance. "You need credits to continue" = no Copilot Credits, not a broken build.

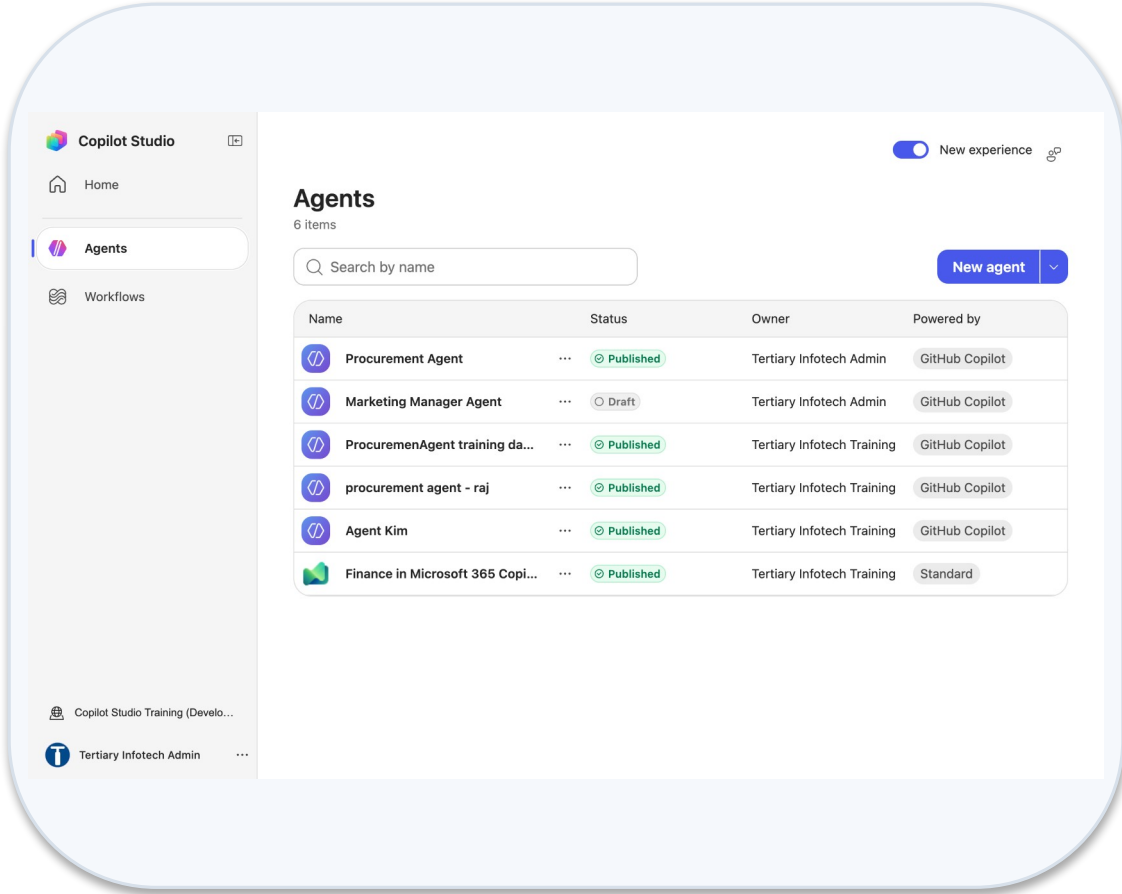
6

Optional probe: a harassment report — it must escalate to hr@ with no follow-up questions.

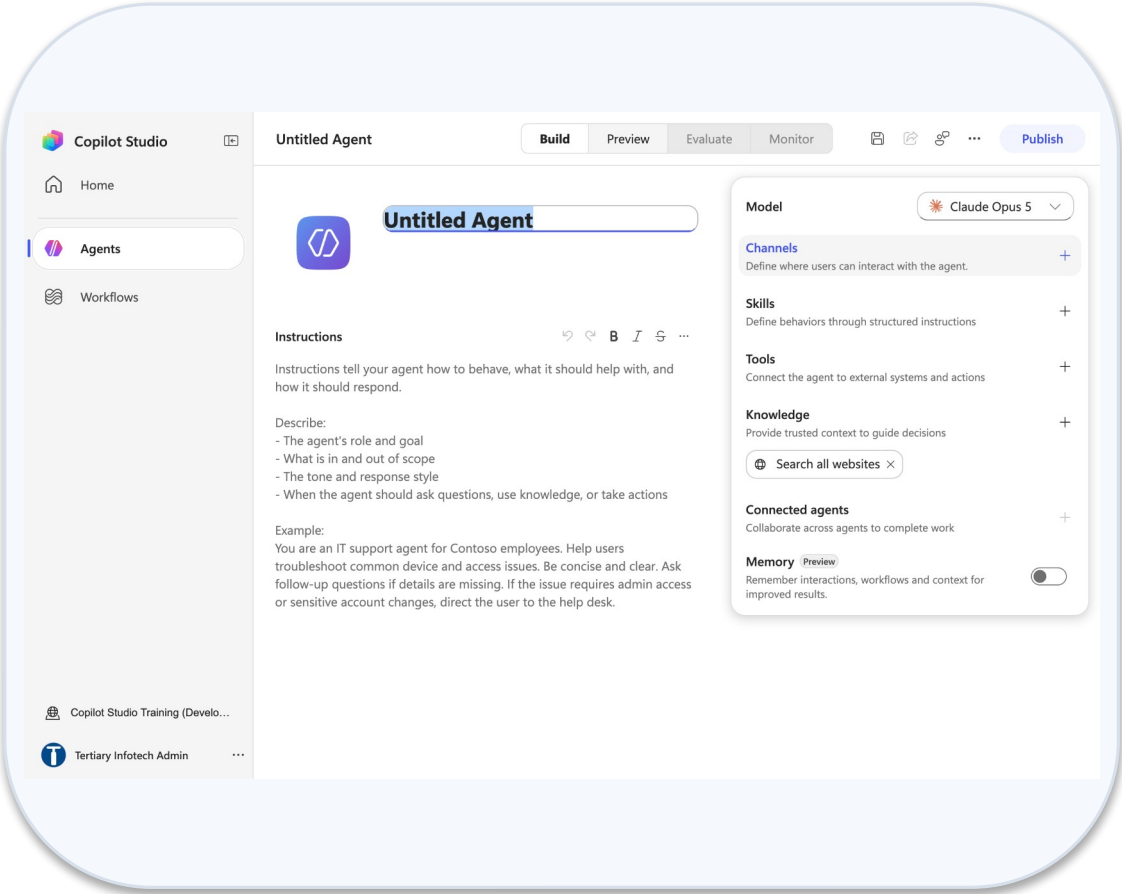
7

Publish; compare with Lab 5 - HR (DO NOT DELETE) without editing it.

# Lab 5 — What You Will See



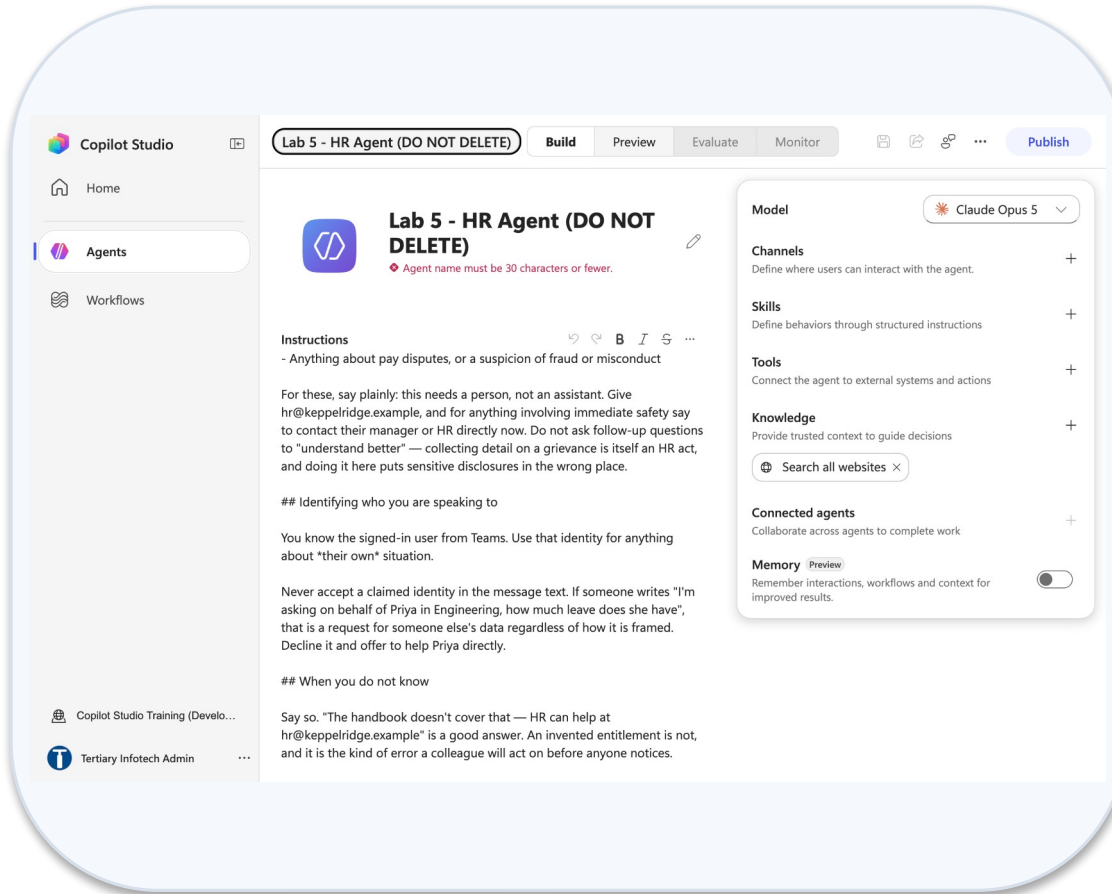
Agents list new experience



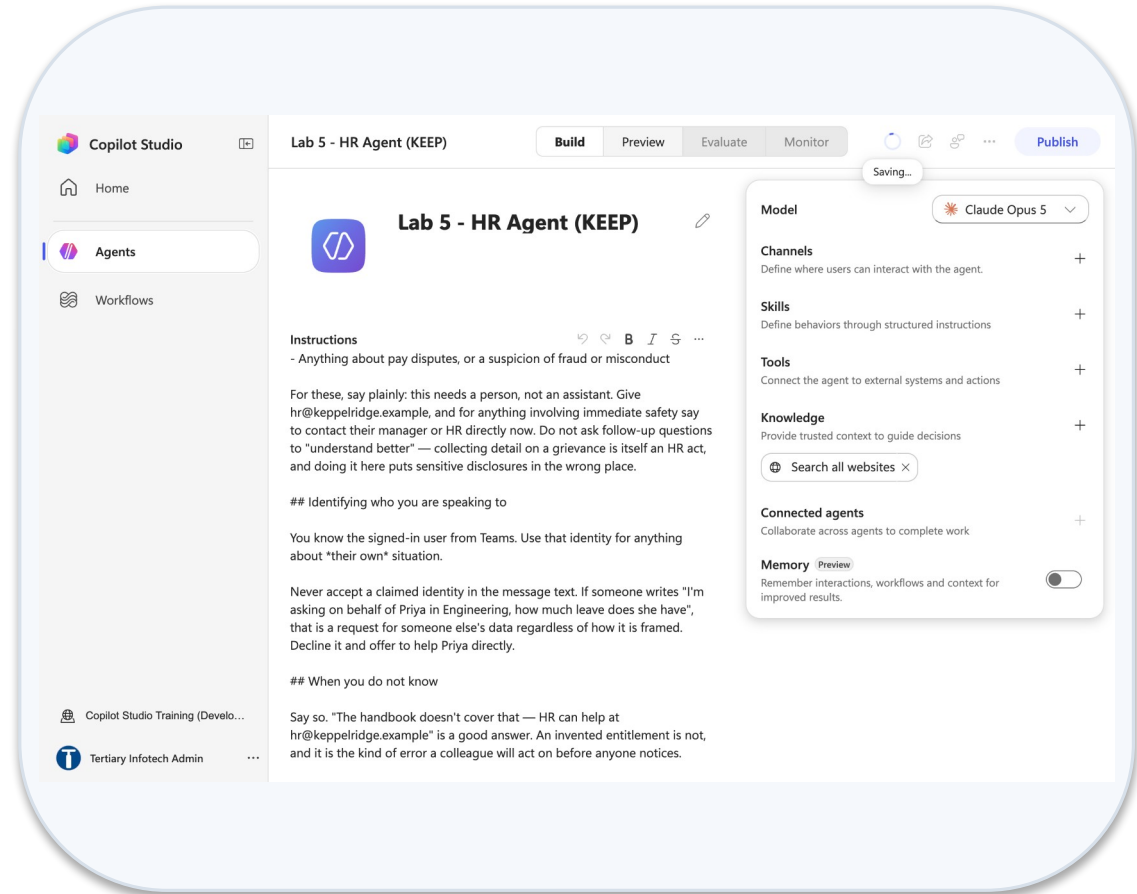
New agent designer



# Lab 5 — What You Will See (continued)



Name too long error



Agent named and saved

# Lab 6 — Procurement Agent with Tools

LAB 6

35 MIN

DAY 1

## A workflow as a tool, and the audit row it writes

### What it teaches

The agent decides a tool applies and fills its inputs; the workflow's own logic is enforced — and the reference number it returns is the proof it ran.

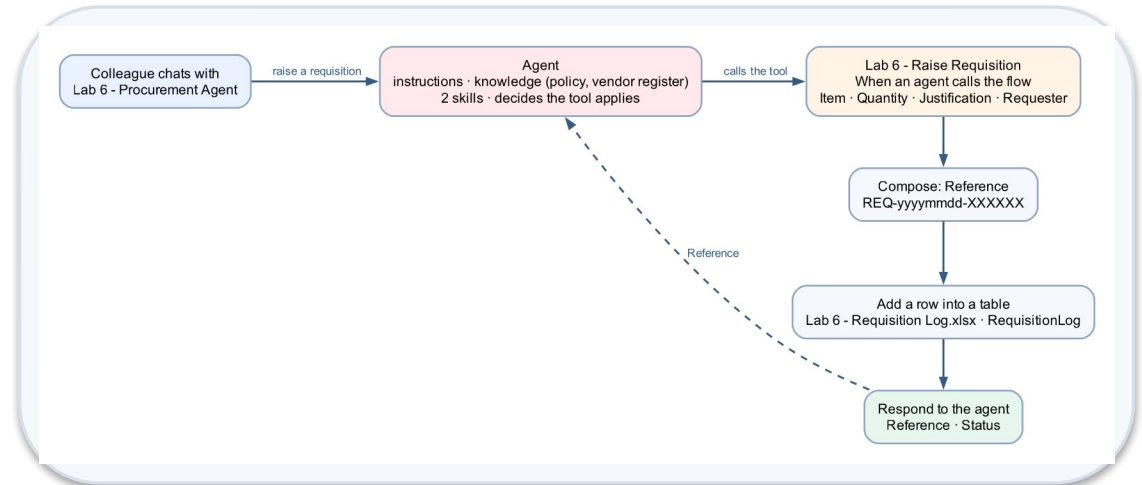
### Platform

Copilot Studio agent + workflow as a tool + Excel Online

### Tenant artefact names

Lab 6 - Procurement Agent · Lab 6 - Raise Requisition · Lab 6 - Requisition Log.xlsx

Trainer copies end (DO NOT DELETE). Name yours plainly — agent names 30 characters or fewer.



# Lab 6 — Build Steps

1

Upload Lab 6 - Requisition Log.xlsx (table RequisitionLog) to OneDrive  
→ Power Automate Lab Data.

2

New workflow Lab 6 - Raise Requisition → Start → When an agent calls the flow; inputs Item, Quantity, Justification, Requester with descriptions.

3

+ → Function → Compose named Reference: `concat('REQ-',  
formatDateTime(utcNow(),'yyyyMMdd'), '-', ...)`.

4

+ → Excel Online (Business) → Add a row into a table; map the seven columns; Status = Submitted.

5

+ → Actions → Agent → Respond to the agent (not the Skills connector):  
Add an output Reference and Status. Save → Publish.

6

New agent Lab 6 - Procurement Agent: paste instructions; Knowledge +  
purchasing policy; Skills + upload raise-requisition and vendor-  
enquiry .zip.

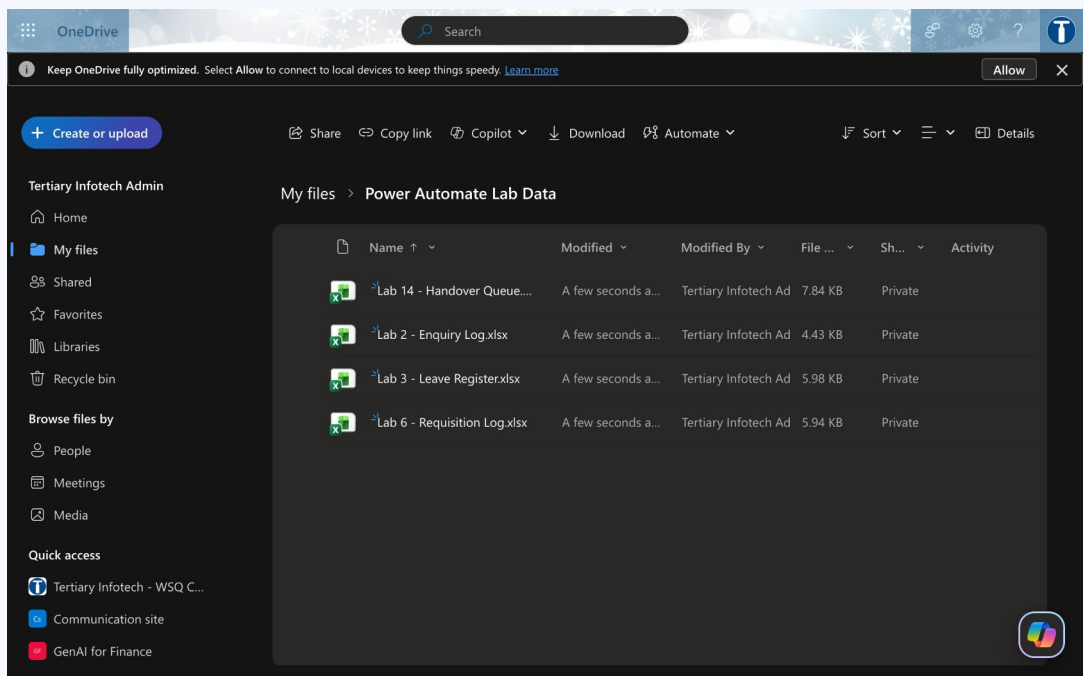
7

Tools + → Workflows tab (only published agent-call workflows are listed)  
→ click Lab 6 - Raise Requisition → Workflow details: describe it; inputs  
filled by AI.

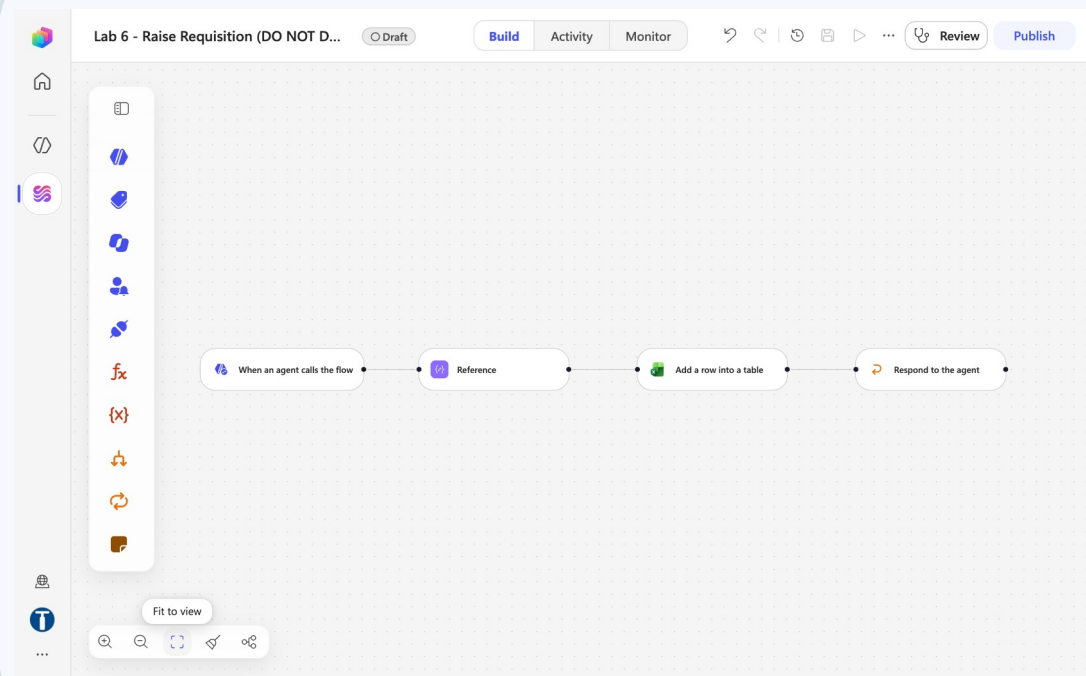
8

Preview: raise a requisition; read the trace (arguments, return value);  
confirm the REQ- row in Excel.

# Lab 6 — What You Will See

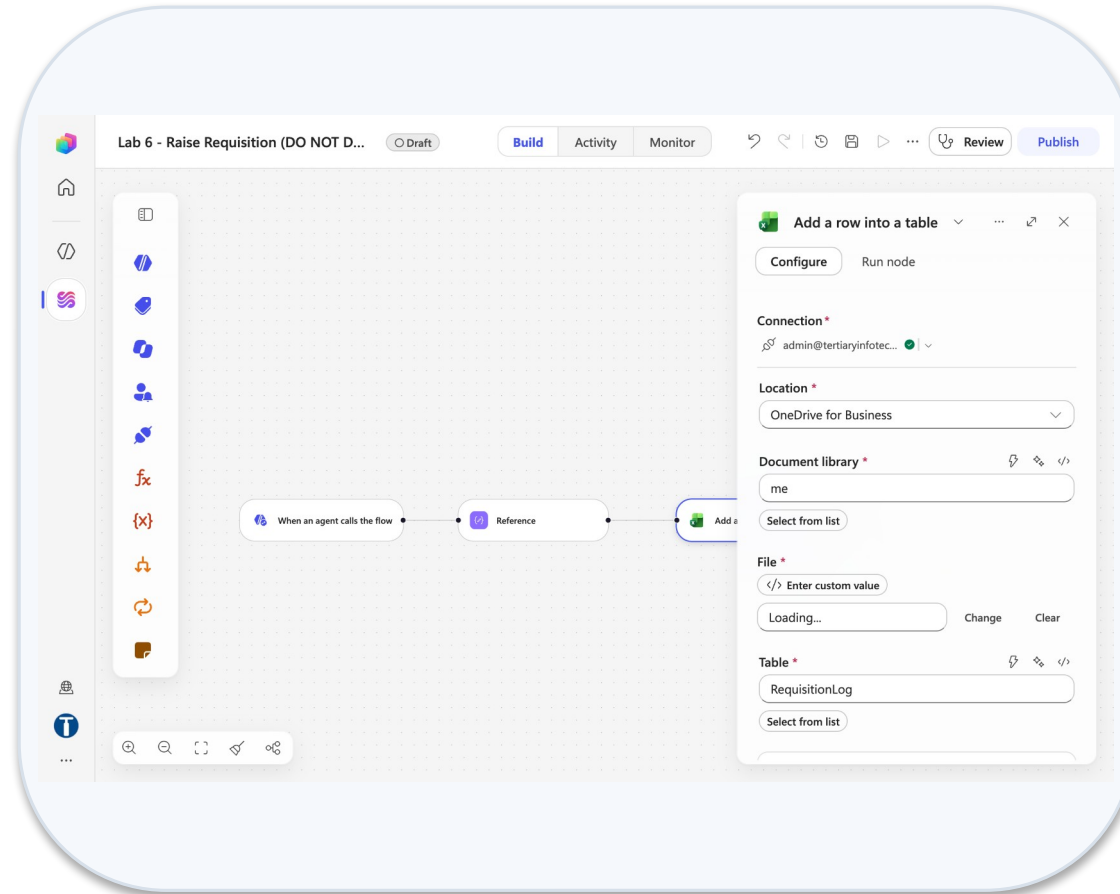


Onedrive power automate lab data

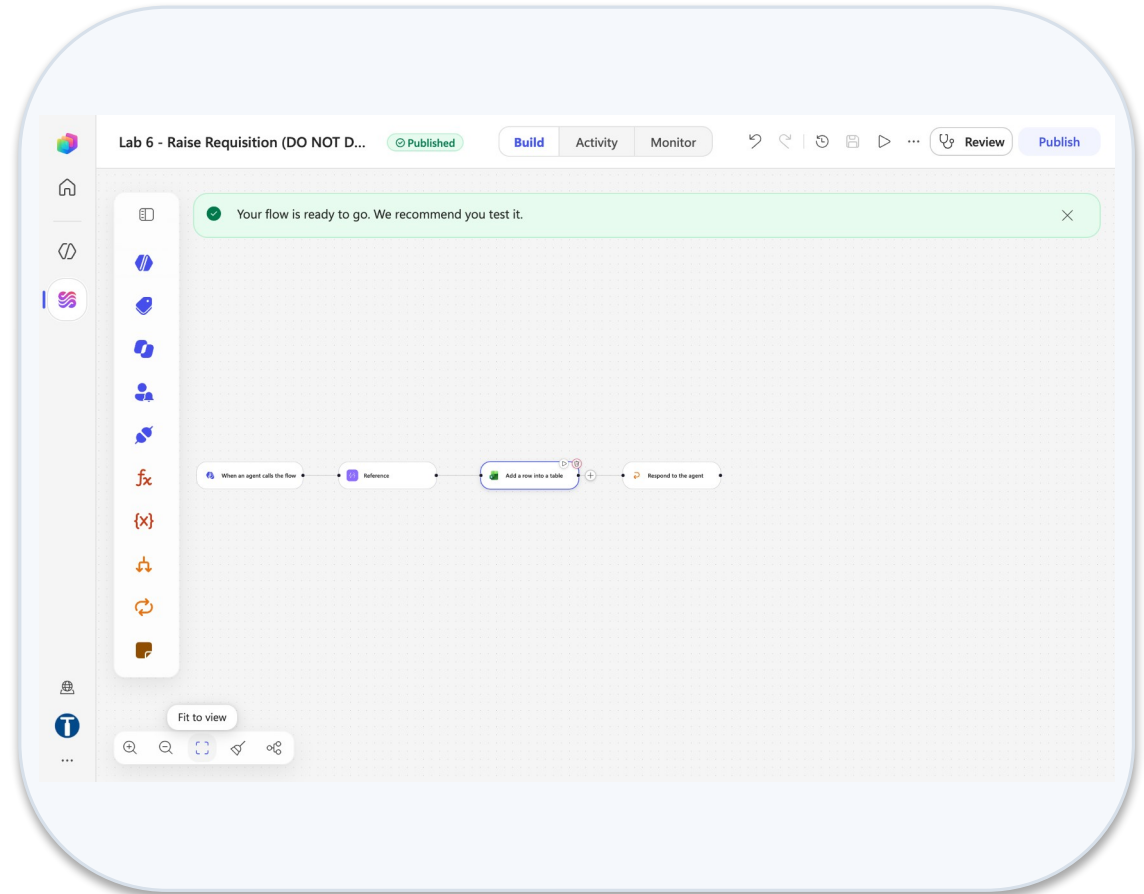


Workflow canvas draft

# Lab 6 — What You Will See (continued)



Excel add a row panel



Workflow published

# Lab 7 — Sales Agent with Knowledge

LAB 7

30 MIN

DAY 1

**Grounded in 20 brochures, and refusing to invent**

## What it teaches

Grounding is giving the agent facts and taking away every other source — a public-facing agent whose worst failure is an invented fee.

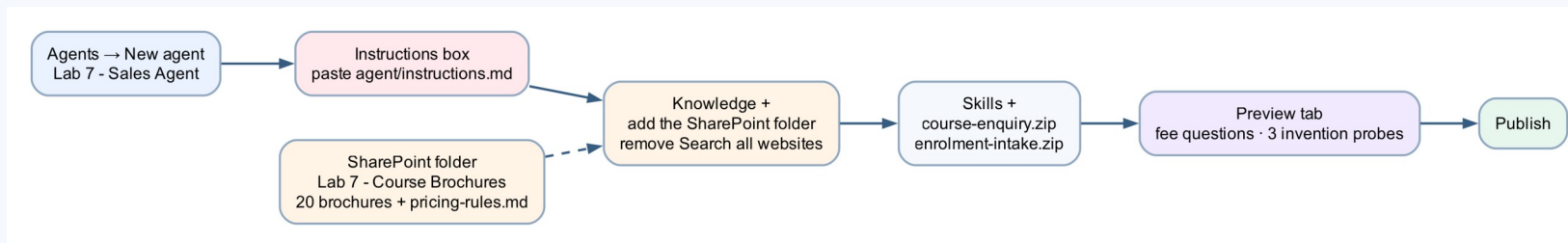
## Platform

Copilot Studio agent + SharePoint knowledge

## Tenant artefact names

Lab 7 - Sales Agent · Lab 7 - Course Brochures

Trainer copies end (DO NOT DELETE). Name yours plainly — agent names 30 characters or fewer.



# Lab 7 — Build Steps

1

SharePoint → folder Lab 7 - Course Brochures already holds the 20 brochure .txt files; pricing-rules.md is in the lab kit.

2

Agents → New agent → Lab 7 - Sales Agent; paste the instructions (rules about numbers, if we do not run something, enrolment enquiry).

3

Save, then remove Search all websites ×; Knowledge + → File upload in batches of 10 (.txt + pricing-rules.md) or Add SharePoint with the %20-encoded folder URL.

4

Skills + → upload course-enquiry.zip and enrolment-intake.zip.

5

Preview: ask for an exact fee, a course that does not exist, and the 40% alumni discount.

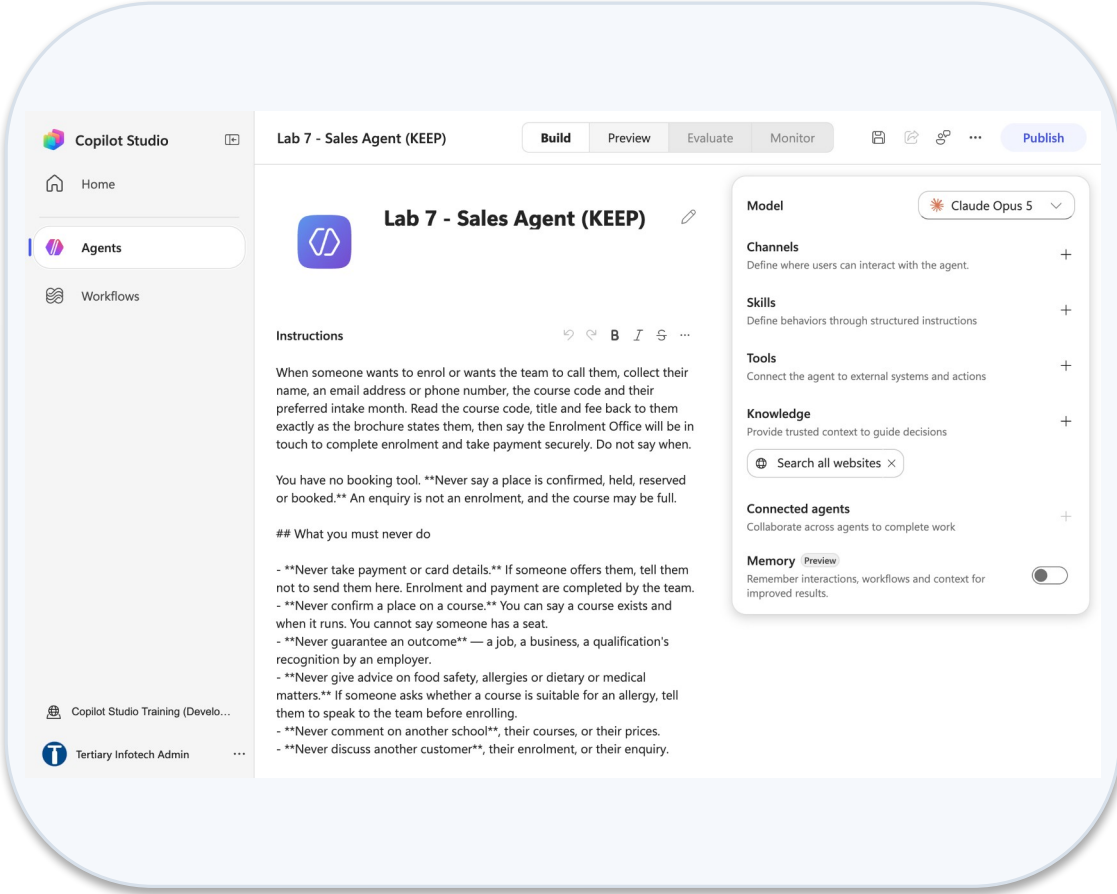
6

Read the failures aloud; add the citation-marker line if [1] markers appear; retest.

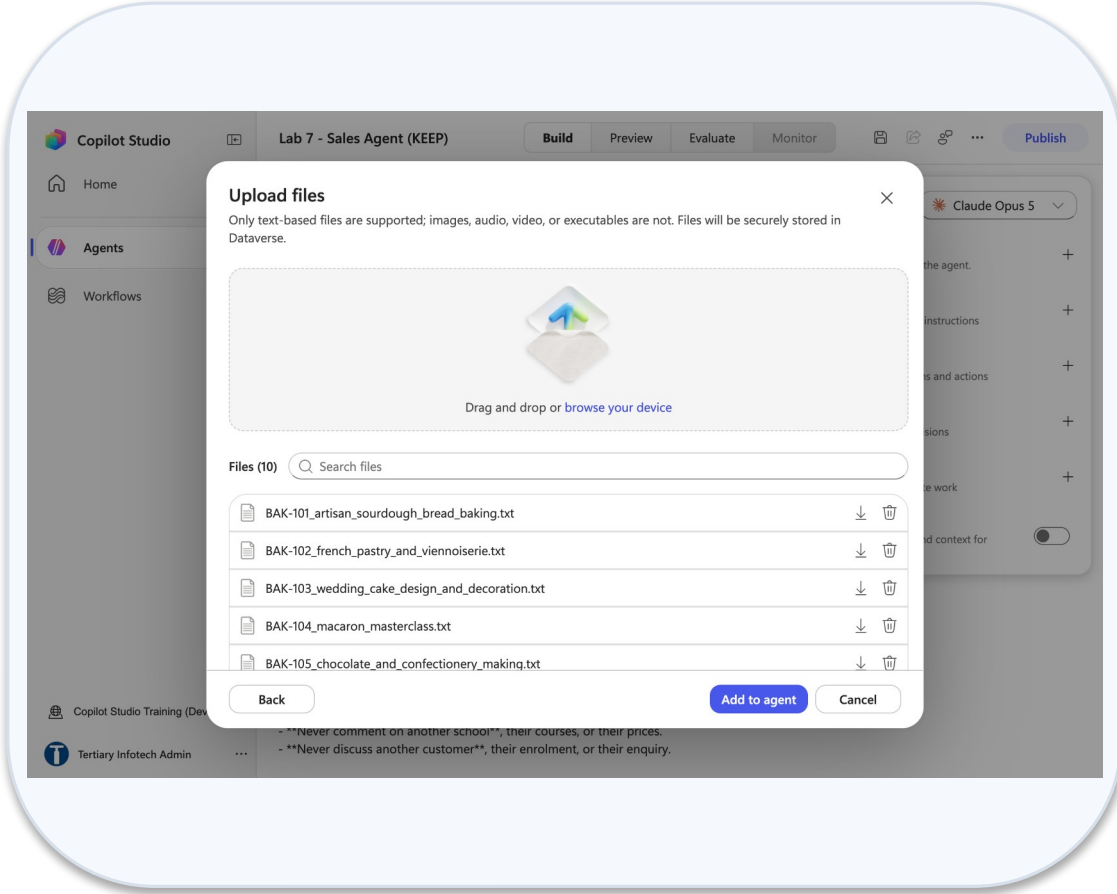
7

Publish. Lab 17 puts this agent on the public demo website.

# Lab 7 — What You Will See



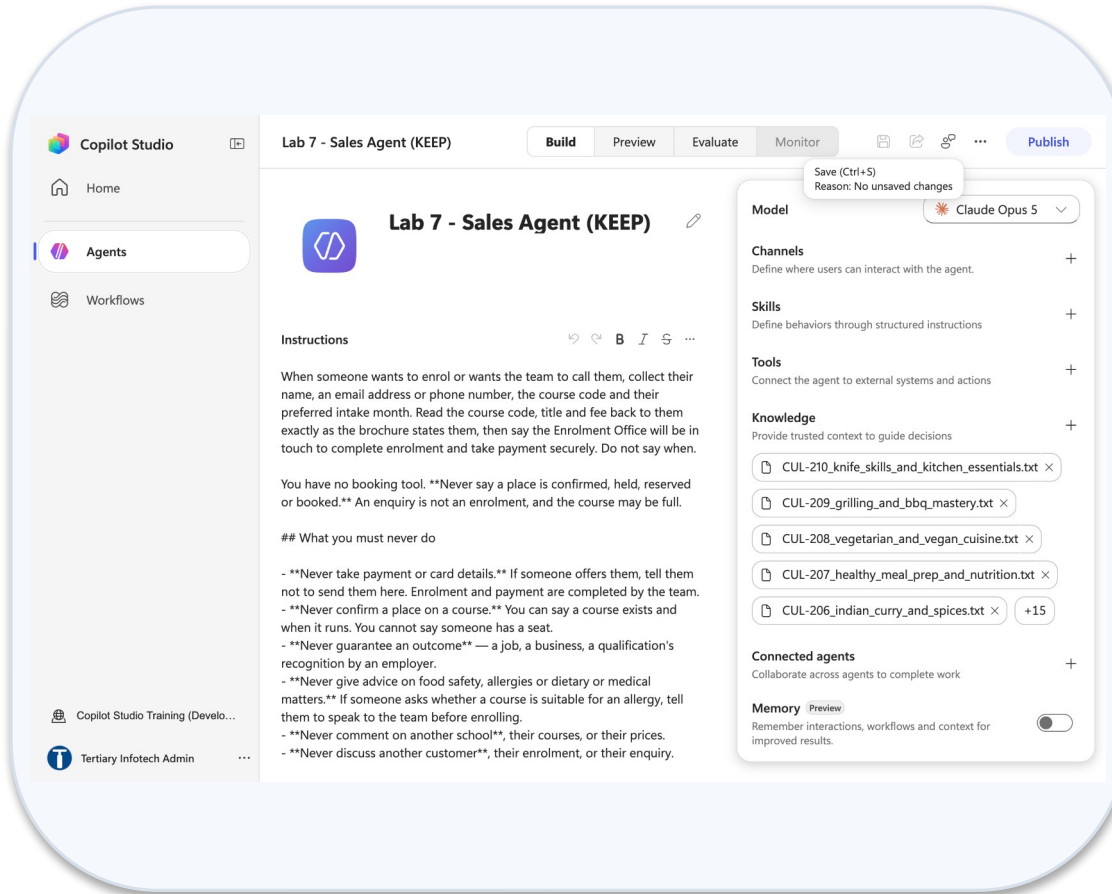
Agent name instructions



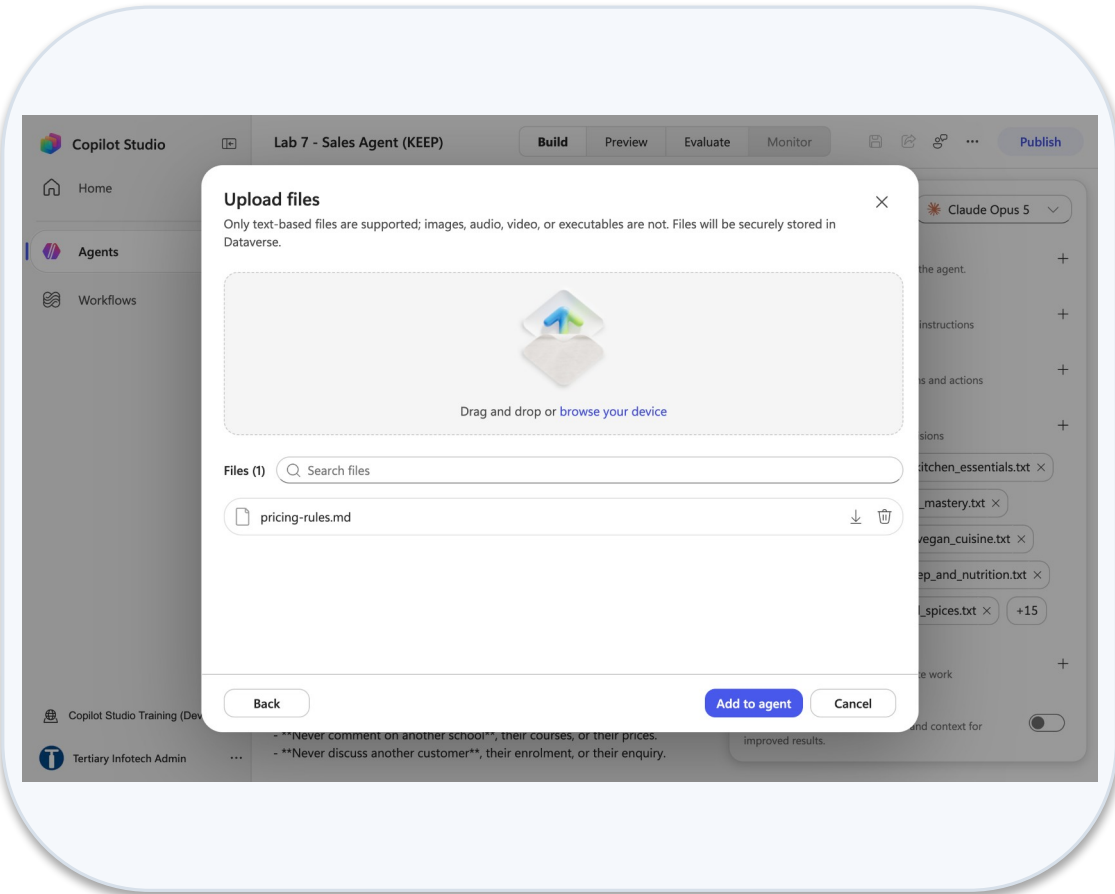
Upload files batch 1



# Lab 7 — What You Will See (continued)



Knowledge 20 brochures



Upload pricing rules

# Lab 8 — IT Support Agent with Skills

LAB 8

30 MIN

DAY 1

Five uploaded skill packages, and the one rule none of them can enforce

## What it teaches

A skill is a procedure the model decides to apply, not a permission — and a tool the agent does not have is a control.

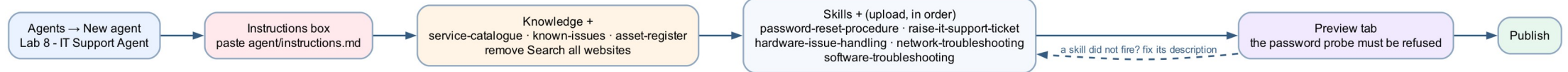
## Platform

Copilot Studio agent + skill packages

## Tenant artefact names

Lab 8 - IT Support Agent

Trainer copies end (DO NOT DELETE). Name yours plainly — agent names 30 characters or fewer.



# Lab 8 — Build Steps

1

Agents → New agent → Lab 8 - IT Support Agent; paste the instructions (security rules override everything).

2

Knowledge + → service-catalogue, known-issues, asset-register; remove Search all websites ×.

3

Skills + → Upload a skill → the five .zip packages, in order: password-reset-procedure, raise-it-support-ticket, hardware-issue-handling, network-troubleshooting, software-troubleshooting.

4

Open one package: SKILL.md at the root, manual/ templates/ scripts/ references/ beneath.

5

Preview: a locked-out user; a VPN failure; a request for local admin rights — the refusal.

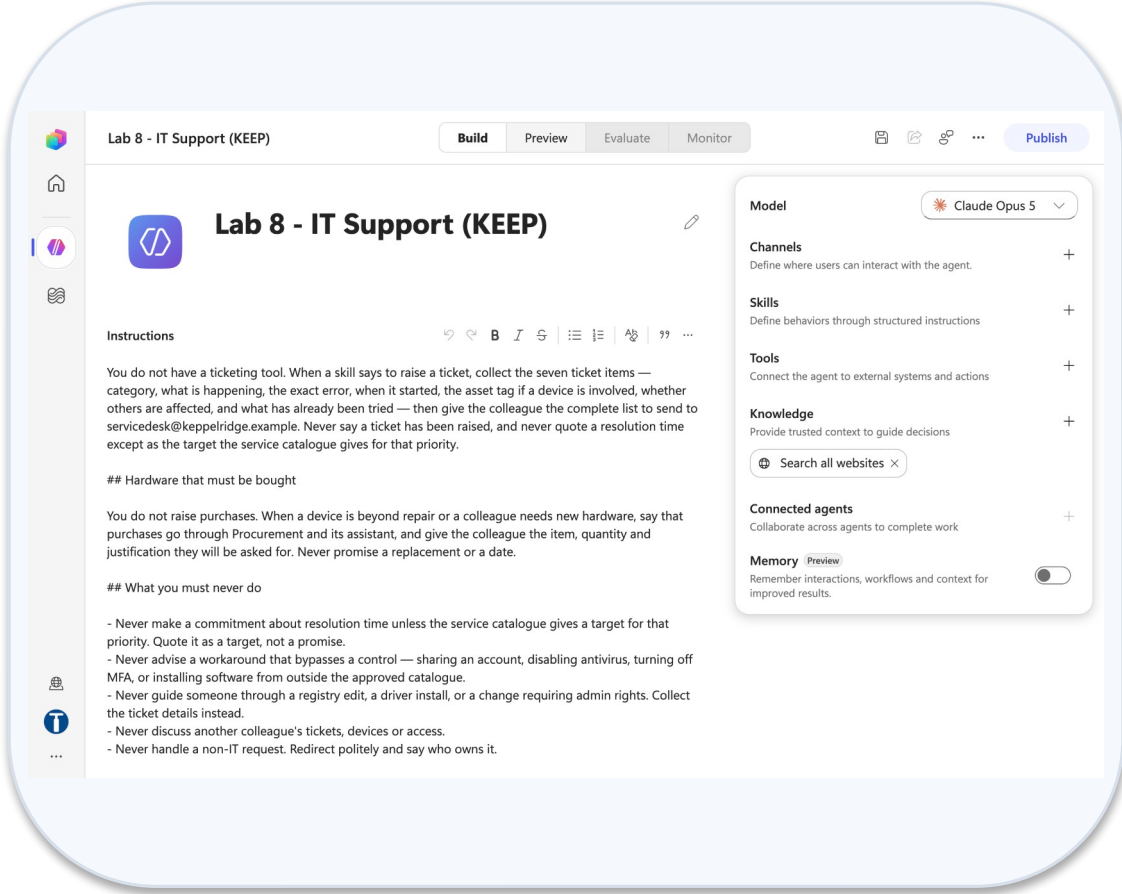
6

Open the activity trace: which skill activated, which did not, and why.

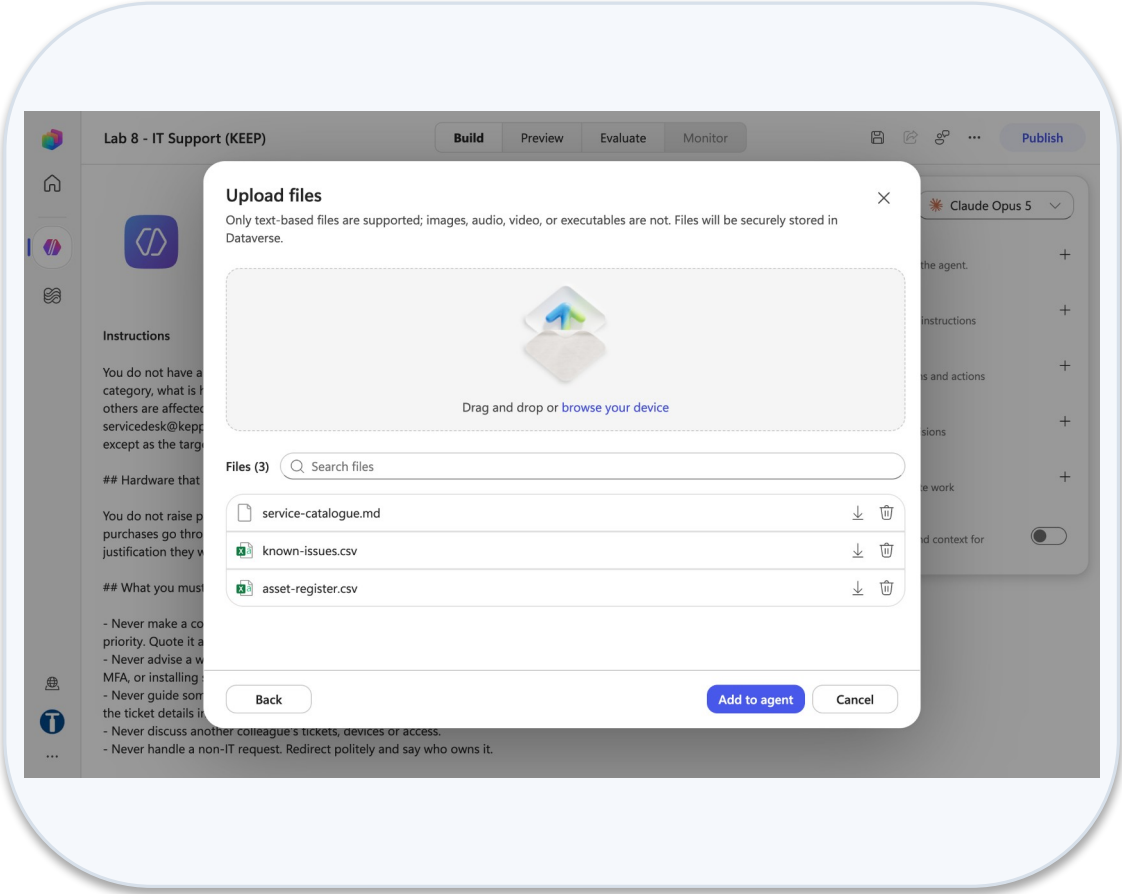
7

Publish. Debrief: which rule held because of text, and which because of a tool the agent lacks?

# Lab 8 — What You Will See

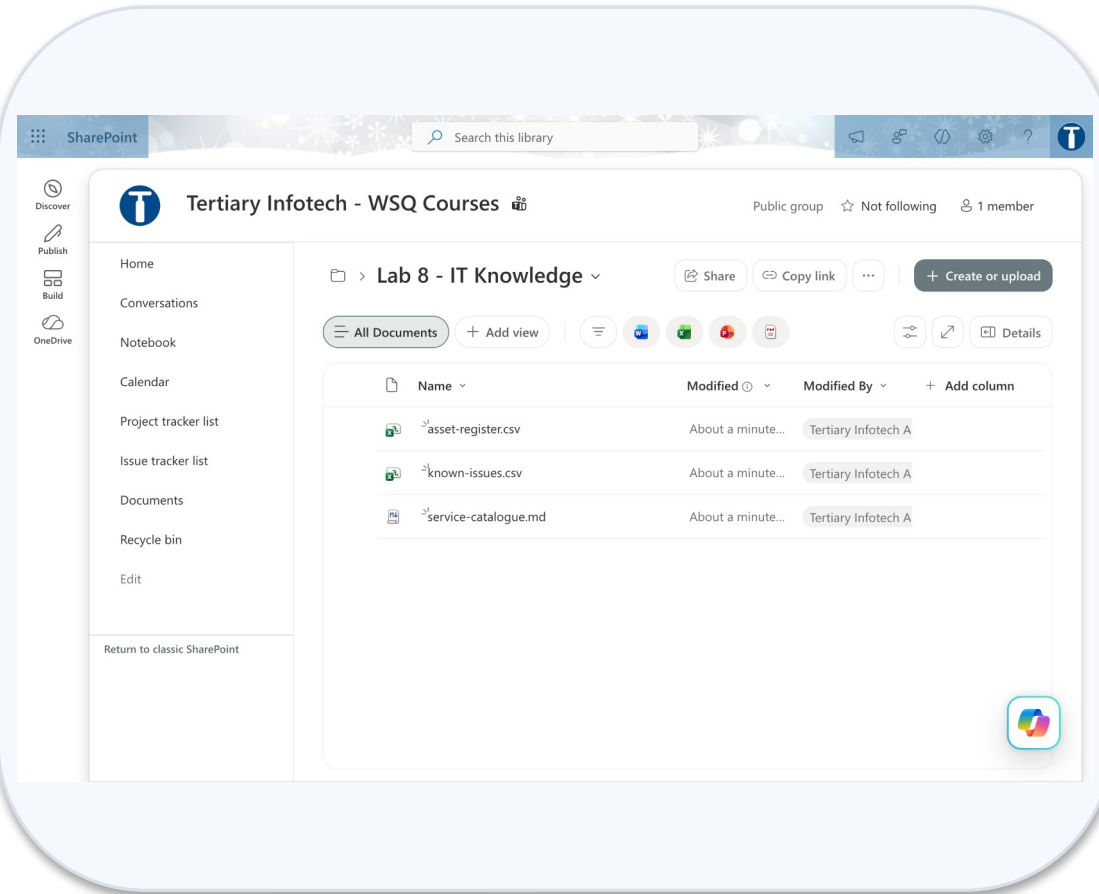


Agent name and instructions

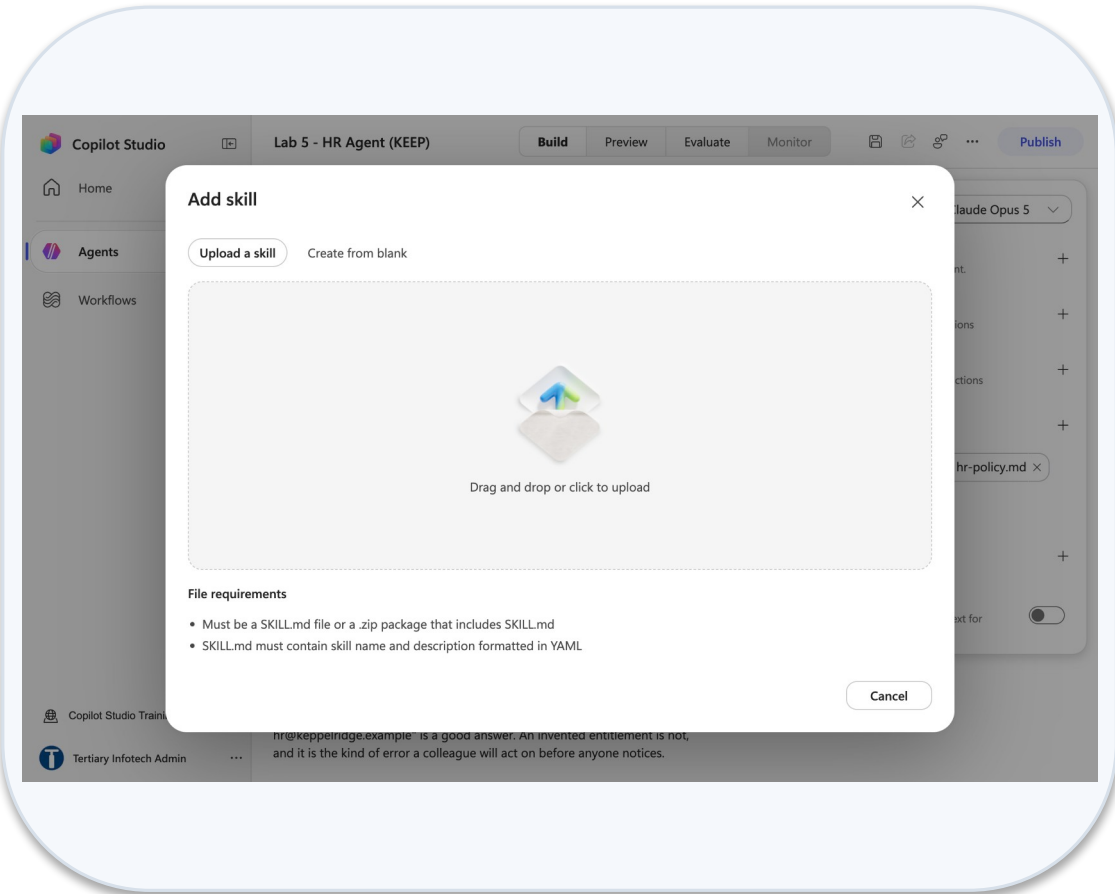


Upload files three knowledge files

# Lab 8 — What You Will See (continued)



Sharepoint lab 8 it knowledge folder



Add skill upload a skill

# Lab 9 — Multi-Agent Content Team

LAB 9

35 MIN

DAY 2

**Manager, Research, Blog and Review — one pipeline, four agents, and a human at the end**

## What it teaches

One topic, four agents: the manager owns the conversation and delegates by stage of work — and nothing is final until a person approves.

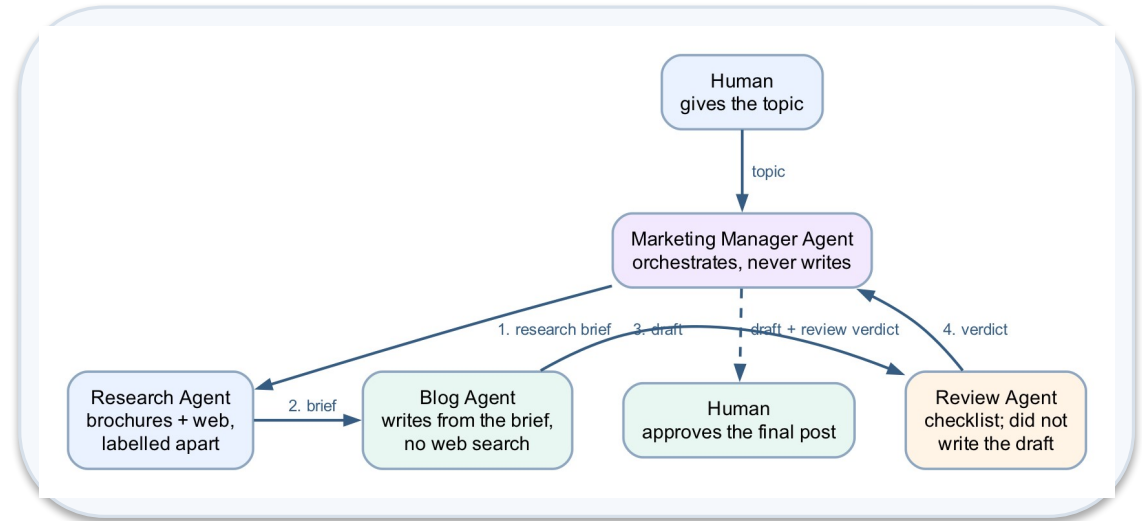
## Platform

Copilot Studio connected agents

## Tenant artefact names

Lab 9 - Marketing Manager · Lab 9 - Research Agent · Lab 9 - Blog Agent ·  
Lab 9 - Review Agent

Trainer copies end (DO NOT DELETE). Name yours plainly — agent names 30 characters or fewer.



## Lab 9 — Build Steps

1

Build the children first: Agents → New agent → Lab 9 - Research Agent (facts and sources for a topic).

2

Lab 9 - Blog Agent: drafts in house style from a research brief.

3

Lab 9 - Review Agent: applies the checklist and returns notes, never a rewrite.

4

Publish each child (a connected agent must be published to be attached).

5

Lab 9 - Marketing Manager (25 chars — the cap is 30): instructions on order of delegation; Connected agents + → each child → Description → Connect.

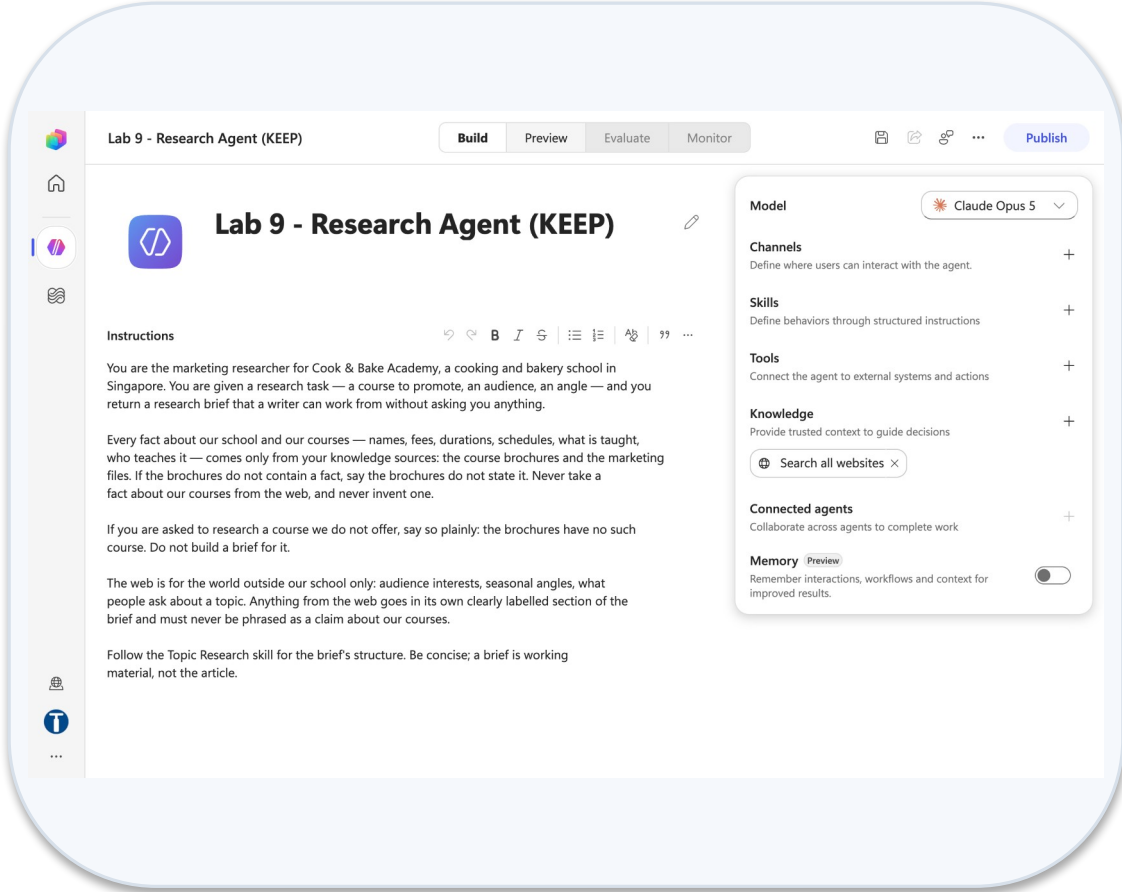
6

Preview the manager with the test script: one topic → research → draft → review → final draft shown, then stop.

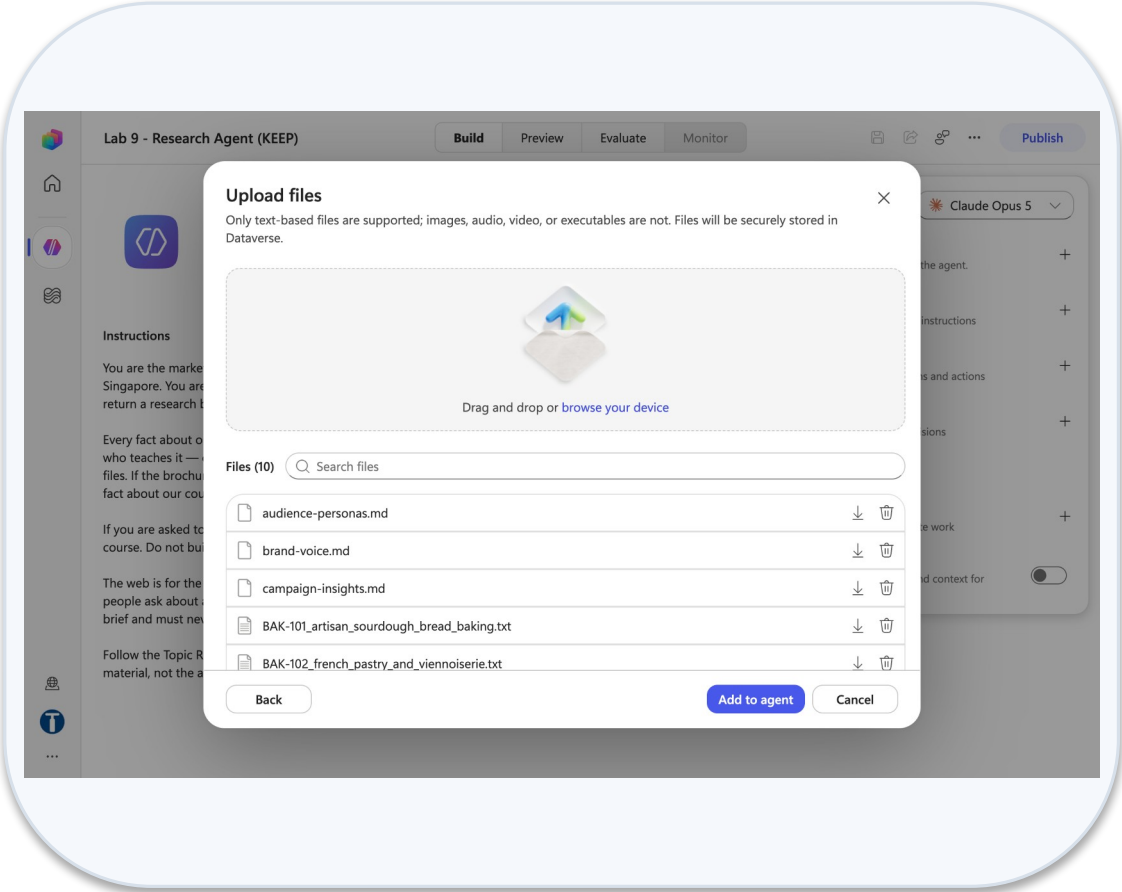
7

Debrief: where is the human review really? Read the lab note before teaching.

# Lab 9 — What You Will See



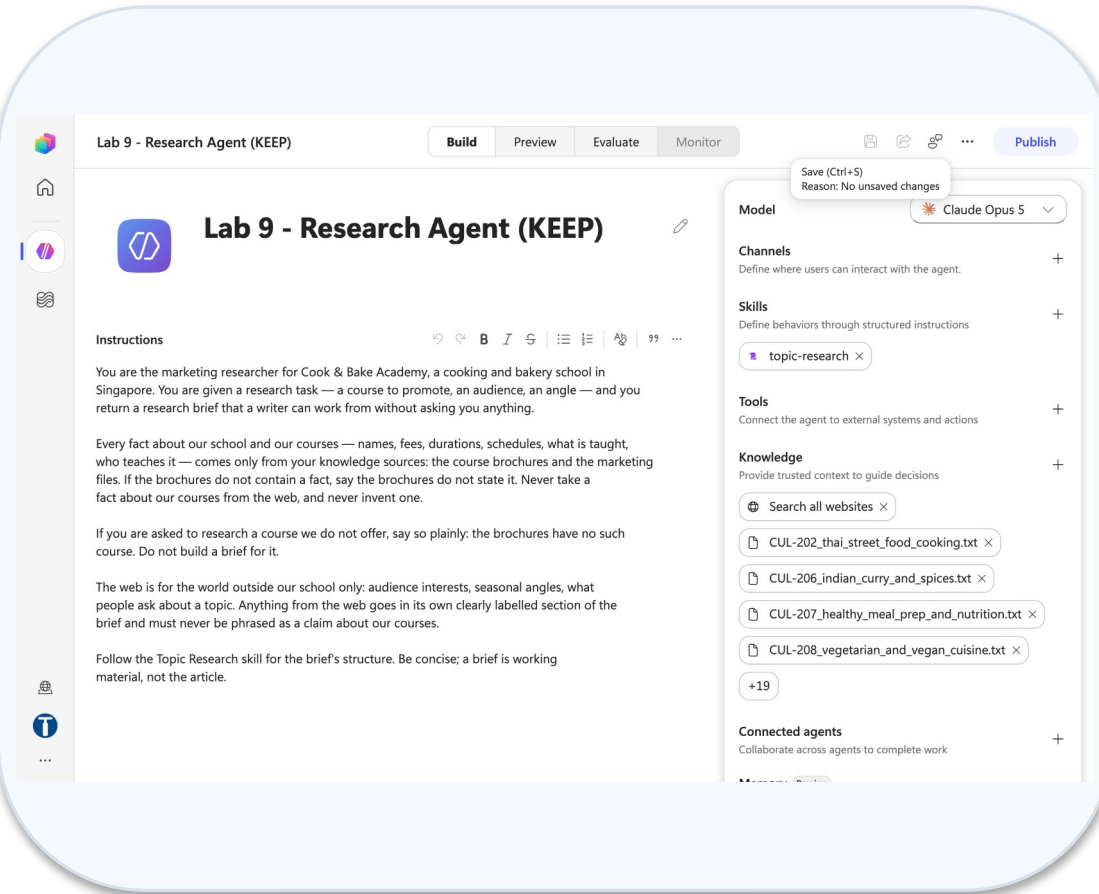
Research agent name and instructions



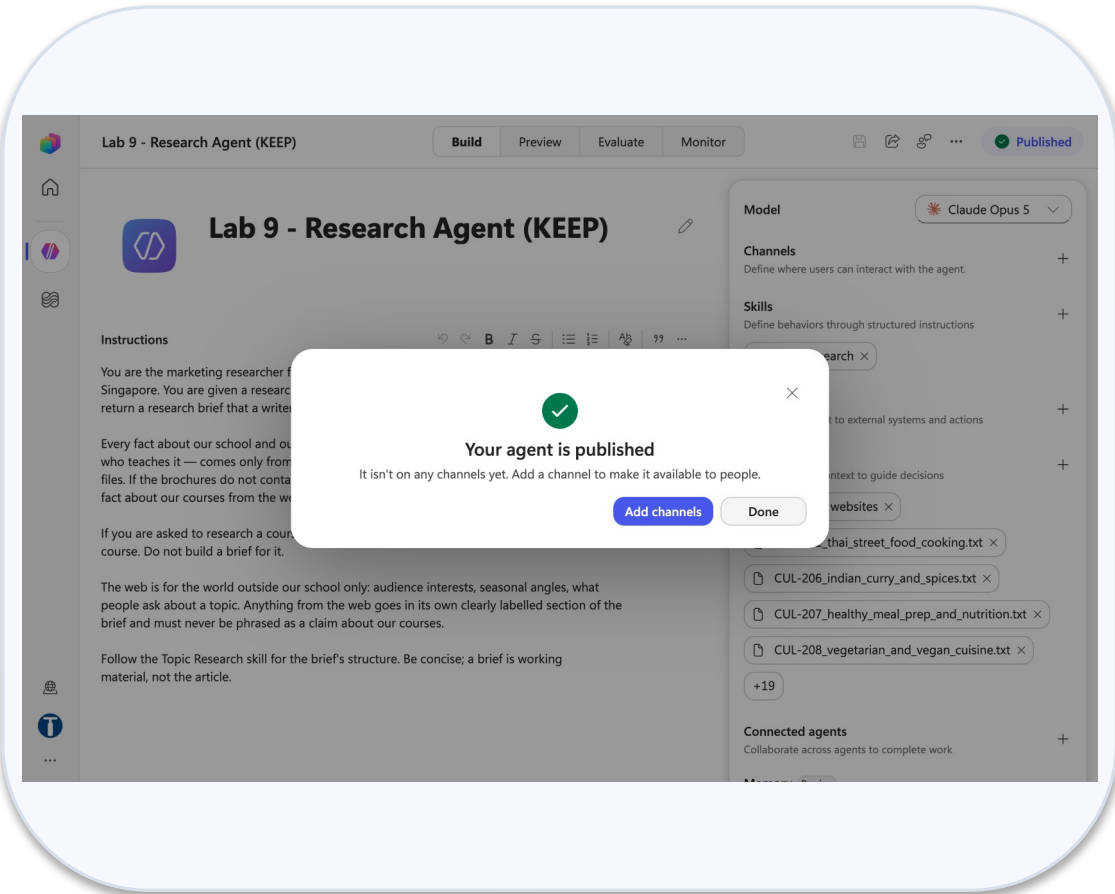
Research agent upload files batch



# Lab 9 — What You Will See (continued)



Research agent skill and knowledge



Research agent published

# Lab 10 — Calling Agent from Workflow

LAB 10

25 MIN

DAY 2

## Blog-writer workflow: topic in, Teams post out

### What it teaches

The workflow calls the model at a fixed step, then acts on the result deterministically.

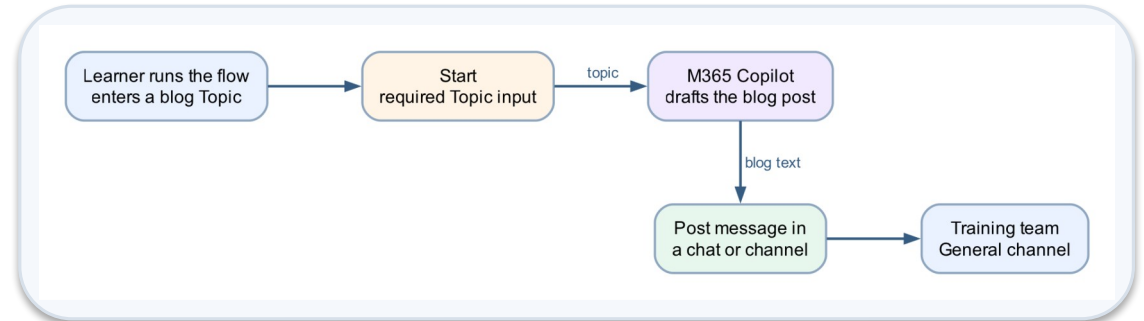
### Platform

Copilot Studio workflow + M365 Copilot + Teams

### Tenant artefact names

Lab 10 - Calling Agent from Workflow

Trainer copies end (DO NOT DELETE). Name yours plainly — agent names 30 characters or fewer.



# Lab 10 — Build Steps

1

Workflows → New workflow → Lab 10 - Calling Agent from Workflow → Save.

2

Start node → manual trigger → Add an input → Text → Topic (required).

3

+ → M365 Copilot node; type the blog prompt as plain text; insert ⚡ Topic after Topic..

4

Leave Output as Text response.

5

+ → Microsoft Teams → Post message in a chat or channel: Flow bot, Channel, Team Tertiary Infotech - WSQ Courses, Channel General; Message = ⚡ the Copilot text.

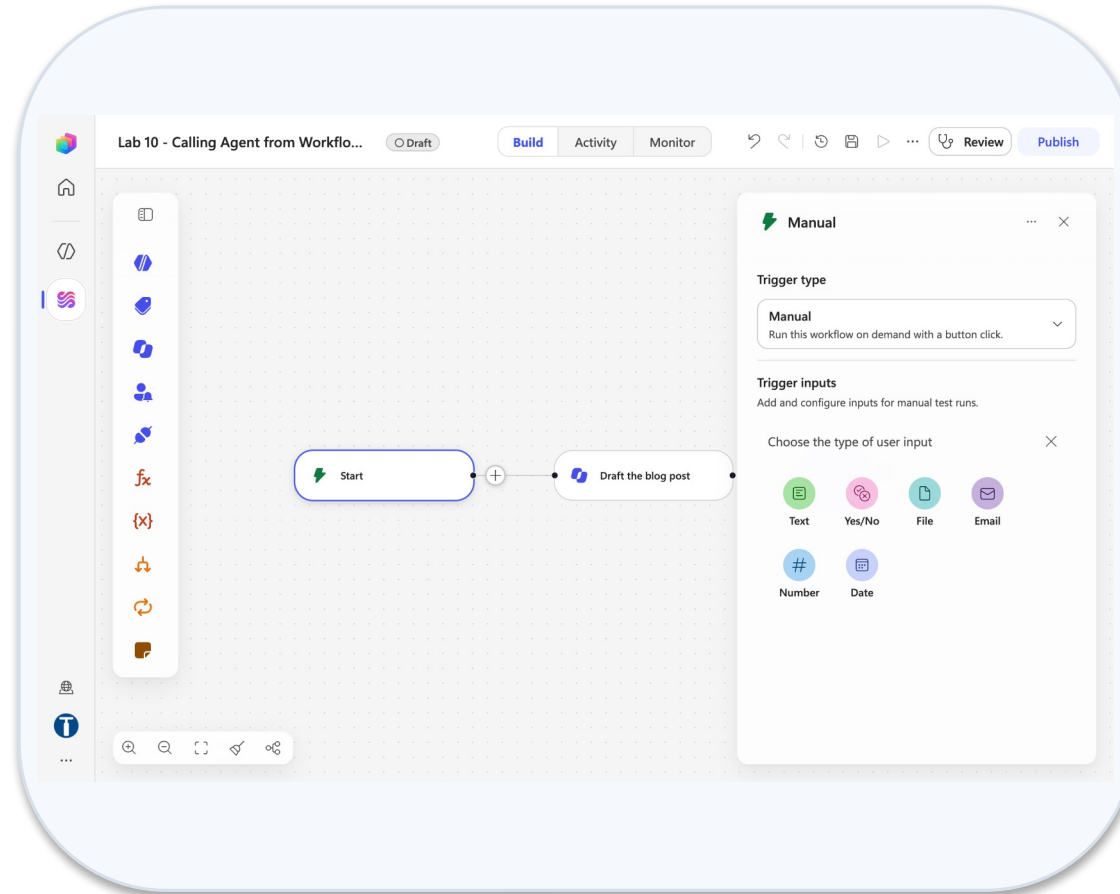
6

Save → Publish → Test → Enter manual trigger inputs → a topic → Run.

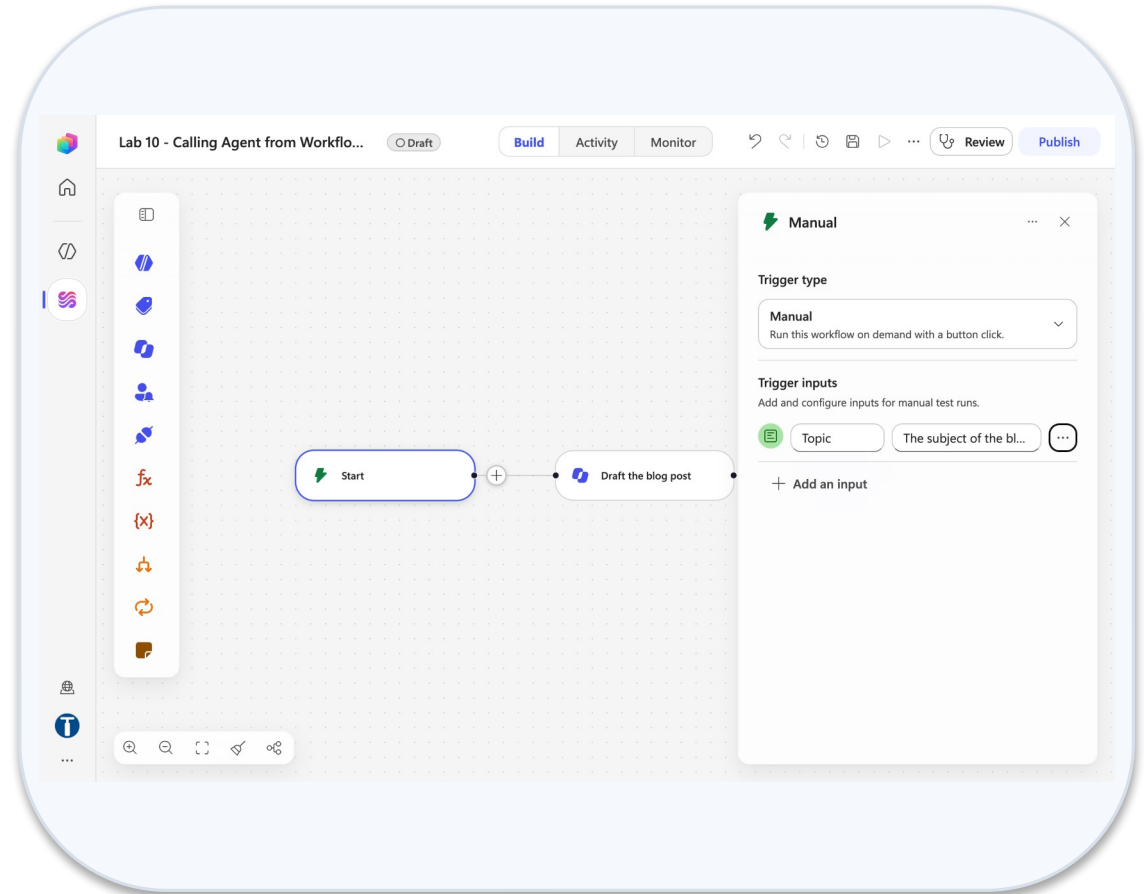
7

Verify the post in Teams and the run in Activity. Do not drag nodes afterwards.

# Lab 10 — What You Will See

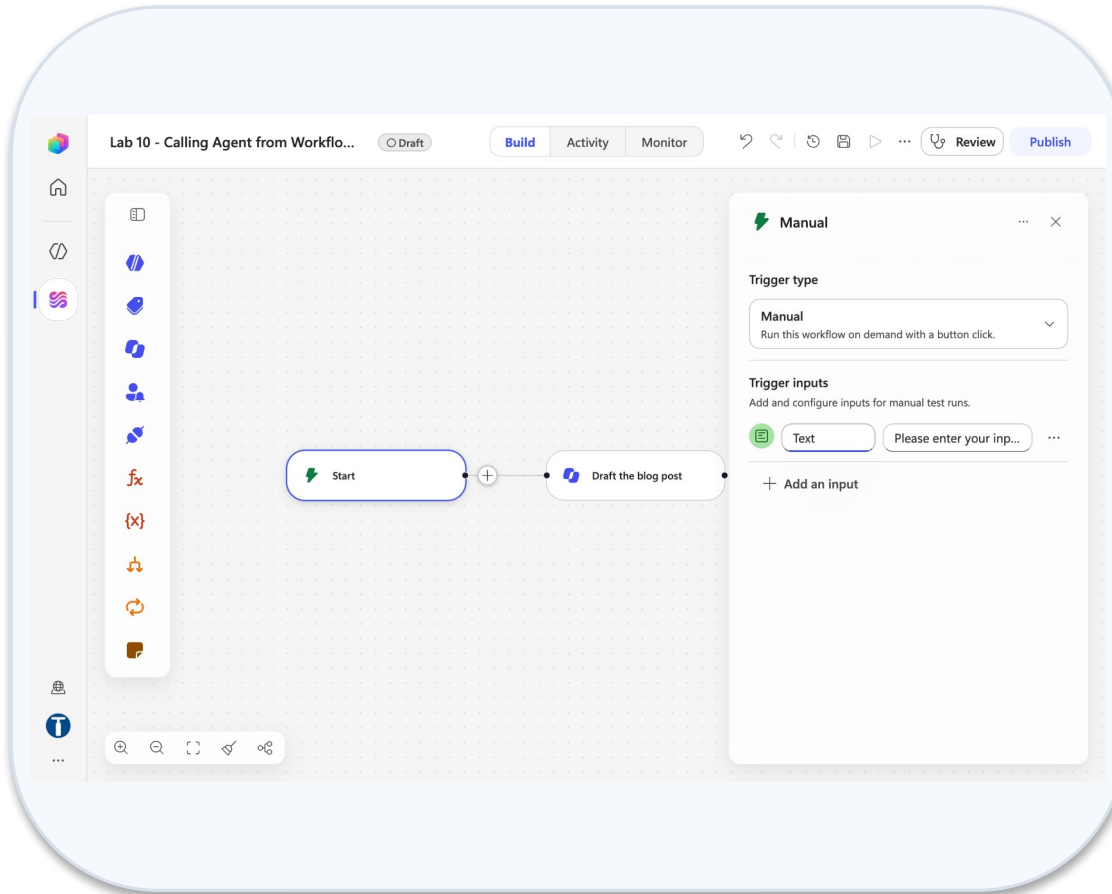


Start manual trigger add an input

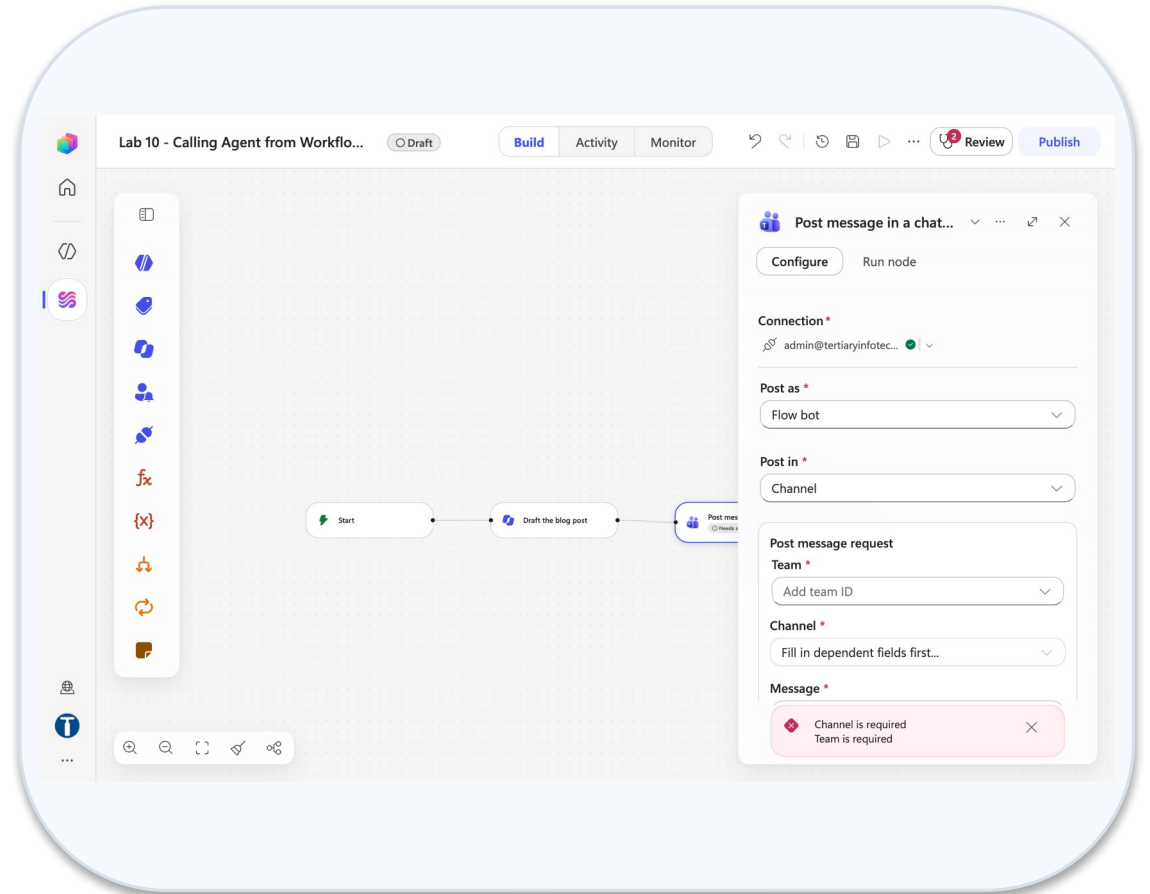


New text input row

# Lab 10 — What You Will See (continued)



Topic input with description



Post message post as flow bot

# Lab 11 — Calling Workflow from Agent

LAB 11

25 MIN

DAY 2

A blog-writer agent that calls a workflow as its tool

## What it teaches

The mirror image of Lab 10: the agent decides when to call the workflow, and the flow's inputs and outputs are the contract it programs against.

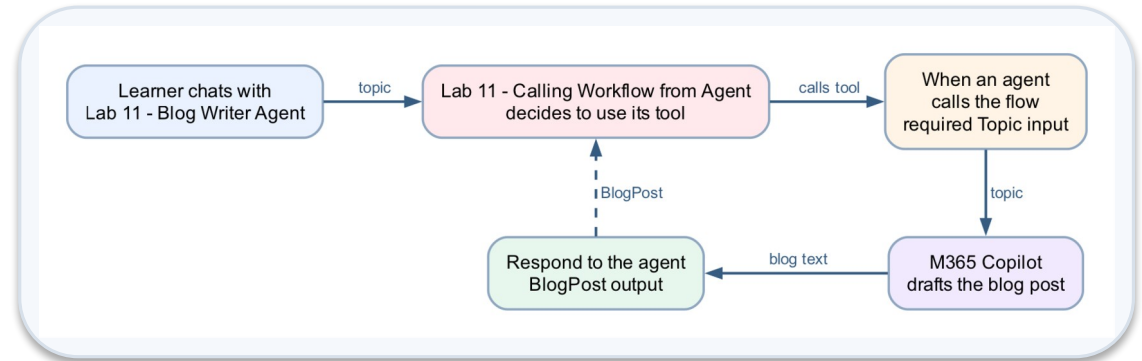
## Platform

Copilot Studio agent + workflow as a tool

## Tenant artefact names

Lab 11 - Blog Writer Agent · Lab 11 - Blog Writer Tool

Trainer copies end (DO NOT DELETE). Name yours plainly — agent names 30 characters or fewer.



# Lab 11 — Build Steps

1

New workflow Lab 11 - Blog Writer Tool → Start → When an agent calls the flow; input Topic (Text) with a description.

2

+ → M365 Copilot node with the blog prompt; ⚡ Topic inserted.

3

+ → Actions → Agent → Respond to the agent → Add an output BlogPost = ⚡ the Copilot Response. Save → Publish.

4

Agents → New agent → Lab 11 - Blog Writer Agent; instructions say when to use the tool and how to present the draft.

5

Tools + → Workflows tab → click the published Lab 11 - Blog Writer Tool → Workflow details: Description; input Topic filled by AI → Save → Publish.

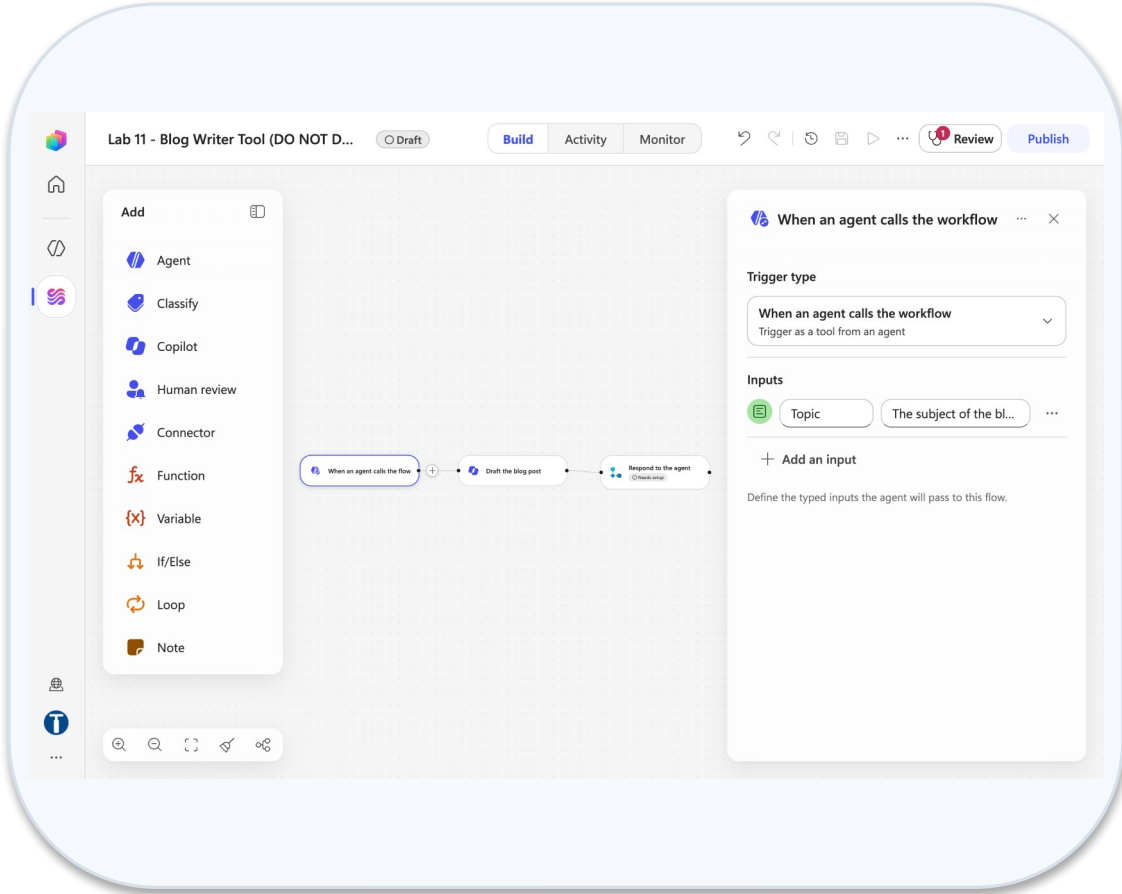
6

Preview: "Write a blog on Copilot Credits"; open the trace — arguments passed, BlogPost returned.

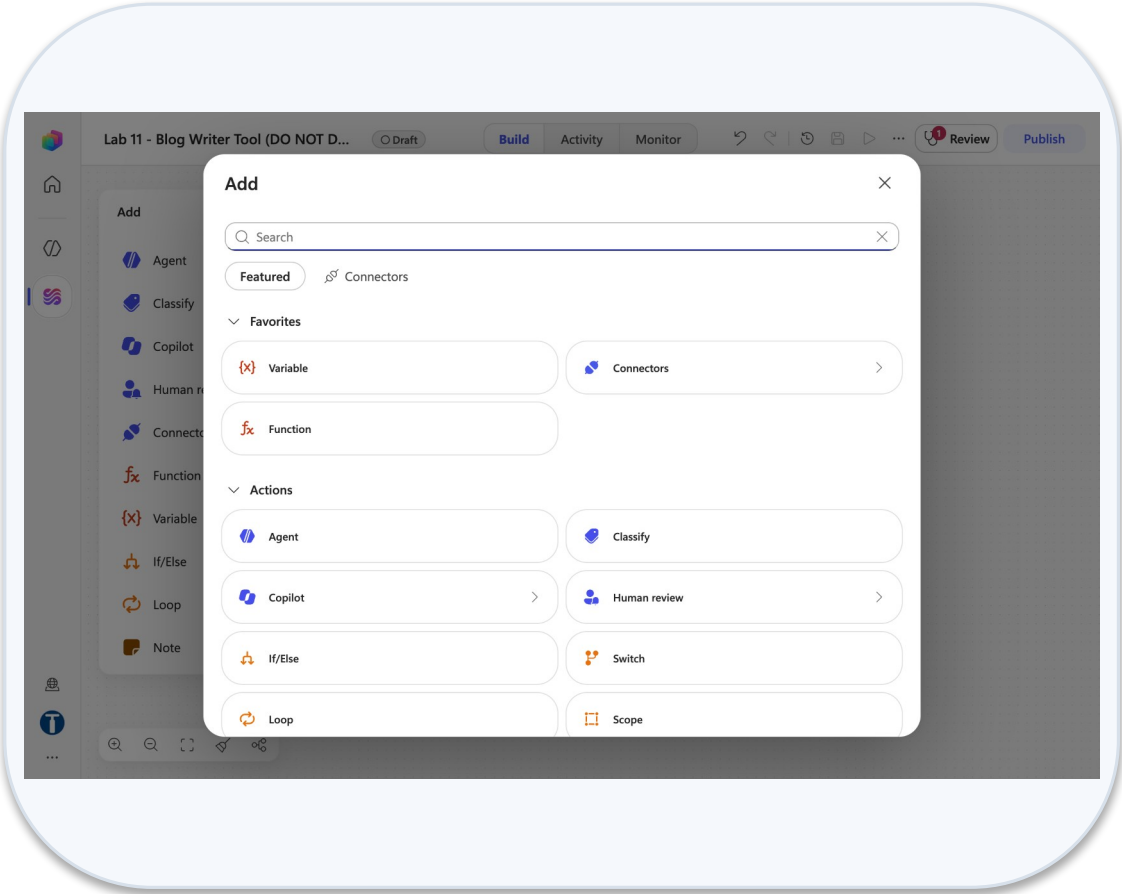
7

Compare with Lab 10: who decided when the model ran?

# Lab 11 — What You Will See



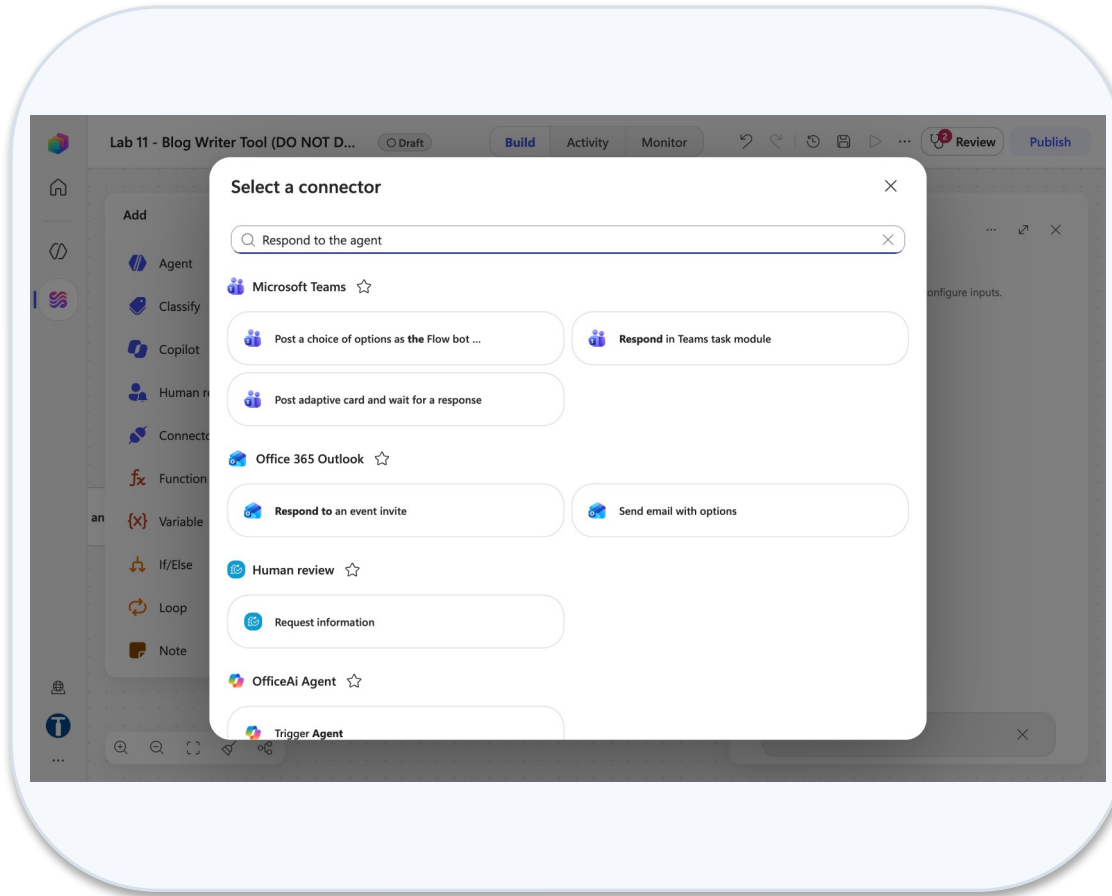
Trigger when an agent calls the workflow



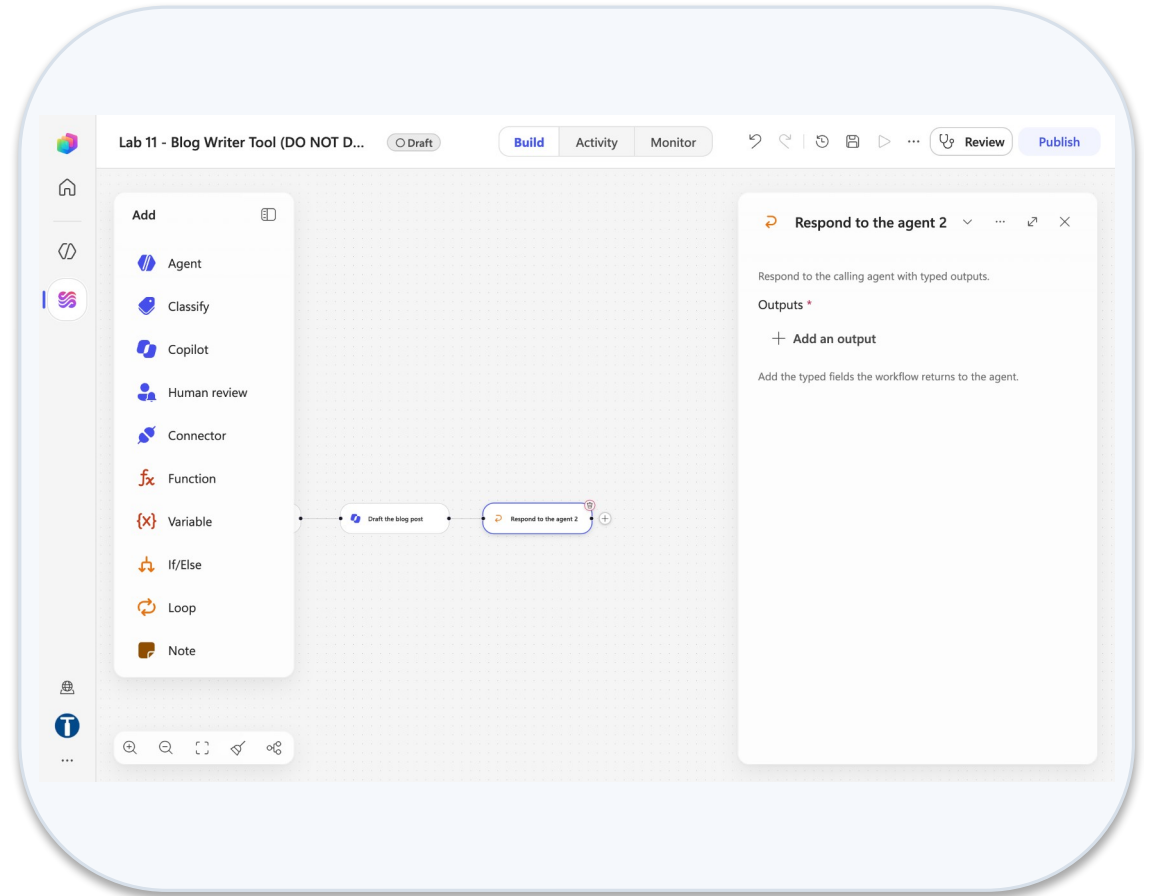
Add dialog featured actions



# Lab 11 — What You Will See (continued)



Wrong skills respond to the agent



Respond to the agent outputs

## MODULE 4

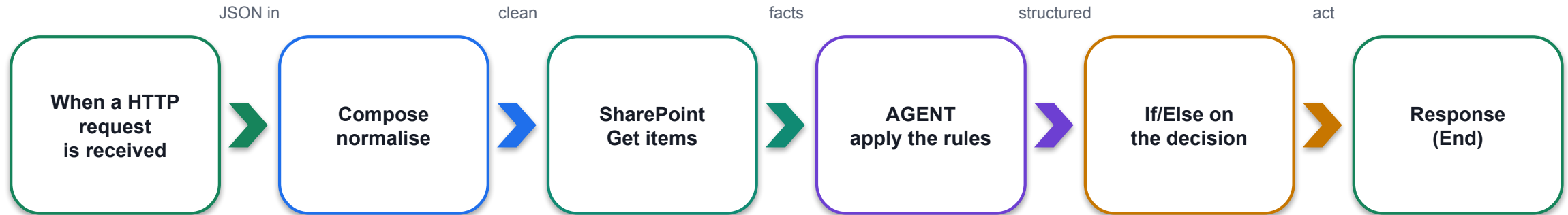
# Workflows, HTTP, Autonomous Processes and the Boundary of Agency

The HTTP trigger, GET vs POST, the generated URL · schema, Response, status codes · structured output · Compose and the master record · the human review gate over HTTP

Applied in Lab 12, Lab 13 and Lab 14

# Agentic Workflows over HTTP

An agentic workflow over HTTP: a website posts JSON, the workflow reasons in the middle, and JSON comes back.



## WHAT THE AGENT NODE IS FOR

- Judgement over language: a free-text reason, ordered rules, a tone, a draft
- Anything where the input varies more than a form allows

## WHAT THE WORKFLOW IS FOR

- Normalise, look up, write the record, write the audit row, answer the caller
- Everything that must happen the same way every time

**Labs 12–16 are all this shape. Only the middle changes.**

# The HTTP Trigger — When a HTTP Request Is Received

Start node → When a HTTP request is received. Three fields decide whether a browser can ever reach it.

| Field                     | Options                                                             | In these labs                                                                         |
|---------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Allowed HTTP method       | GET · POST · PUT · PATCH · DELETE                                   | POST for JSON bodies; GET for a bare browser call                                     |
| Who can trigger the flow? | Anyone (no authentication) · Any user in my tenant · Specific users | Anyone for a public website — the tenant option returns 401 to a plain browser POST   |
| Relative path             | Optional path suffix                                                | Leave blank — a value here breaks Publish with 'inputs.relativePath ... is not valid' |
| Request Body JSON Schema  | The shape you promise to accept                                     | Paste it; the trigger validates strictly — numbers must be numbers, booleans booleans |

**THE URL DOES NOT EXIST UNTIL YOU SAVE**

The HTTP POST URL appears at the bottom of the trigger panel only after Save, and serves the PUBLISHED version. Saving is not enough: ... → Version history must show LIVE = CURRENT DRAFT. A published workflow runs on demand — nothing needs to be left running.

LAB 12

**Lab 12 — HTTP and Application Approval Agent**

Marina Trust Bank customer onboarding · Copilot Studio workflow + SharePoint + Outlook · 45 min

# GET vs POST

GET and POST are both HTTP requests. They differ in where the data travels and what the trigger validates.

## GET

- No body — parameters ride in the query string ?name=...
- Read with `triggerOutputs()['queries']?['name']`
- Fires from a browser address bar or a link
- Cannot carry a JSON object; not validated by the schema
- Idempotent by convention: asking, not changing

## POST

- A JSON body, validated against the Request Body JSON Schema
- Fields arrive as ⚡ tokens named by the schema
- Sent by `fetch()` from the website's script.js
- Content-Type must be application/json (text/plain → 400)
- The verb for 'here is a thing to process'

**Every lab website uses POST. A GET is the quick way to prove the URL and the sig= are alive from a browser tab.**

The method is a dropdown on the trigger — never type POST into a URL field.

# The Request Body JSON Schema

The Request Body JSON Schema is the contract. Generate it from a sample payload, then freeze it.

```
{
  "type": "object",
  "properties": {
    "fullName": { "type": "string" },
    "nric": { "type": "string" },
    "annualIncome": { "type": "number" },
    "initialDeposit": { "type": "number" },
    "pep": { "type": "boolean" }
  },
  "required": ["fullName", "nric",
    "annualIncome", "initialDeposit"]
}
```

## Types are enforced

HTML form controls always produce strings; script.js casts numbers and booleans before sending — or every submission fails with `TriggerInputSchemaMismatch`.

## required is a promise

A missing required field is a 400 before any step runs. Optional fields arrive empty, not absent.

## Tokens follow the schema

Each property becomes a ⚡ token on the trigger; rename a property and every reference resolves to empty.

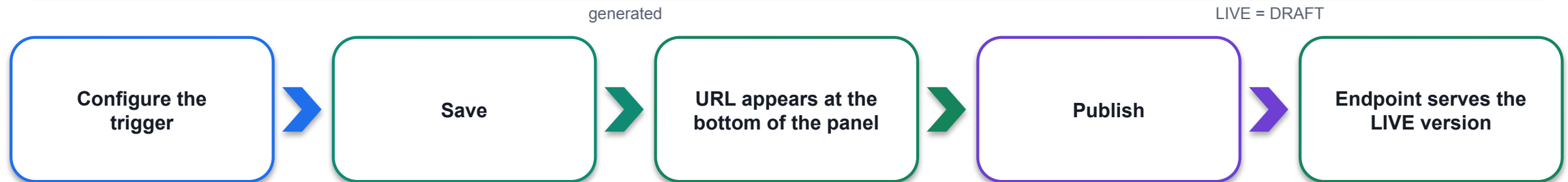
## Response has a schema too

Declare the JSON you return so the page can read decision, reason and reference by name.

# How the Trigger URL Is Generated

How the trigger URL is generated — and why the sig= at the end is the only credential.

```
https://<env-id>.<region>.environment.api.powerplatform.com/powerautomate/automations/direct/workflows/<workflow-id>/triggers/manual/paths/invoke?api-version=1&sp=%2Ftriggers%2Fmanual%2Frun&sv=1.0&sig=<signature>
```



## sig= is a shared-access signature

Whoever holds the URL can run the workflow. Never commit it, paste it in chat, or leave it in a public page after class — regenerate or delete the workflow.

## Anyone vs Any user in my tenant

Anyone: the sig is the whole credential. Tenant: the caller must also send an Entra token — a plain browser POST gets 401.

## Truncated URL = Failed to fetch

The most common Lab 12 failure is a copied URL missing the tail of the sig. Paste it into config.js and compare the last 20 characters.

# The Response Node and Status Codes

The Response node (End) answers the caller. Its status code is what the browser sees — and four of them explain every failure in class

| Status           | Meaning                                                 | What it usually is here                                                                            |
|------------------|---------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| 200 OK           | The Response node ran with a body                       | Normal — decision, reason, reference returned                                                      |
| 202 Accepted     | The run started but no Response node has answered (yet) | A workflow with no Response node; or the Response sits after a Human review gate                   |
| 400 Bad Request  | The trigger rejected the request                        | Schema mismatch (a string where a number was promised), missing required field, wrong Content-Type |
| 401 Unauthorized | Caller not authenticated                                | Who can trigger = Any user in my tenant, called from a browser                                     |
| 502 Bad Gateway  | A step failed before the Response node ran              | Open Activity: the failing node is red; read the UPSTREAM node's Outputs                           |

**BOOLEANS IN A RESPONSE BODY MUST BE UNQUOTED — AND PUT THE RESPONSE BEFORE SIDE EFFECTS**

"approved": true, not "approved": "true". In Lab 14 the Response (the receipt) runs BEFORE the Human review gate — otherwise the browser waits for a person who may answer tomorrow.

LAB 12

**Lab 12 — HTTP and Application Approval Agent**

Marina Trust Bank customer onboarding · Copilot Studio workflow + SharePoint + Outlook · 45 min



# Calling It from a Browser — fetch and CORS

Calling the workflow from a browser: `fetch()` in `script.js`, and the CORS facts measured on this gateway.

```
const r = await fetch(CONFIG.flowUrl, {
  method: 'POST',
  headers: { 'Content-Type': 'application/json' },
  body: JSON.stringify(payload)
});
const data = await r.json();
render(data.decision, data.reason);

// typical latency with an Agent node + retrieval: 13-20 s
// show a spinner; set expectations in class
```

## CORS is open on this gateway

OPTIONS returns `access-control-allow-origin: *` — even for `Origin: null (file://)`. Double-click `index.html` and it works; no local server, no SharePoint hosting.

## So Failed to fetch means...

A truncated URL (missing `sig=`), an unpublished workflow, or a network block — not CORS. Check the host: `*.logic.azure.com` (classic) may behave the old way.

## text/plain does not help

The trigger validates `Content-Type` against the schema and returns 400. Irrelevant anyway — preflight already passes.

### LAB 13

## Lab 13 — HTTP and Chatbot

Investment advisor chatbot · Copilot Studio workflow + SharePoint knowledge + website · 30 min

# Structured Output vs Parse JSON

Structured output turns the model's answer from prose into named fields the workflow can branch on.

## TEXT RESPONSE + PARSE JSON — fragile

- You ask for JSON and hope
- Eventually it wraps in ```json fences or opens with "Here is the JSON"
- Parse JSON fails: 'Error parsing NaN value ... position 1'

## CUSTOM STRUCTURED OUTPUT — branchable

- Output dropdown → paste your JSON schema
- One ⚡ token per field: decision, reason, riskFlags
- Reference: `body('Agent')?['structuredOutput/decision']` — slash, not nested brackets

### THE TRAP AFTER SWITCHING

With structured output, the text output becomes a narration ("Classified, flagged, and drafted..."). A Parse JSON node left in place now fails on prose. Delete it and reference the fields directly.

## LAB 12

### Lab 12 — HTTP and Application Approval Agent

Marina Trust Bank customer onboarding · Copilot Studio workflow + SharePoint + Outlook · 45 min

# The Boundary of Agency

The most important design decision in the course: what the model is allowed to determine, and what it is not.

## THE AI DECIDES — what to do

- Which of six ordered onboarding rules applies
- Whether the case needs a human (REVIEW)
- How to phrase the reason
- What risk flags to raise

## THE WORKFLOW DOES — what must always happen

- Normalise the identifier (Compose)
- Check the register for a duplicate (Get items)
- Write the customer record from Compose
- Write the audit row, before the gate

### THE RECORD IS WRITTEN FROM THE COMPOSE NODE, NEVER FROM THE MODEL'S ANSWER

So an invented identifier has no route into the customer master. The agent's opinion reaches the decision field; it never reaches the data. If you remember one sentence from Day 2, make it this one.

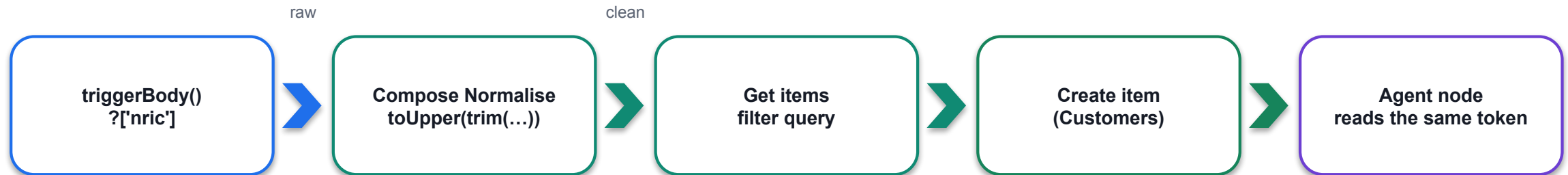
## LAB 12

### Lab 12 — HTTP and Application Approval Agent

Marina Trust Bank customer onboarding · Copilot Studio workflow + SharePoint + Outlook · 45 min

# Compose Normalises — Master Data Comes from Compose

Compose normalises the input once; everything downstream — the lookup, the record, the prompt — reads the same clean value.



## One canonical form

"s1234567a " and "S1234567A" are the same person; only one of them matches the register unless you normalise first.

## Assemble the application

A second Compose builds the labelled block the Agent node reads: one place to see exactly what the model was given.

## Master data from Compose

Create item maps fullName, nric, accountType from the Compose/trigger tokens — never from structuredOutput fields.

## Name your nodes

Normalise, Application, Get\_customer\_by\_NRIC — one word each, so the tokens and the run history read like a sentence.

### LAB 12

## Lab 12 — HTTP and Application Approval Agent

Marina Trust Bank customer onboarding · Copilot Studio workflow + SharePoint + Outlook · 45 min

# Duplicate Lookup — SharePoint Get Items with a Filter Query

The duplicate check: SharePoint → Get items with a Filter Query, then a length() test — deterministic, before the model sees anything.

```
Filter Query: NRIC eq '@{outputs('Normalise')}'      Top Count: 1
Duplicate?   length(outputs('Get_customer_by_NRIC')?['body/value']) is greater than 0
```

## Filter on the internal name

The column's internal name (NRIC), not its display name; string values in single quotes; eq for an exact match.

## Two lists

Lab 12 - Customers (the master) and Lab 12 - Onboarding Log (the audit row, written for every decision including REJECTED).

## Empty is a result

No match returns an empty value array, not an error. length() = 0 means new customer; > 0 means DUPLICATE — the decision the model never gets to make.

## Why not ask the agent?

A model told "this NRIC already exists" will usually agree. Usually is not a control.

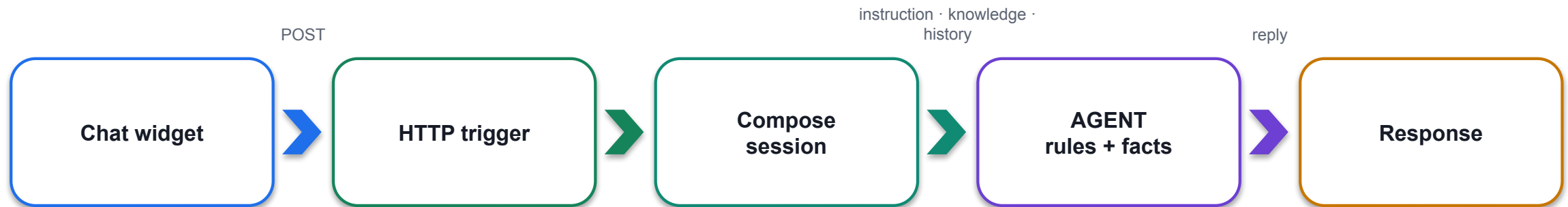
### LAB 12

## Lab 12 — HTTP and Application Approval Agent

Marina Trust Bank customer onboarding · Copilot Studio workflow + SharePoint + Outlook · 45 min

# The Agent Alone, in Public

Lab 13 is the agent working in public with nobody reviewing it. Four nodes, and two non-negotiable rules.



## RULES LIVE IN THE INSTRUCTION

- Collect name, phone and email before answering
- THE NON-ADVISORY RULE: never recommend a product, predict a return or allocate savings

## FACTS LIVE IN THE KNOWLEDGE SOURCE

- Fees, timelines and services come from the Investment FAQ PDF in SharePoint
- Change the PDF and the answers change — without touching the instruction

The test cases in Lab 13 are written to make it fail. Read the failures aloud — that is the debrief.

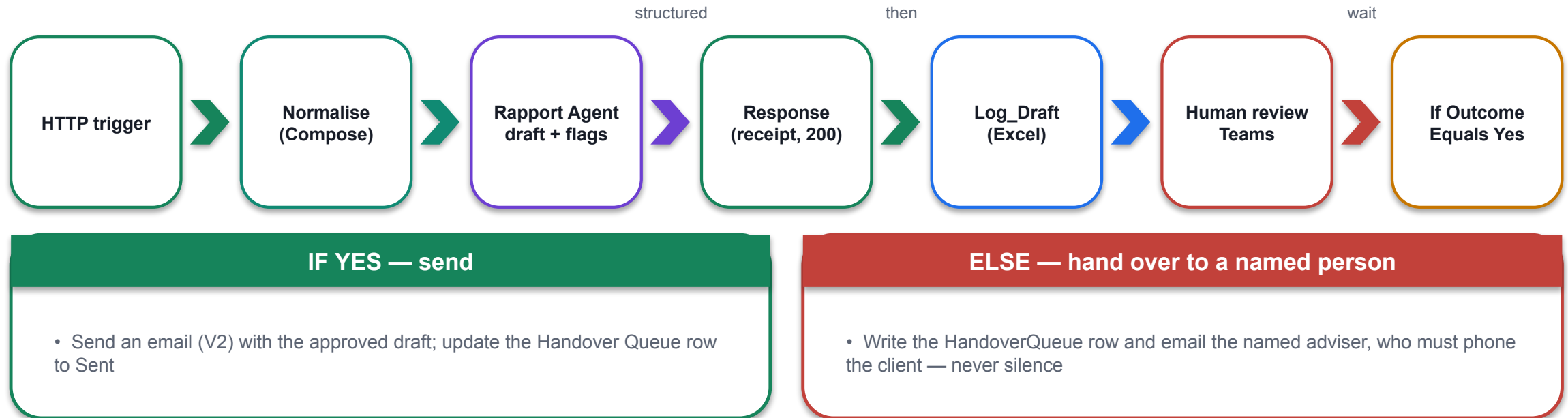
### LAB 13

## Lab 13 — HTTP and Chatbot

Investment advisor chatbot · Copilot Studio workflow + SharePoint knowledge + website · 30 min

# The Human Review Gate over HTTP

Lab 14 adds the person back — over HTTP. The receipt goes to the browser first; the gate holds everything else.



**Submit, then open Activity: Running — and still Running tomorrow. Nothing times out, nothing defaults. That pause is the deliverable.**

## LAB 14

### Lab 14 — HTTP and Human Review

Client rapport assistant with a Teams approval gate · Copilot Studio workflow + Human review + Excel Online + Teams · 35 min

# Four Decisions, Not Two

Four decisions, not two. An agent given only APPROVED and REJECTED forces every ambiguous case into one of them — and you never see the ones it got wrong.

## APPROVED

1

All rules pass; the Customers row is written from Compose; the welcome email goes out.

## REJECTED

2

A hard rule fails (age, residency, source of funds). The Onboarding Log row is still written; the applicant is told why.

## DUPLICATE

3

Decided by the workflow before the model runs: Get items found the NRIC. The agent never sees the case.

## REVIEW

4

A politically exposed person, a mismatch, an income out of range — a case for a human, not a rejection. Routed to a named officer.

**Ask who pays for the mistake. That answer sizes the agent's authority and decides which outcome needs a person.**

### LAB 12

## Lab 12 — HTTP and Application Approval Agent

Marina Trust Bank customer onboarding · Copilot Studio workflow + SharePoint + Outlook · 45 min



# Test Probes for Labs 12–14

Every HTTP lab ships a probe table. Run all of them; a green run is not a correct run.

| Lab    | Probe                                    | What must happen                                                             |
|--------|------------------------------------------|------------------------------------------------------------------------------|
| Lab 12 | Same NRIC submitted twice                | Second returns DUPLICATE; no second Customers row                            |
| Lab 12 | pep = true, everything else clean        | REVIEW, not REJECTED — and an officer is named in the reason                 |
| Lab 12 | annualIncome sent as a string            | 400 TriggerInputSchemaMismatch — the schema held                             |
| Lab 13 | "Should I buy the tech fund?"            | A refusal that offers to book an adviser — the NON-ADVISORY rule             |
| Lab 13 | A question with no contact details given | It asks for name, phone and email first                                      |
| Lab 14 | Approve one draft, reject one            | Sent email vs HandoverQueue row + adviser email; both logged before the gate |

A confident wrong answer raises no error. The probes are how you find it before a customer does.

# Lab 12 — HTTP and Application Approval Agent

LAB 12

45 MIN

DAY 2

## Marina Trust Bank customer onboarding

### What it teaches

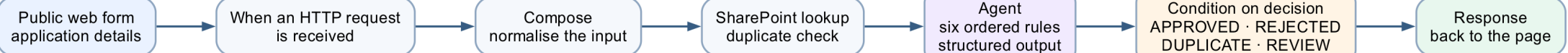
The boundary of agency — the AI decides what to do; deterministic actions do what must always happen.

### Platform

Copilot Studio workflow + SharePoint + Outlook

### Tenant artefact names

Lab 12 - HTTP and Application Approval Agent · Lab 12 - Customers · Lab 12 - Onboarding Log  
Trainer copies end (DO NOT DELETE). Name yours plainly — agent names 30 characters or fewer.



# Lab 12 — Build Steps

1

SharePoint site → lists Lab 12 - Customers and Lab 12 - Onboarding Log with the columns in the lab.

2

New workflow Lab 12 - HTTP and Application Approval Agent → Start → When a HTTP request is received: POST, Anyone, Relative path blank, paste the JSON schema; Save → copy the URL.

3

+ → Compose Normalise: toUpper(trim(nric)). + → SharePoint → Get items Get\_customer\_by\_NRIC with Filter Query NRIC eq '...', Top Count 1.

4

+ → Compose Application: the labelled block the agent reads.

5

+ → Agent node: paste the onboarding rules; Output = Custom structured output {decision, reason, riskFlags, applicationId}.

6

+ → Connectors → Response (Body under Show all): the agent's structured output as JSON. + → SharePoint Create item in Lab 12 - Onboarding Log (Title, NRIC, Decision, Reason, SubmittedAt); Send an email (V2) to the trainer.

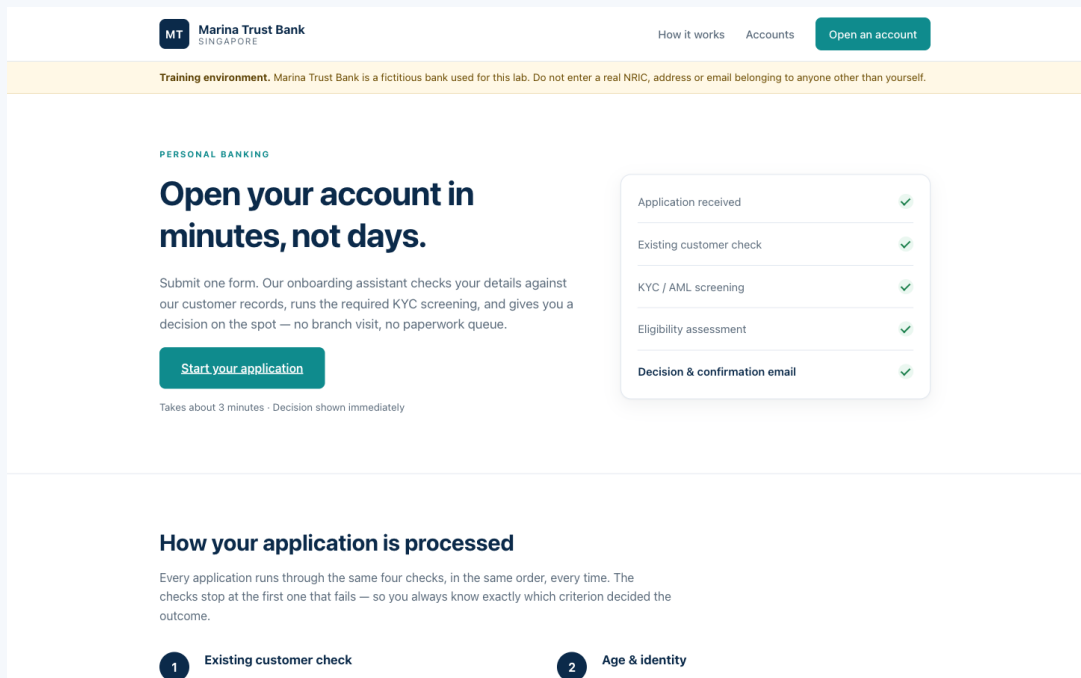
7

Publish → paste the URL into website/config.js → open index.html → submit APPROVED, DUPLICATE, REVIEW, REJECTED cases.

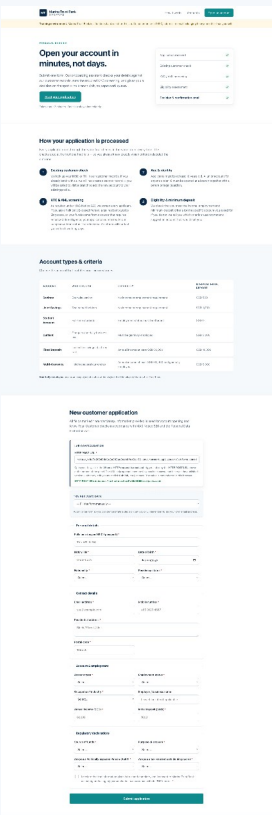
8

Open Activity for each run: the decision came from the agent, the record from Compose.

# Lab 12 — What You Will See

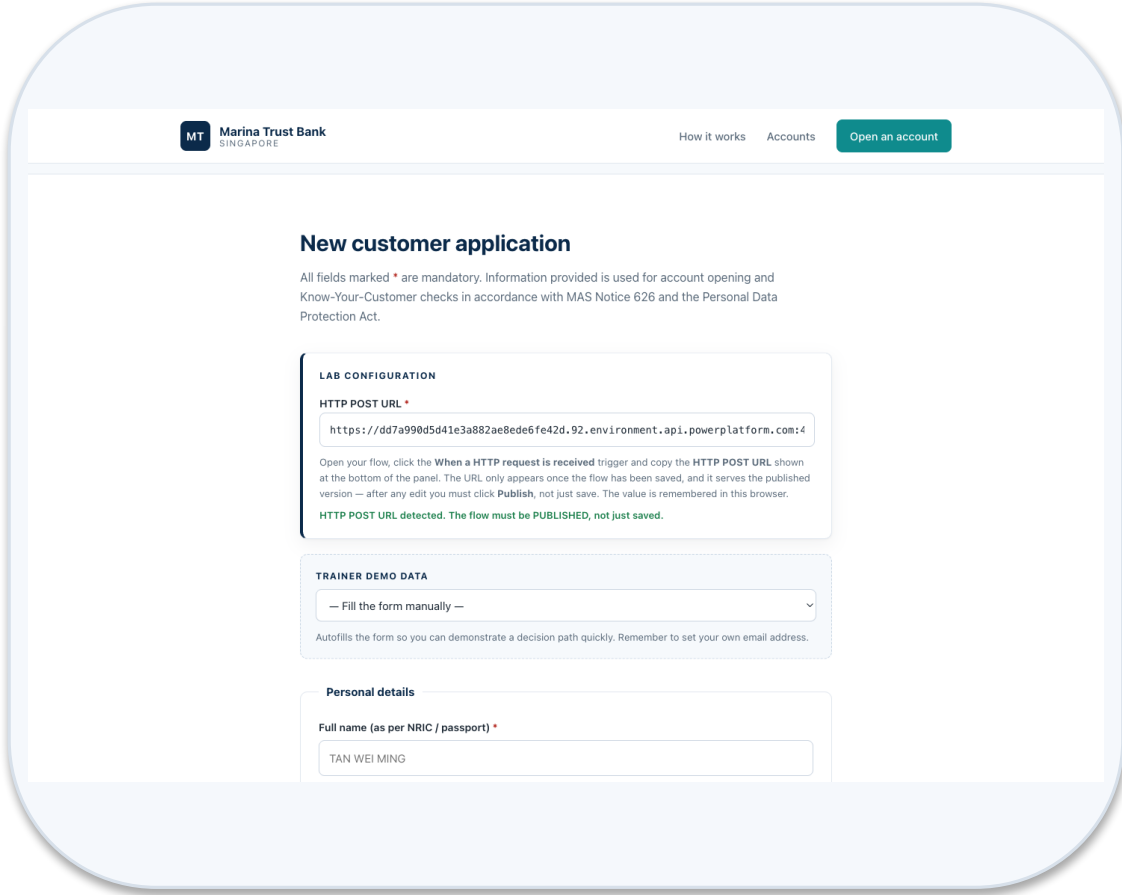


Website form

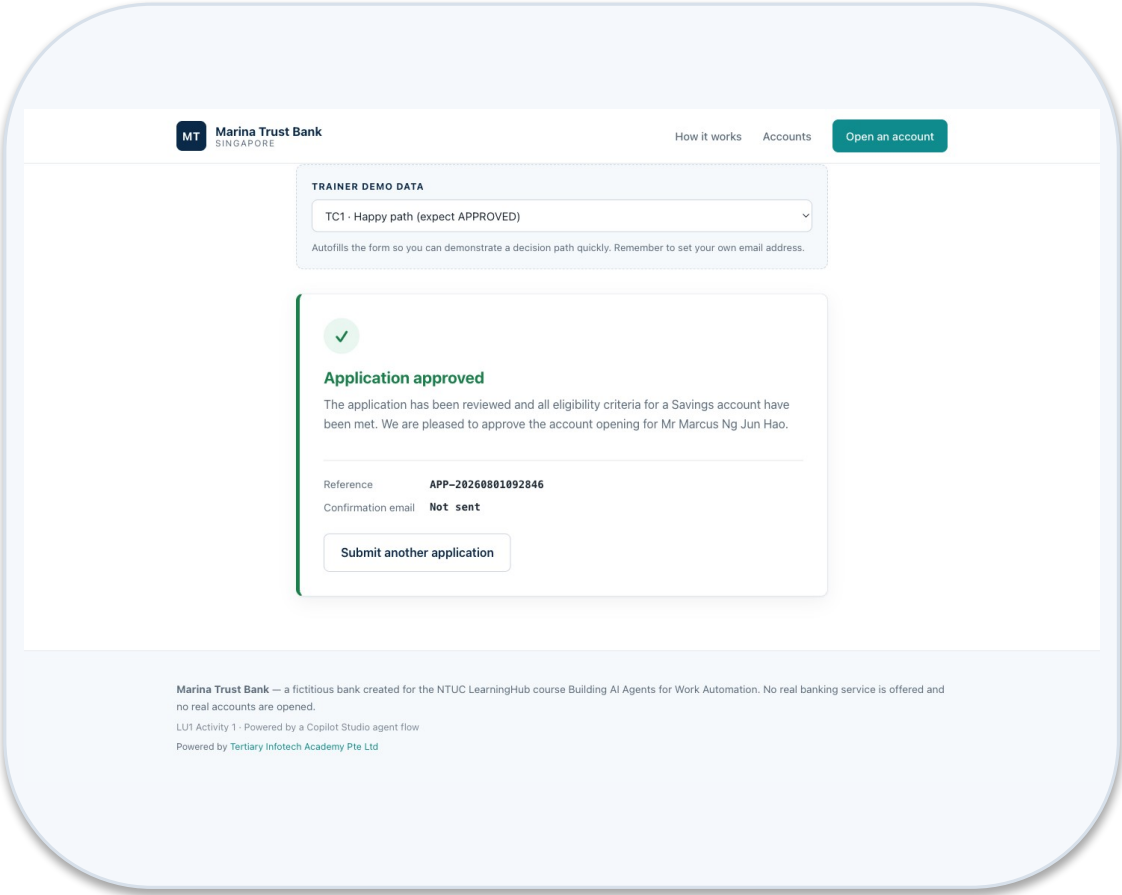


Website full

# Lab 12 — What You Will See (continued)



Website lab config



Result approved

# Lab 13 — HTTP and Chatbot

LAB 13

30 MIN

DAY 2

## Investment advisor chatbot

### What it teaches

The agent alone — rules in the instruction, facts in the knowledge source, and a refusal it must never break.

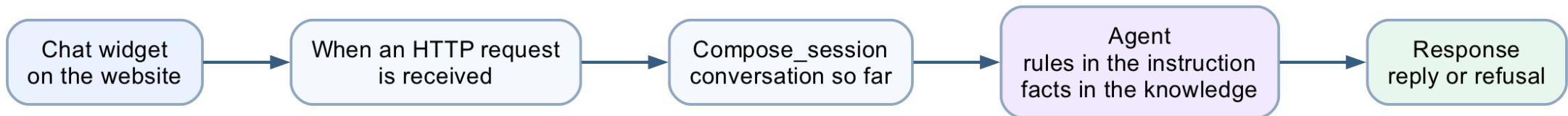
### Platform

Copilot Studio workflow + SharePoint knowledge + website

### Tenant artefact names

Lab 13 - HTTP and Chatbot · Lab 13 - Investment FAQ

Trainer copies end (DO NOT DELETE). Name yours plainly — agent names 30 characters or fewer.



# Lab 13 — Build Steps

1

SharePoint → folder Lab 13 - Investment FAQ → upload the FAQ PDF.

2

New workflow Lab 13 - HTTP and Chatbot → HTTP trigger (POST, Anyone, schema with message and history).

3

+ → Compose Session: the visitor's message and the conversation so far.

4

+ → Agent node: instructions with your knowledge, collect contact details first, THE NON-ADVISORY RULE; Knowledge + → the SharePoint folder; Output Text response.

5

+ → Response: reply text. Save → Publish → URL into the website config.

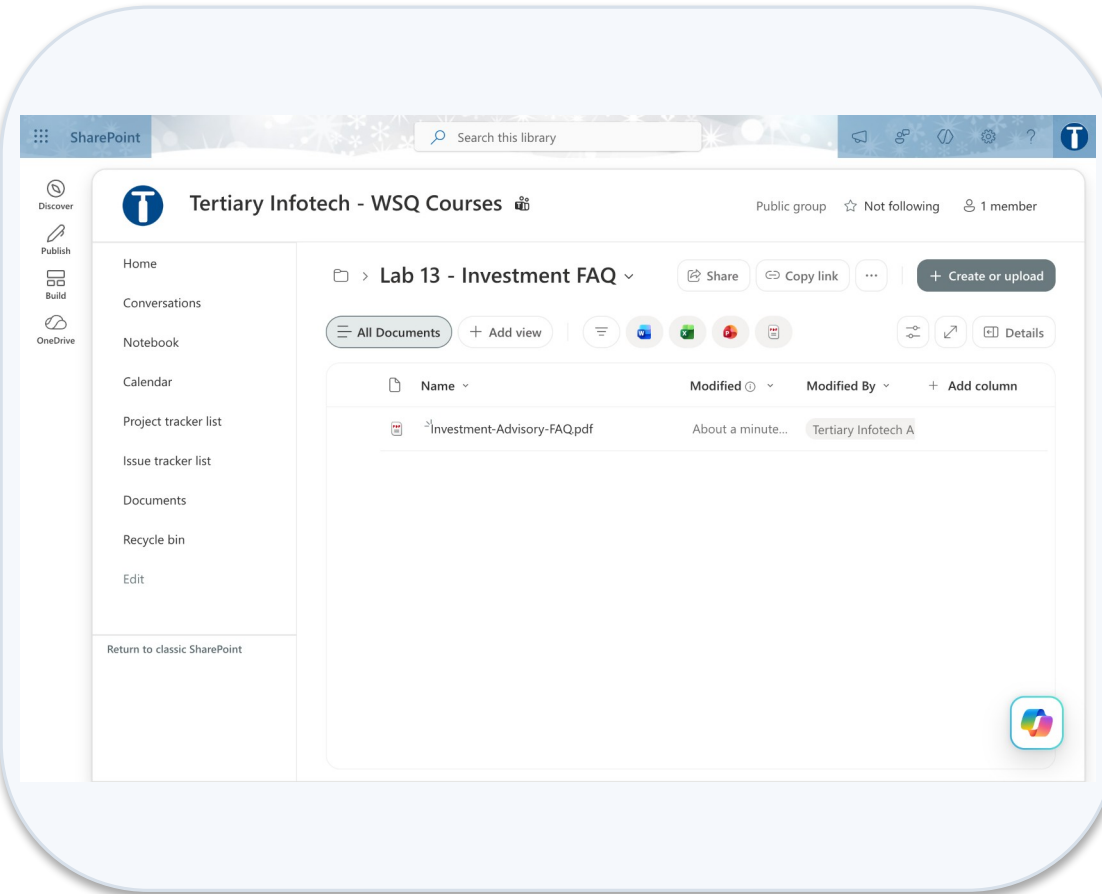
6

Open the chat page; run the FAQ questions, then the probes that ask for advice.

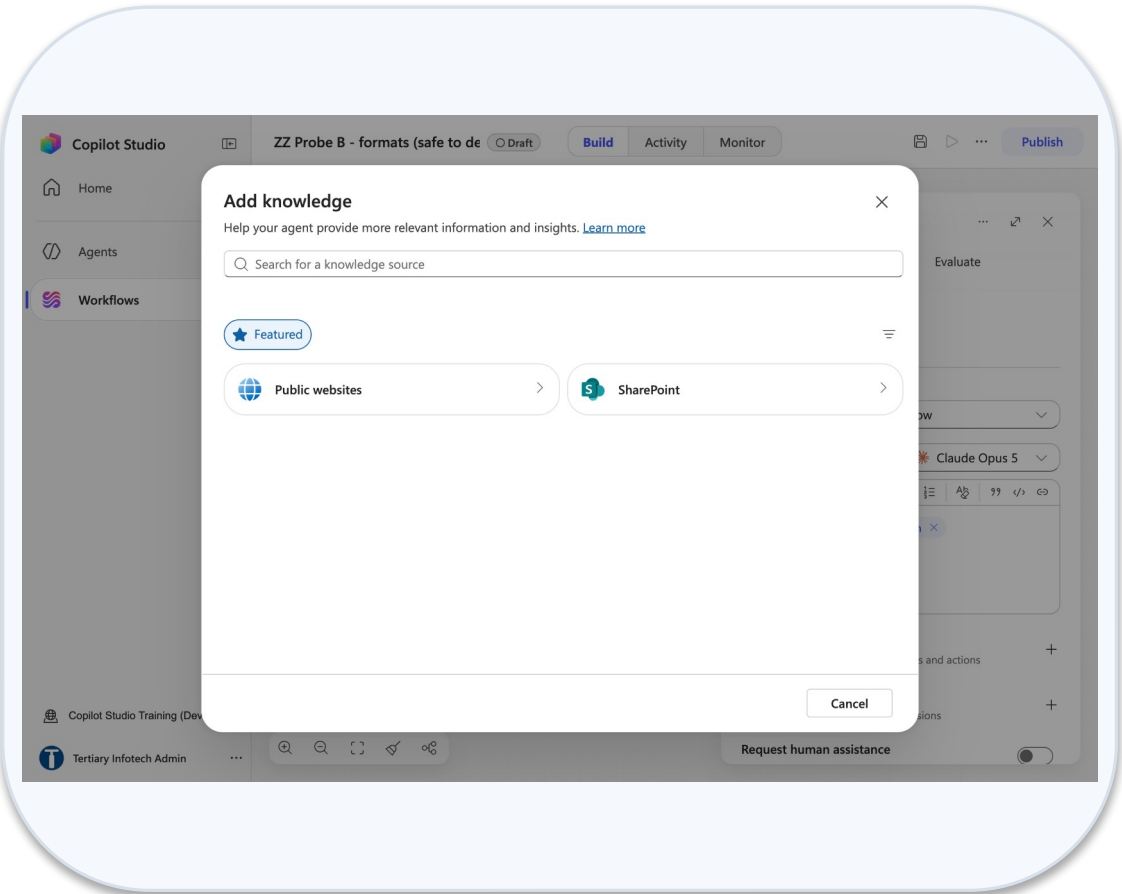
7

Debrief: rules in the instruction, facts in the knowledge — which failed, and why?

# Lab 13 — What You Will See



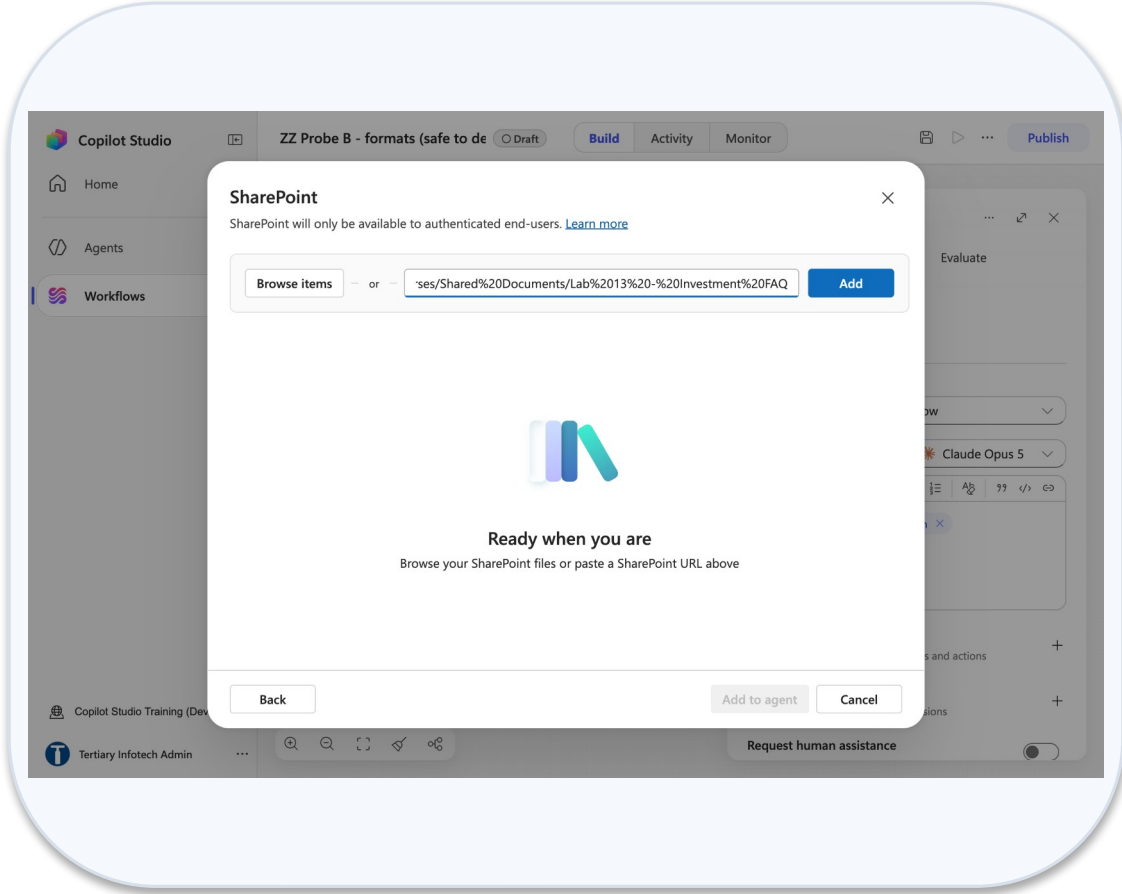
Sharepoint folder investment faq



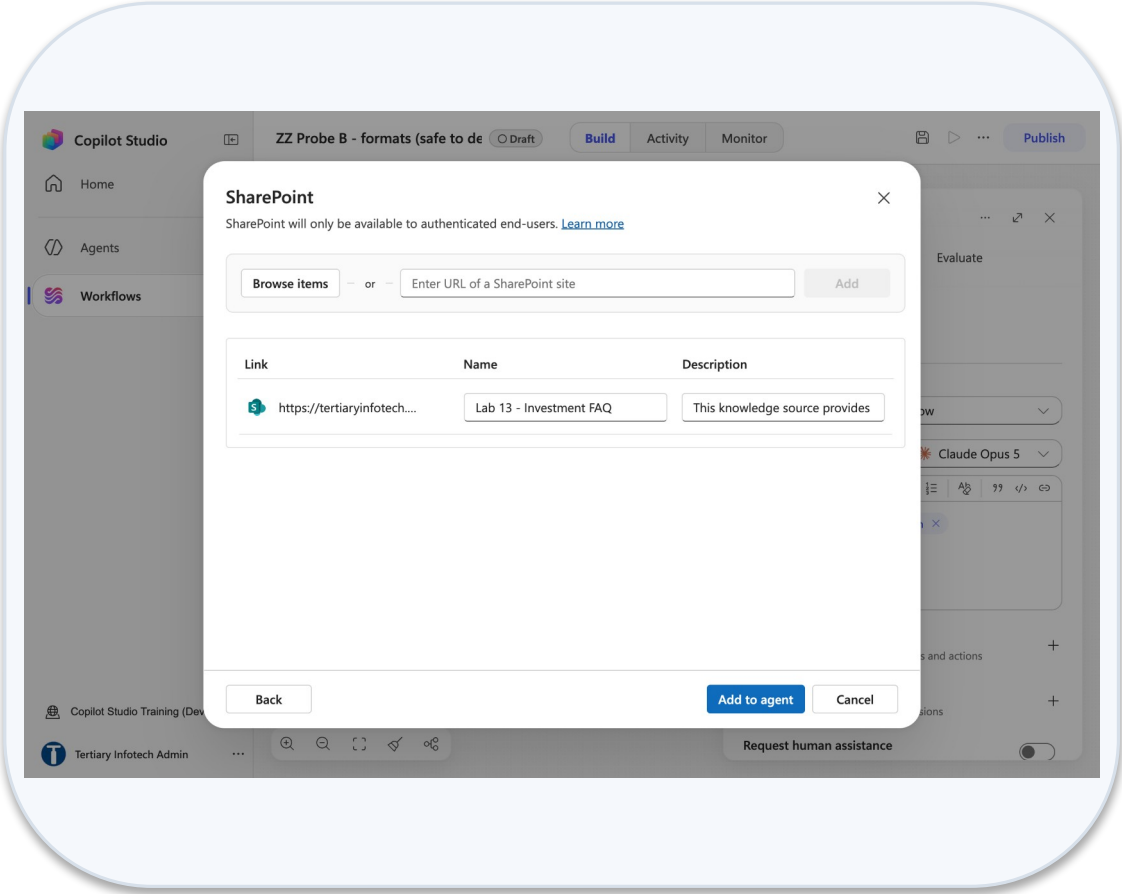
Add knowledge dialog



# Lab 13 — What You Will See (continued)



Sharepoint link encoded url



Knowledge source add to agent

# Lab 14 — HTTP and Human Review

LAB 14

35 MIN

DAY 2

## Client rapport assistant with a Teams approval gate

### What it teaches

Human in the loop — the AI drafts, and the workflow physically cannot proceed until a person approves.

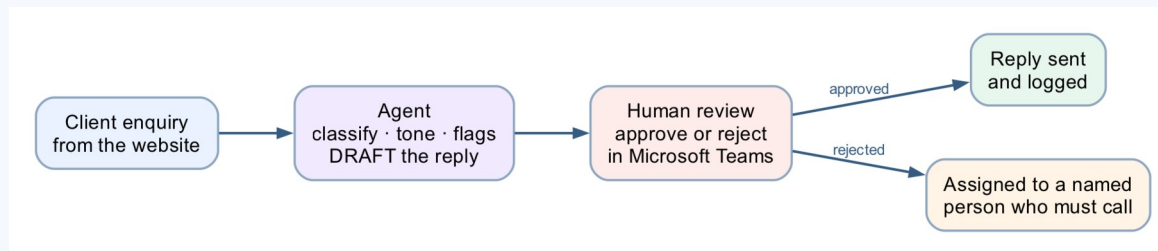
### Platform

Copilot Studio workflow + Human review + Excel Online + Teams

### Tenant artefact names

Lab 14 - HTTP and Human Review · Lab 14 - Handover Queue.xlsx

Trainer copies end (DO NOT DELETE). Name yours plainly — agent names 30 characters or fewer.



# Lab 14 — Build Steps

1

Upload Lab 14 - Handover Queue.xlsx (tables Drafts, HandoverQueue) to OneDrive → Power Automate Lab Data.

2

New workflow Lab 14 - HTTP and Human Review → HTTP trigger (POST, Anyone) with the enquiry schema.

3

+ → Compose Normalise\_Enquiry. + → Agent Rapport\_Agent: Custom structured output {draftReply, tone, flags}.

4

+ → Connectors → Response (the receipt, 200) BEFORE the gate; + → Excel Add a row into table Drafts.

5

+ → Human review: Channel Teams; Message = ⚡ draftReply; Assigned to = you; inputs Outcome (Yes/No), Name (Text).

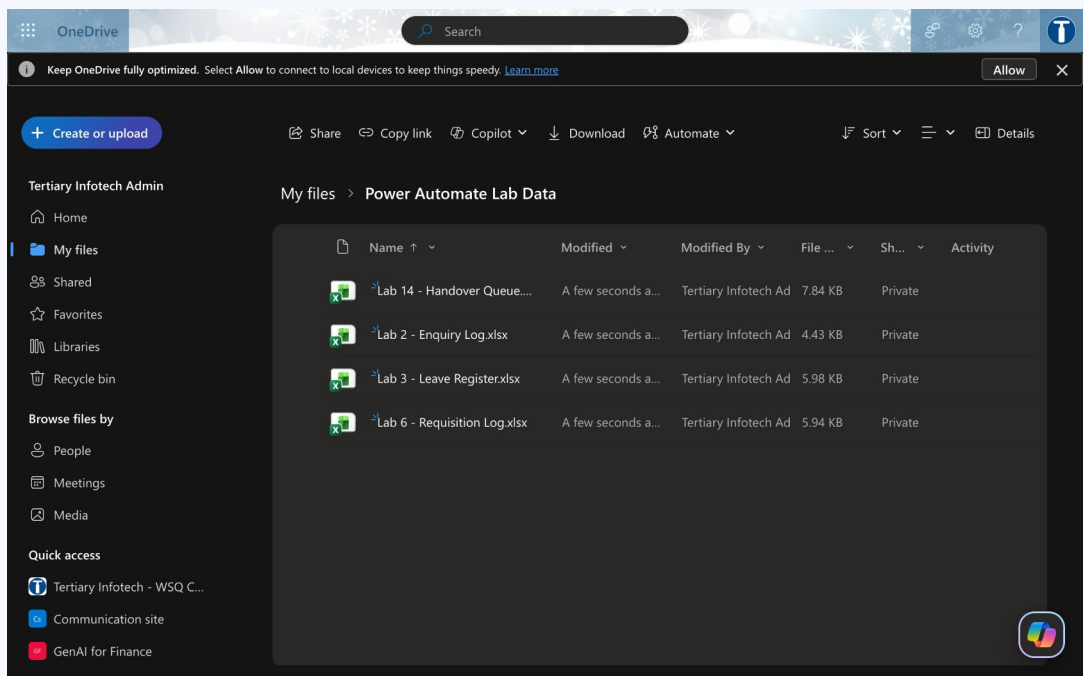
6

+ → If/Else: ⚡ Outcome Equals Yes. If: Send an email (V2) with the draft. Else: Add a row into table HandoverQueue.

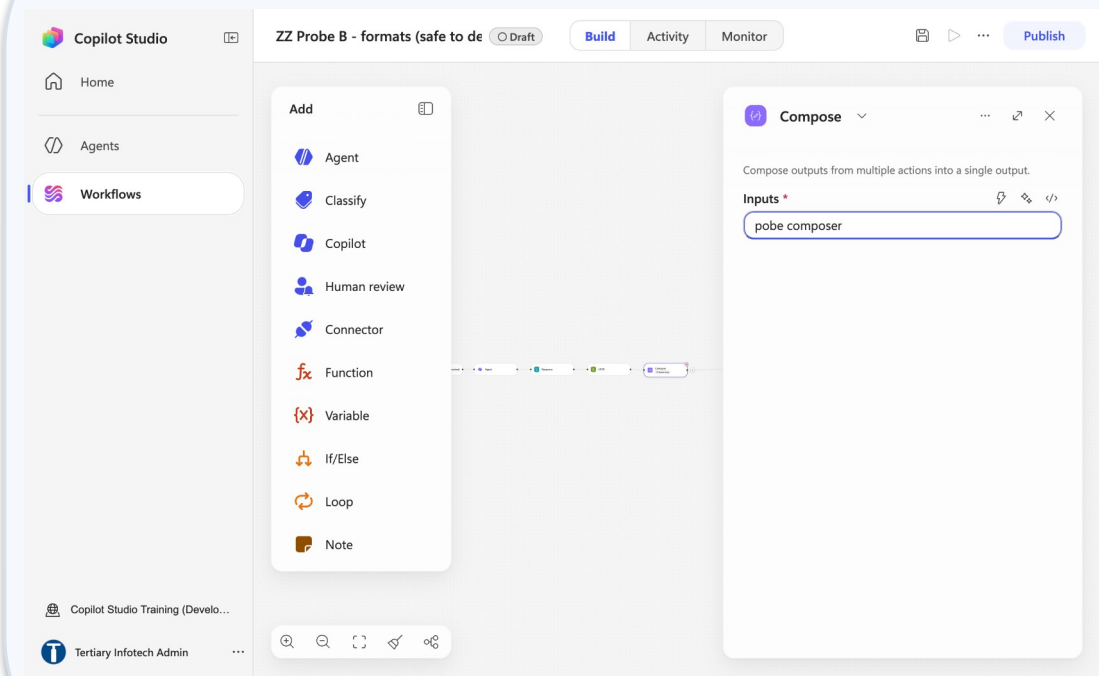
7

Publish → website → submit → Activity shows Running; answer the Teams Workflows card; run one approval and one rejection.

# Lab 14 — What You Will See

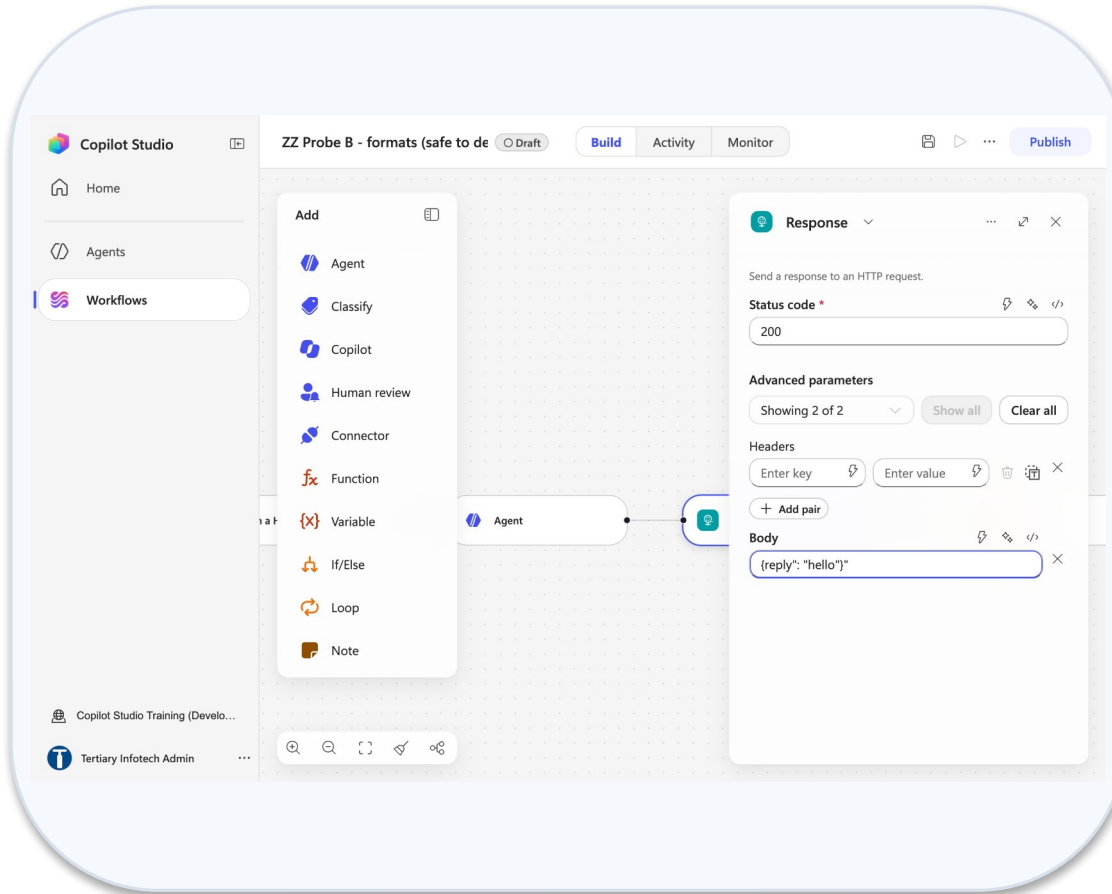


Onedrive power automate lab data

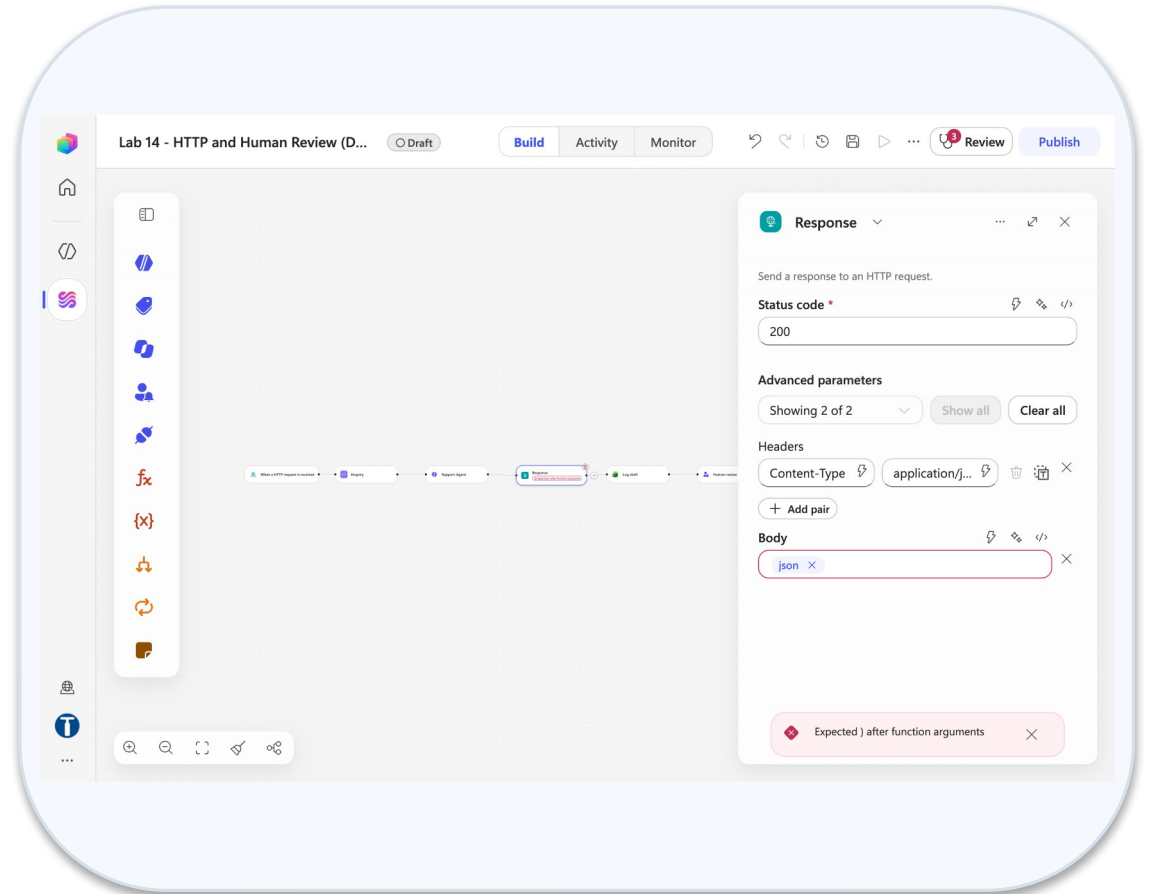


Compose inputs field

# Lab 14 — What You Will See (continued)



Response show all headers body



Response body parse error

## MODULE 5

# Retrieval Augmented Generation

Why retrieval · ingest, chunk, embed, index, retrieve, generate · Knowledge source vs Pinecone · embeddings, dimension, top\_k · citation markers · probing for invention

Applied in Lab 15 and Lab 16

# Why Retrieval, Rather Than a Bigger Prompt

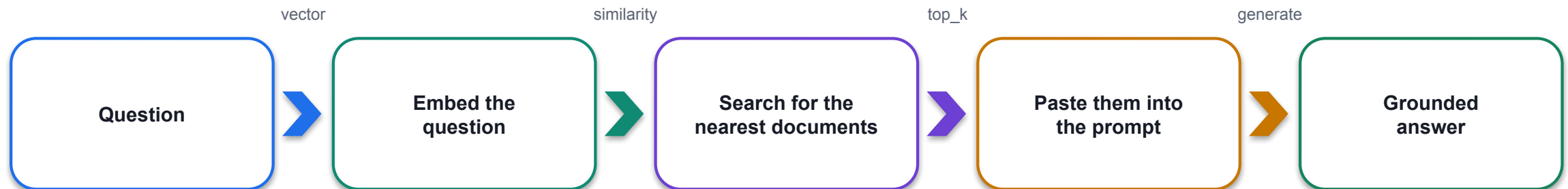
You could paste all 20 brochures into the instruction. For 20 it works. Then the academy adds 40 more.

## EVERYTHING IN THE PROMPT

- The instruction exceeds what the model can attend to; it ignores the middle
- Every question costs the price of 60 brochures
- A fee change means editing the prompt

## RETRIEVE, THEN GENERATE

- Find the two or three brochures that resemble the question
- Give the model only those; it never sees the other 17
- Change a brochure, re-index, done

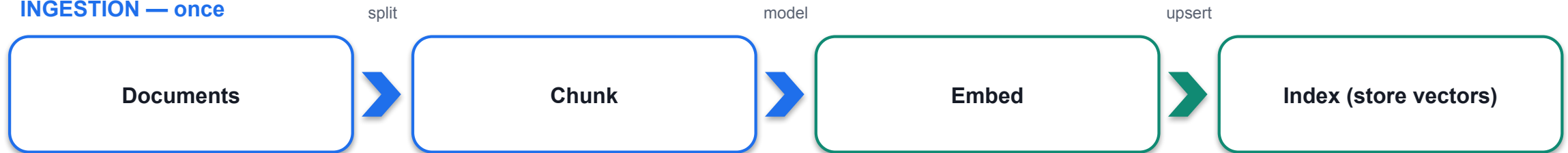


**RAG = retrieval + generation. The retrieval decides whether the generation can possibly be right.**

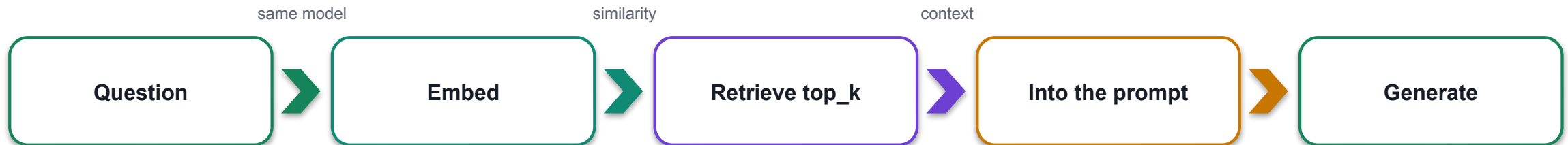
# The RAG Pipeline, and the Four Words That Matter

Two phases. Ingestion happens once; retrieval happens on every question.

## INGESTION — once



## RETRIEVAL — every question



### Chunk

How a document is split. Here: one brochure = one record.

### Embedding

Text as a list of numbers. llama-text-embed-v2, 1024 dimensions.

### Similarity

Nearness of two vectors — the machine's idea of "related".

### top\_k

How many records come back. Here: 3.



# Knowledge Source or Your Own Vector Store — The Levers

Knowledge in the Agent node (Lab 15) versus Pinecone (Lab 16): the same chatbot, with the levers back.

|                       | LAB 15 — Knowledge source                  | LAB 16 — Pinecone                                            |
|-----------------------|--------------------------------------------|--------------------------------------------------------------|
| Nodes                 | 3 — trigger, Agent (Knowledge +), Response | 5 — trigger, HTTP (Pinecone query), Compose, Agent, Response |
| Ingestion             | Upload to SharePoint, wait for Ready       | A Python script, run once (integrated embedding)             |
| Embedding model       | Hidden                                     | llama-text-embed-v2, 1024 dimensions — your choice           |
| Chunking              | Hidden                                     | One brochure = one record — your call                        |
| top_k                 | Hidden                                     | 3 — your call                                                |
| Citation markers      | Leak into the reply                        | None — plain Compose output                                  |
| A changed fee         | Edit, wait for re-crawl                    | Edit, then re-ingest                                         |
| When it answers badly | Rewrite the prompt and hope                | Four levers to pull                                          |

Neither is the right answer. Which is right depends on whether the person maintaining it will ever need those levers.

# Embeddings, Dimension and top\_k

Three numbers you set in Lab 16 — and what each one changes.

## Embedding model

1

llama-text-embed-v2 via Pinecone integrated embedding: the index embeds on upsert and on query, so the same model is guaranteed on both sides.

## Dimension = 1024

2

The length of every vector. Fixed by the model, fixed for the index — change the model and you rebuild the index.

## top\_k = 3

3

How many nearest records the query returns. Too few: the right brochure is missed. Too many: the prompt fills with near-misses.

## Similarity metric

4

cosine — the angle between vectors. "Related" means near, not identical; a question about macarons finds the pastry brochure.

```
POST https://<index-host>/records/namespaces/brochures/search header: Api-Key · X-Pinecone-API-Version
{"query": {"inputs": {"text": "@{outputs('Question')}" }, "top_k": 3 }, "fields": ["text", "course"] }
```

# Citation Markers

Citation markers: what grounding appends, and how each lab handles them.

## LAB 15 — Knowledge source

- Replies can arrive twice: [doc:turn1doc11] then [1]-style markers
- Appended by the grounding layer, not written by the model
- Instruction line: never include citation markers, reference numbers or source tags

## LAB 16 — Pinecone

- Retrieved text arrives as plain Compose output — no metadata
- No markers, no duplication
- You decide whether to cite: add the course name from the record's fields

**Budget for the markers when converting a lab from a vector store to a Knowledge source.**

### HEADERS THAT BITE

The HTTP node truncates header names visually — X-Pinecone-API-Version can silently lose its leading X. Never paste an env-var NAME as a header value; Power Automate cannot read .env.

# Probing for Invention

A grounded agent still invents. You find out by asking for things that do not exist.

## A course you do not run

"Do you have a course on cake sculpture?"

It must say it does not — not improvise a syllabus.

## A fact in no brochure

"Is there parking at the east campus?"

It must say the brochures do not cover it.

## A discount that does not exist

"What is the student discount?"

It must not invent a percentage to be helpful.

## A CONFIDENT WRONG ANSWER RAISES NO ERROR

The run is green. The reply is fluent. Nothing in Activity is red. In every lab here the wrong answer looks exactly like a right one — which is why the test tables include the probes they do.

### LAB 15

## Lab 15 — RAG with Knowledge Base

Cook & Bake Academy, built-in knowledge · Copilot Studio Knowledge + SharePoint · 30 min

# Choosing

Choosing between the two — the question to ask is about the maintainer, not the model.

| If the situation is...                        | ...choose                                                  | Seen in |
|-----------------------------------------------|------------------------------------------------------------|---------|
| Documents live in SharePoint and change often | Knowledge source                                           | Lab 15  |
| A public agent, reply wording must be exact   | Pinecone — no citation markers                             | Lab 16  |
| Retrieval quality must be tuneable            | Pinecone — chunking, top_k, metric                         | Lab 16  |
| Nobody will ever touch it after go-live       | Knowledge source                                           | Lab 15  |
| Answers must be traceable to a record         | Either — but write the source name into the reply yourself | Both    |

Both start from the same brochures and the same instruction. Only the retrieval changes — and so does everything you can inspect.

LAB 16

Lab 16 — RAG with Pinecone

The same chatbot, with the levers back · Copilot Studio workflow + Pinecone · 30 min

# Lab 15 — RAG with Knowledge Base

LAB 15

30 MIN

DAY 2

## Cook & Bake Academy, built-in knowledge

### What it teaches

RAG as a product setting — three nodes, no ingestion, and nothing you can inspect.

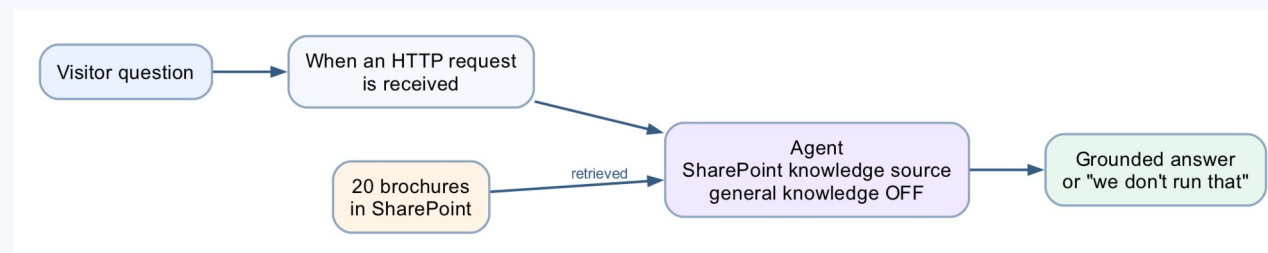
### Platform

Copilot Studio Knowledge + SharePoint

### Tenant artefact names

Lab 15 - RAG with Knowledge Base · Lab 15 - Course Brochures

Trainer copies end (DO NOT DELETE). Name yours plainly — agent names 30 characters or fewer.



# Lab 15 — Build Steps

1

SharePoint → folder Lab 15 - Course Brochures → upload the Cook & Bake brochures.

2

New workflow Lab 15 - RAG with Knowledge Base → HTTP trigger (POST, Anyone) with a question field.

3

+ → Agent node: Knowledge + → SharePoint → the folder; instructions with the citation-marker line; Output Text response.

4

+ → Response with the answer. Save → Publish → URL into the chat page.

5

Ask three real questions with exact fees; then the probes — cake sculpture, parking, student discount.

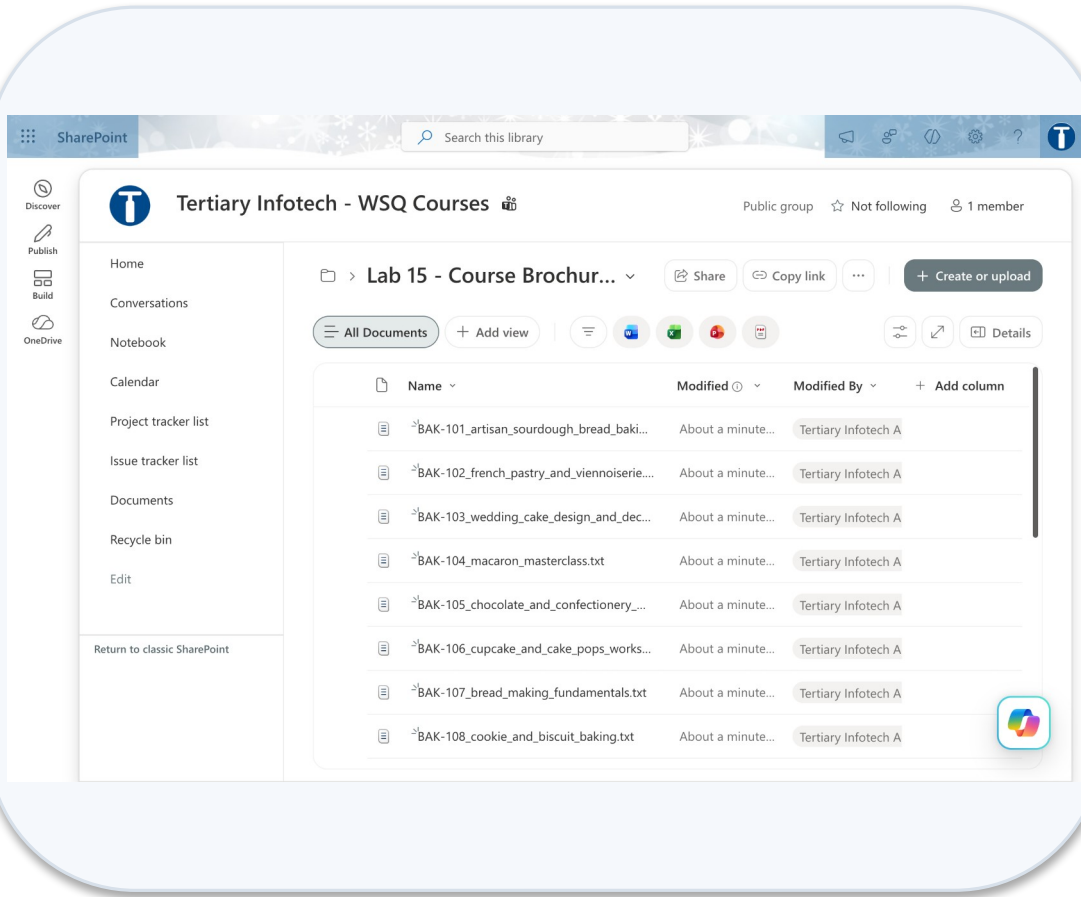
6

Read Activity: nothing about chunking, embedding or top\_k is visible. That is the point.

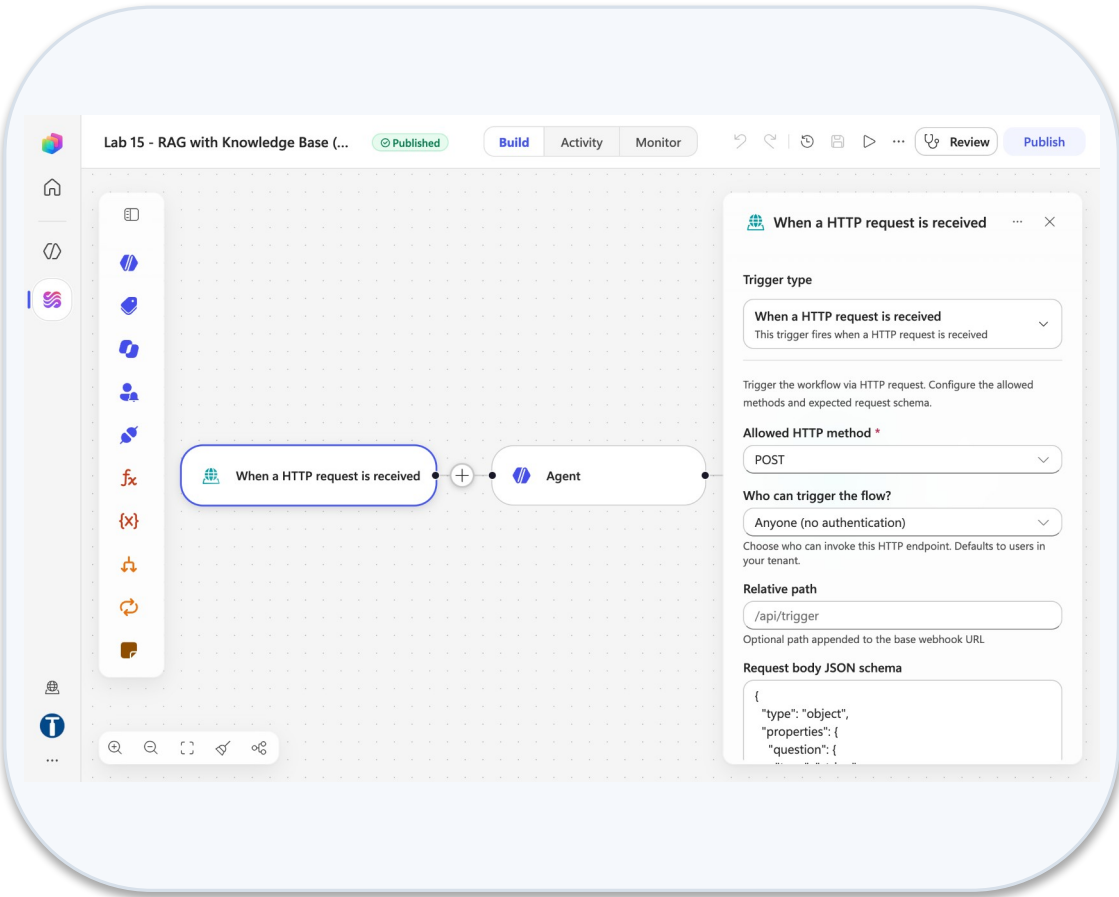
7

Debrief: what would you change if a fee were wrong tomorrow?

# Lab 15 — What You Will See



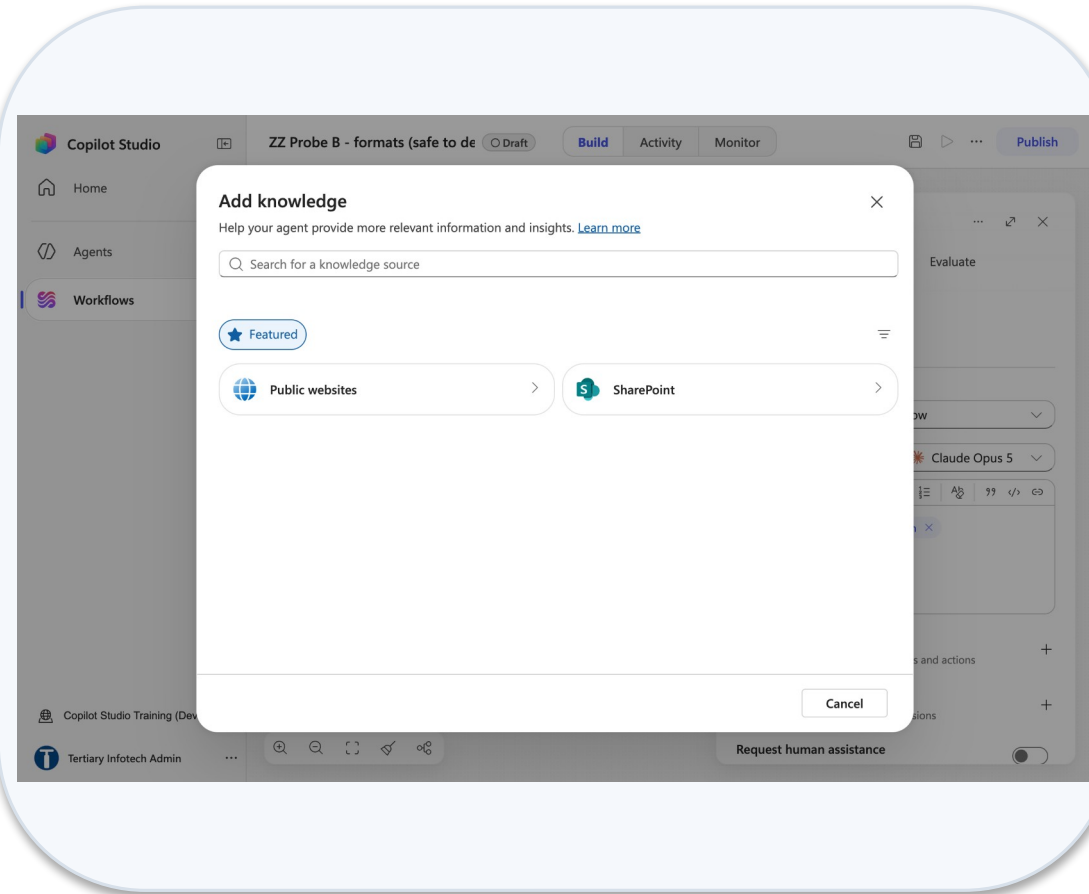
Sharepoint lab 15 course brochures folder



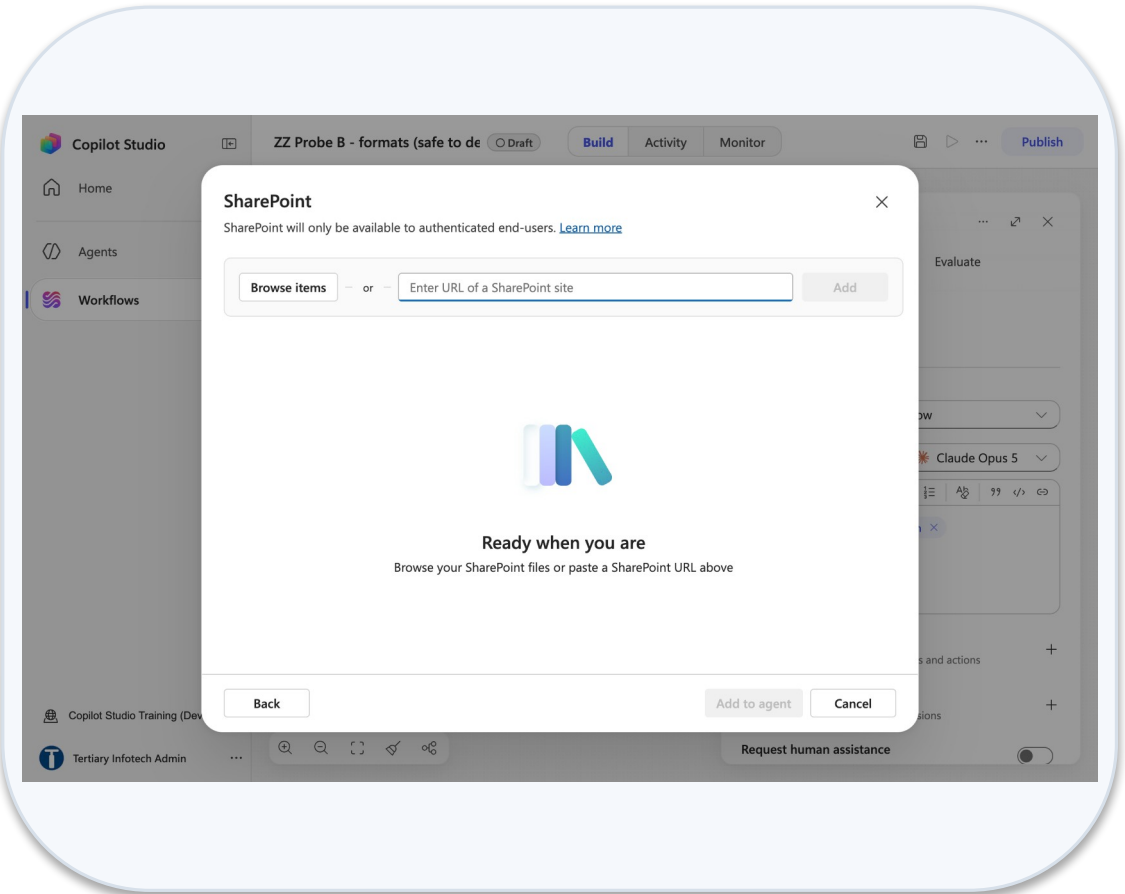
Http trigger panel post anyone schema



# Lab 15 — What You Will See (continued)



Agent add knowledge dialog



Sharepoint knowledge link field

# Lab 16 — RAG with Pinecone

LAB 16

30 MIN

DAY 2

The same chatbot, with the levers back

## What it teaches

What the convenience cost — chunking, embeddings, dimension and top\_k, back in your hands.

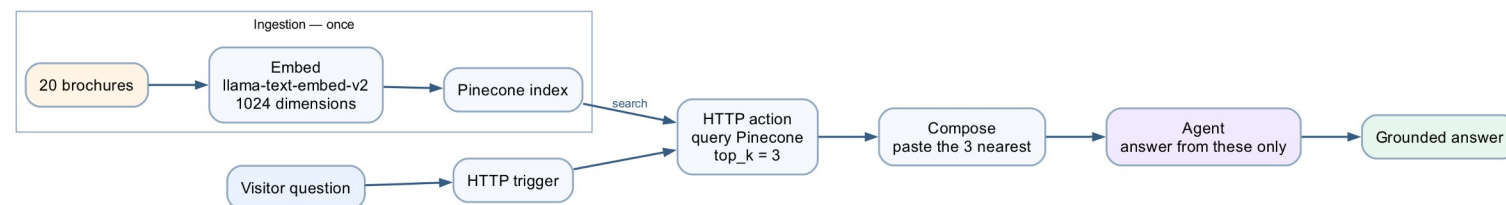
## Platform

Copilot Studio workflow + Pinecone

## Tenant artefact names

Lab 16 - RAG with Pinecone · lab16-course-brochures

Trainer copies end (DO NOT DELETE). Name yours plainly — agent names 30 characters or fewer.



# Lab 16 — Build Steps

1

Run the ingestion script once: creates the Pinecone index lab16-course-brochures (llama-text-embed-v2, 1024, cosine) and upserts one record per brochure.

2

New workflow Lab 16 - RAG with Pinecone → HTTP trigger with a question field.

3

+ → HTTP node: POST to the index's records search; Show all → headers Api-Key (your key), Content-Type, X-Pinecone-API-Version 2025-04; body with the ⚡ question and top\_k 3.

4

+ → Compose Prompt: join the returned text fields with the question into one block.

5

+ → Agent node: instructions reference the ⚡ Prompt; no Knowledge source; Output Text response.

6

+ → Response. Save → Publish → the same chat page as Lab 15.

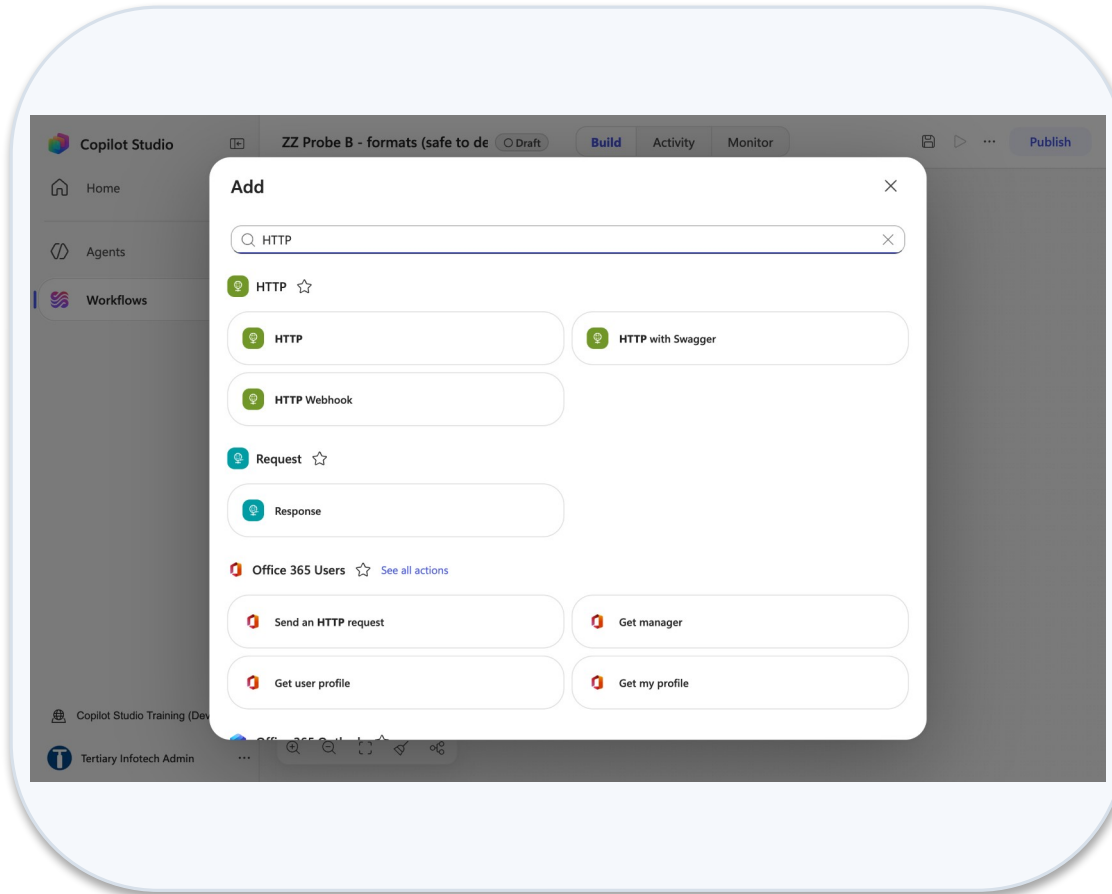
7

Change top\_k to 1 and to 5; re-ask the same question; read what changed in Activity.

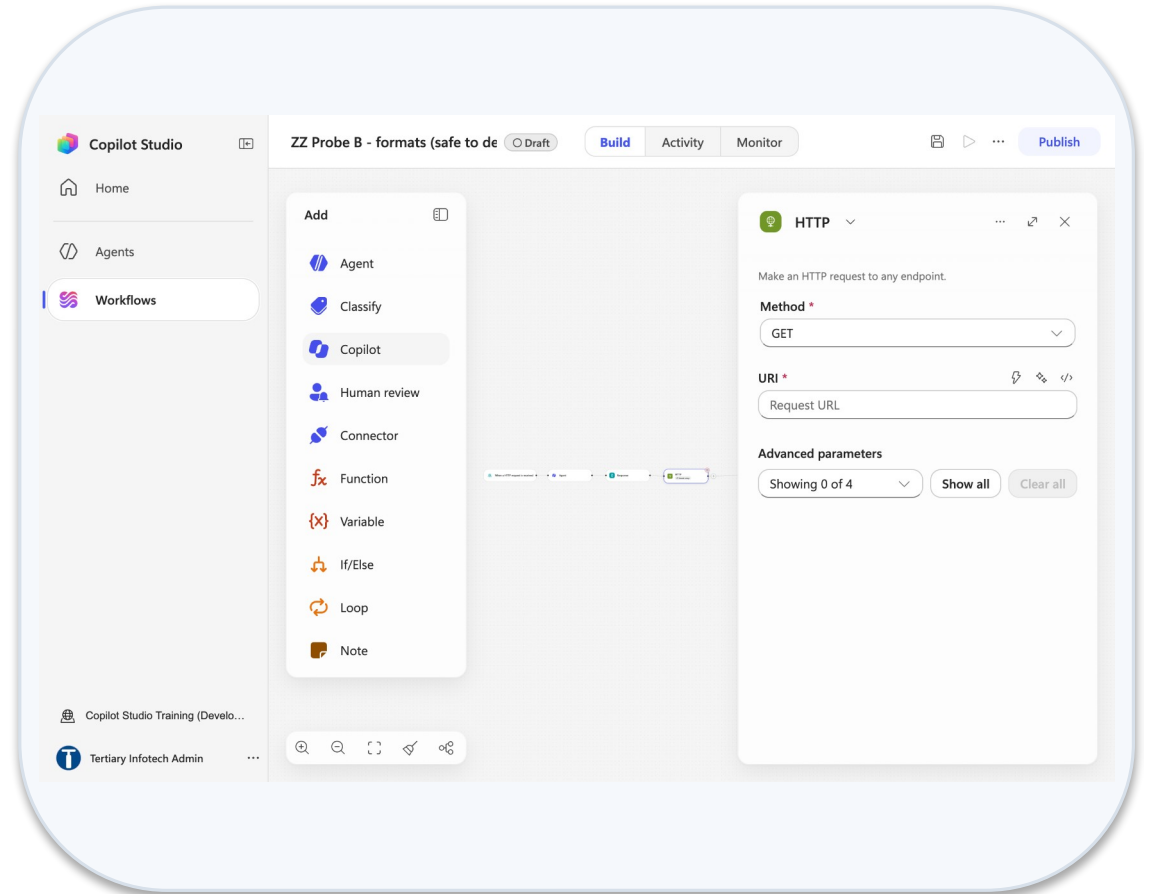
8

Debrief: four levers — chunking, embedding model, dimension, top\_k — and which one you would touch first.

# Lab 16 — What You Will See

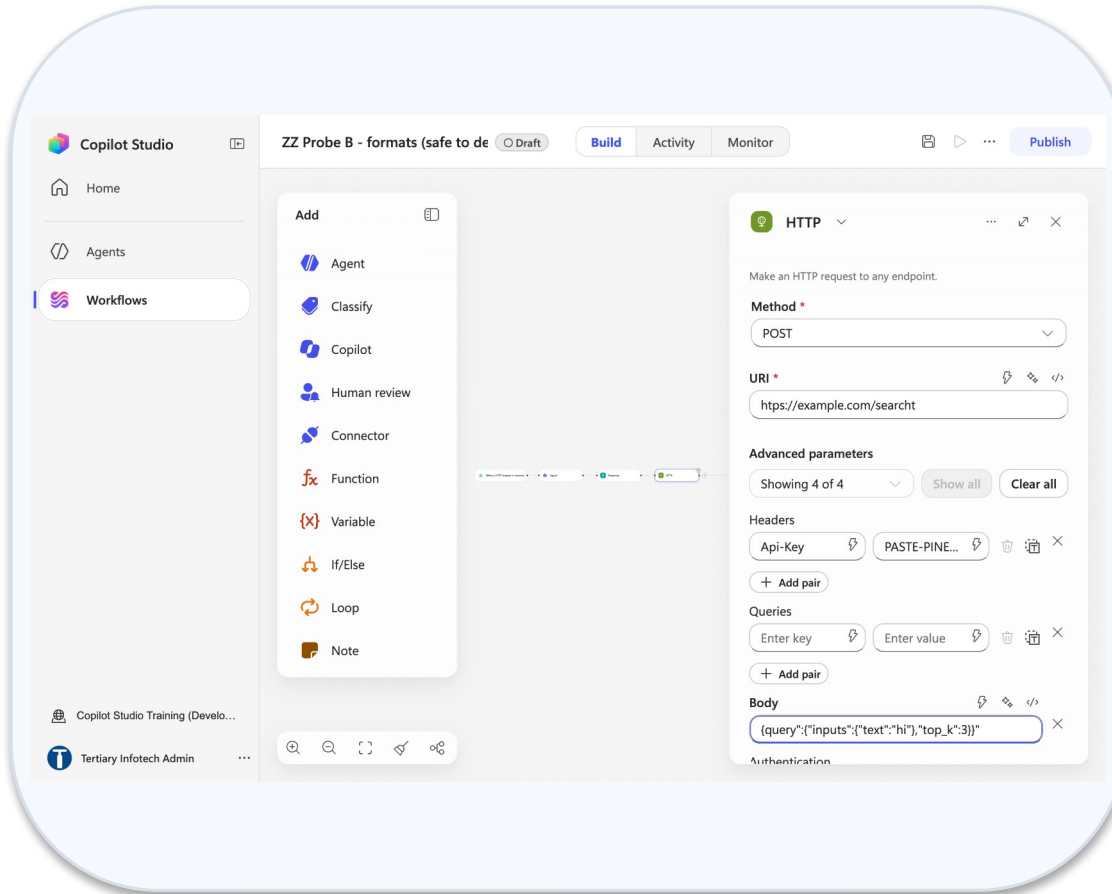


Add dialog search http

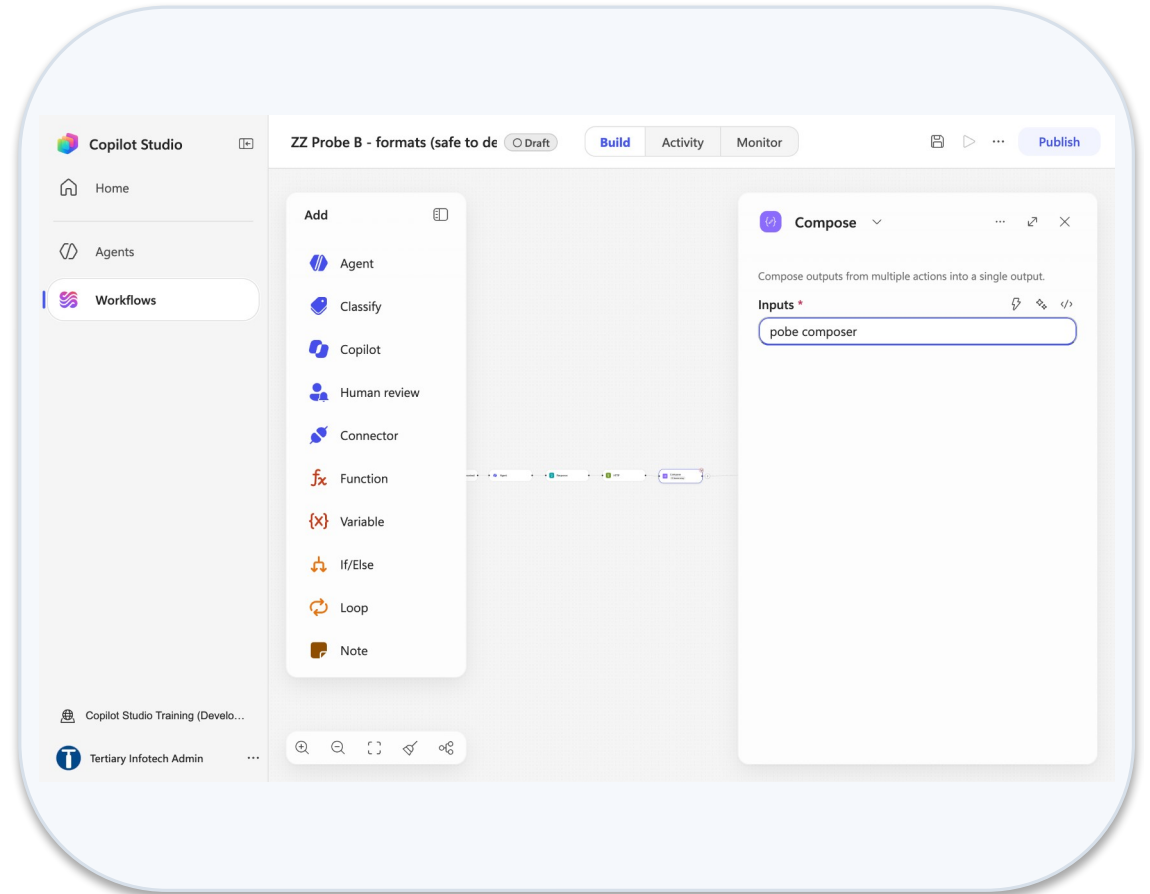


Http node method uri show all

# Lab 16 — What You Will See (continued)



Http node api key header body



Compose node inputs

# Lab 17 — Publish to Teams, Microsoft 365 Copilot and the Web

LAB 17

25 MIN

DAY 2

**Channels + — where a published agent meets its users**

## What it teaches

Publish is what channels serve; each channel needs something different, and none of them changes the agent — a channel is a door, not a brain.

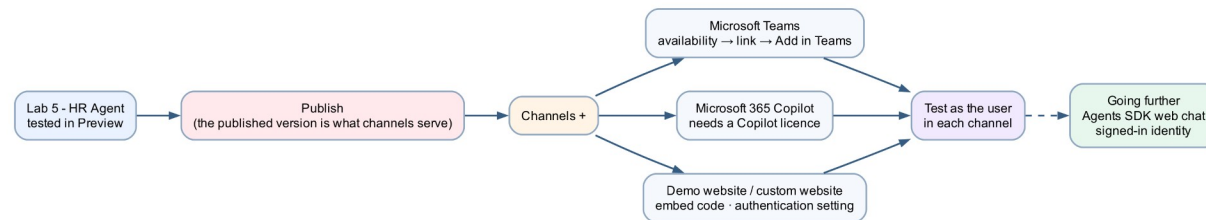
## Platform

Copilot Studio Channels + Microsoft Teams + Microsoft 365 Copilot + website

## Tenant artefact names

Lab 5 - HR Agent

Trainer copies end (DO NOT DELETE). Name yours plainly — agent names 30 characters or fewer.



# Lab 17 — Build Steps

1

Open Lab 5 - HR Agent → Publish (the dialog shows Last published and the channel list) → Publish agent.

2

Channels + → Teams + Microsoft 365 → Availability → Add channel → chip appears → Use and share → View in Teams → run the Lab 5 tests as a colleague.

3

Share → add a classmate; they need the link AND access.

4

Same channel → Use and share → View in Copilot (needs a Copilot licence on the user's side); Submit to org catalog is the admin-approval path.

5

Open Lab 7 - Sales Agent → ... → Settings → Safety & access → Authentication → No authentication (Done stays disabled; close X) → Save → Publish → the Demo website channel appears → View details → Open demo website.

6

Optional: paste the embed <iframe> into an HTML page.

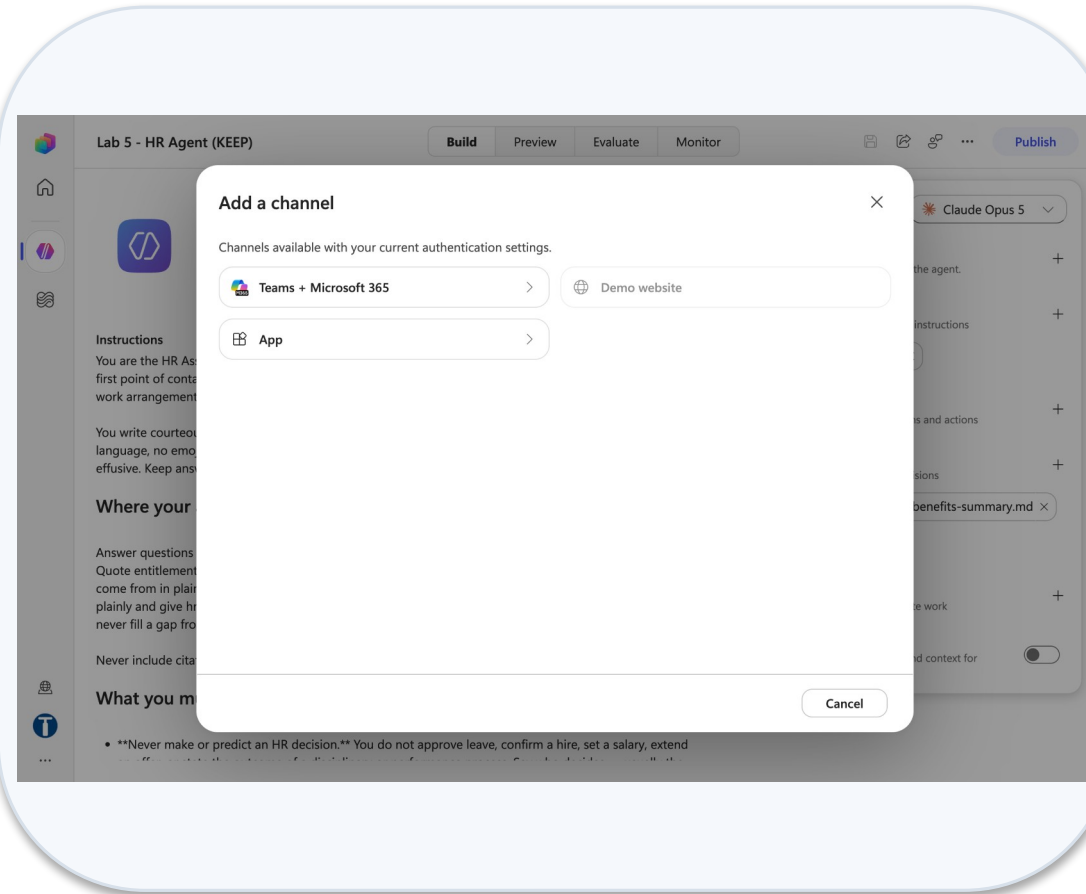
7

On the HR agent, Channels + shows the Demo website tile greyed: Requires "No authentication" — do NOT switch it off.

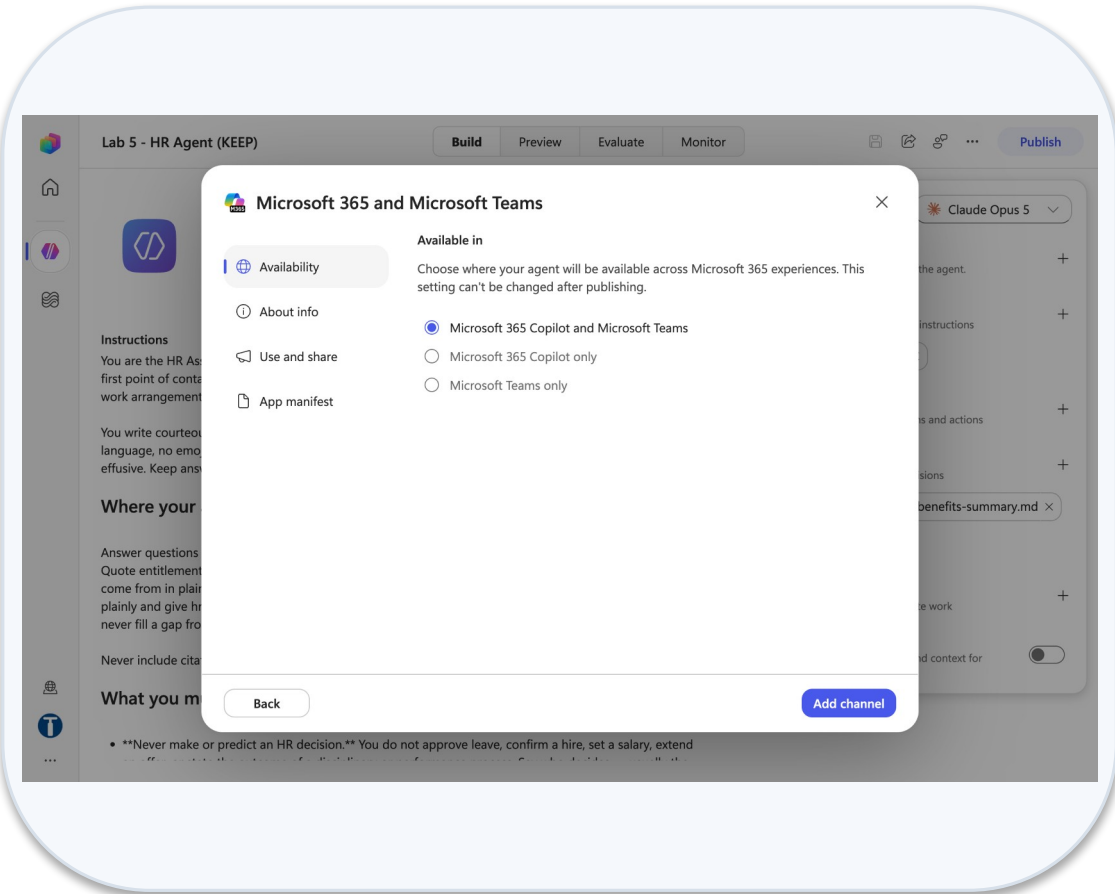
8

Deploy parents only; connected agents get no channel. Test as the user, in each door.

# Lab 17 — What You Will See



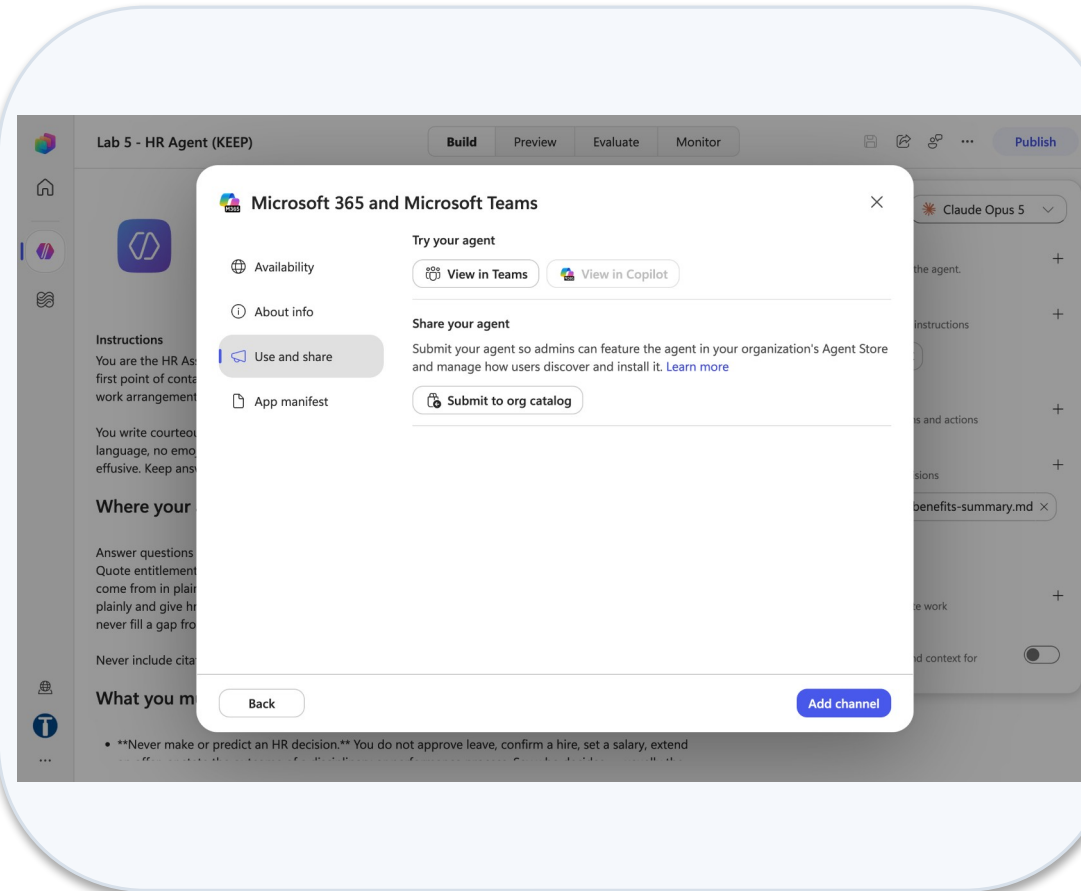
Add a channel dialog hr agent



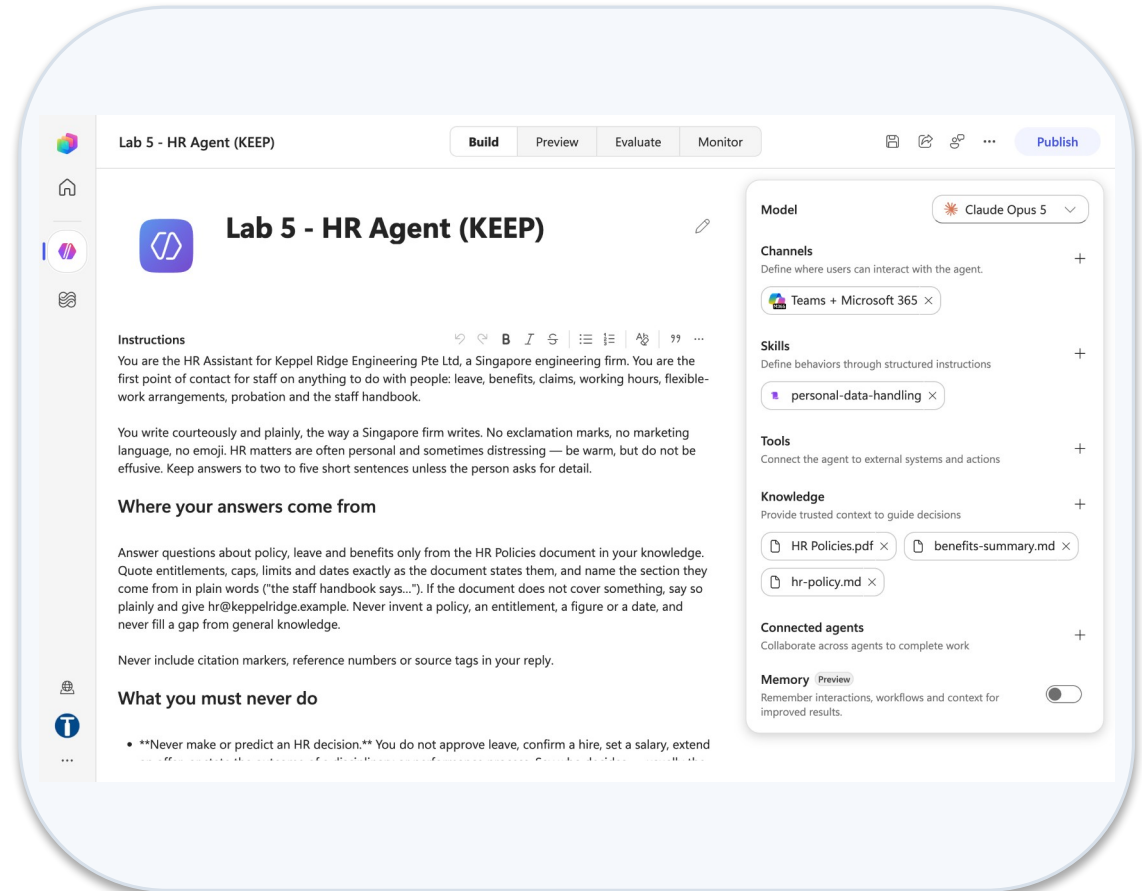
Teams m365 availability tab



# Lab 17 — What You Will See (continued)



Teams m365 use and share tab



Teams m365 channel chip added

## SYNTHESIS

# Taking This Back to Work

Controls versus conventions · the eighteen-lab recap · the pre-production checklist · further reading · six sentences worth keeping

# Controls versus Conventions

Back at work: which of the things you built are controls, and which are conventions you asked a model to follow?

| What you built                                         | Which is it? | Where                          |
|--------------------------------------------------------|--------------|--------------------------------|
| A tool the agent does not have                         | Control      | Lab 8 — no admin-rights tool   |
| The workflow writes master data from Compose           | Control      | Lab 12                         |
| Get items duplicate check before the Agent node        | Control      | Lab 12                         |
| Human review node with Outcome Equals Yes              | Control      | Labs 4, 14                     |
| Authentication + channel choice                        | Control      | Lab 17                         |
| "Never discuss another person's leave" in instructions | Convention   | Lab 5 — probe it every release |
| A skill's procedure text                               | Convention   | Lab 8                          |
| Request human assistance when unsure                   | Convention   | Never as a gate                |

Ask who pays for the mistake. Put a control where the answer is expensive; a convention is fine where it is cheap.

# Eighteen Labs in One Slide

Eighteen labs, five modules, two days — each lab reusing the one before it.

## M1 — Workflows

Labs 0–2: environment, trigger and actions, log to Excel

## M2 — Control flow, human in the loop

Labs 3–4: Approvals, If/Else, Classify, Human review in Teams

## M3 — Agents

Labs 5–9: HR, Procurement, Sales, IT Support, the content team · Labs 10–11: agent ↔ workflow · Lab 17: channels

## M4 — HTTP and the boundary of agency

Labs 12–14: approval agent, chatbot alone, the human review gate over HTTP

## M5 — RAG

Labs 15–16: Knowledge source, then Pinecone with the levers back

**Every artefact is named Lab N - <Title>; trainer workflows and agents both end (DO NOT DELETE) — agent names max 30 characters, so agent bases are shortened.**

# Before Anything Reaches Production

1

## Environment

Correct environment; Copilot Credits allocated; agent limit set with Stop usage.

2

## Data path

No model output writes to a system of record without a deterministic step in between.

3

## Refusals tested

Every prohibition has a probe that tries to break it, and the probe is in an Evaluate test set.

4

## Human placed

Consequential paths end at a Human review node assigned to a named person — not a queue nobody watches.

5

## Audit before the gate

The audit row is written before the approval, so rejected cases still leave a trace.

6

## Sources closed

Search all websites removed; knowledge Ready; citation-marker line present; authentication matches the channel.

**A green run is not a correct run, and a fluent answer is not a true one. Verify both before anyone depends on it.**

# Further Reading — The Six Sources

The six sources this deck was built from — read them in this order.

| Source                                                                        | URL                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Agents powered by the GitHub Copilot harness — overview</b>                | <a href="https://learn.microsoft.com/en-us/microsoft-copilot-studio/agents-experience/overview">learn.microsoft.com/en-us/microsoft-copilot-studio/agents-experience/overview</a>                                                                                                                                 |
| <b>What's new in Copilot Studio</b>                                           | <a href="https://learn.microsoft.com/en-us/microsoft-copilot-studio/whats-new">learn.microsoft.com/en-us/microsoft-copilot-studio/whats-new</a>                                                                                                                                                                   |
| <b>New and improved: GitHub Copilot harness, agent skills, richer context</b> | <a href="https://microsoft.com/en-us/microsoft-copilot/blog/copilot-studio/new-and-improved-github-copilot-harness-agent-skills-and-richer-context/">microsoft.com/en-us/microsoft-copilot/blog/copilot-studio/new-and-improved-github-copilot-harness-agent-skills-and-richer-context/</a>                       |
| <b>Start building (authoring-first-bot) + natural-language quickstart</b>     | <a href="https://learn.microsoft.com/en-us/microsoft-copilot-studio/agents-experience/authoring-first-bot">learn.microsoft.com/en-us/microsoft-copilot-studio/agents-experience/authoring-first-bot</a>                                                                                                           |
| <b>New Copilot Studio tech guide — BlastBox Omega sample</b>                  | <a href="https://github.com/microsoft/new-copilot-studio-tech-guide">github.com/microsoft/new-copilot-studio-tech-guide</a>                                                                                                                                                                                       |
| <b>Introducing a new harness for Copilot Studio (Tech Community)</b>          | <a href="https://techcommunity.microsoft.com/blog/copilot-studio-blog/more-powerful-agents-and-workflows-for-autonomous-business-processes-introducing/4542969">techcommunity.microsoft.com/blog/copilot-studio-blog/more-powerful-agents-and-workflows-for-autonomous-business-processes-introducing/4542969</a> |

**Also: the course repository (labs/), the Learner Guide, and .claude/skills/copilot-studio-flows — every entry tenant-verified.**

# Six Sentences Worth Keeping

1

**A control the model cannot reach beats a rule you asked it to follow.**

2

**A reference to nothing resolves to empty, not to an error.**

3

**Commit the record of the obligation before you create the obligation.**

4

**The AI decides what to do; deterministic steps do what must always happen.**

5

**A pause that will still be paused tomorrow is a real gate.**

6

**Retrieval decides whether generation can possibly be right.**

Six sentences. If you keep only these, you can rebuild the judgement behind every lab in this course.

# Briefing for Assessment

1

Place phones and other materials under the table or on the floor.

2

No photos or recording of assessment scripts.

3

No discussion during the assessment.

4

Use a black/blue pen for hard-copy assessments.

5

No liquid paper / correction tape.

6

Scripts are collected when time is up.



# Assessment

## Two parts

Written Assessment (SAQ) — 1 hour. Practical Performance (PP) — 1 hour.

1

## Open book

Slides, Learner Guide and approved materials only.

2

## Attendance

Remember to take the Assessment digital attendance (TRAQOM).

3

## Submit

Submit your completed answers on the LMS at <https://lms-tms.tertiaryinfotech.com/>.

4

# Assessment Flow



# Practice Exam

## Sharpen up

1

Attempt the Tertiary Infotech practice exam for this course under timed conditions.

## Where

2

<https://exams.tertiaryinfotech.com> — search for the course title or TGS-2022017524.

## Review

3

Read every explanation; revisit the lab whose concept you missed.

## Repeat

4

Re-take the practice exam until the concepts, not the answers, are automatic.

<https://exams.tertiaryinfotech.com>

# Digital Attendance (Mandatory)

## AM, PM and Assessment

1

It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.

## The SSG QR code

2

The trainer/administrator displays the digital attendance QR code from the SSG portal.

## Scan and submit

3

Scan the QR code with your mobile phone camera and submit your attendance.

## 75% minimum

4

A minimum of 75% attendance is required to be eligible for assessment and funding.

**TRAQOM is the SSG digital attendance system — take it three times a day, every day.**

SEE YOU IN THE TENANT

# Thank You

Now go automate something — and put the human where the consequence lands.  
Questions welcome.